

Adaptive Design of Clustered Experiments

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Abstract

In clustered experiments with cluster-level fixed costs and a budget, designers often find themselves trading between the number of clusters and the number of units per cluster to minimize the variance of a population mean or ATE estimator. Sampling additional clusters reduces the statistical dependence between observations, but it is also costly, what creates a trade-off between statistical accuracy and costs mediated by the intra-cluster correlation (ICC). Current cluster designs depend on ICC guesstimates that entail substantial inefficiencies. In this context, I propose two pilot-based algorithms that implement near-optimal designs without previous knowledge on the ICC. Under a symmetry assumption on the errors, I further show that inference on the population mean remains unchanged when conditioning on designs given by my algorithms. In fact, I prove a strong result on this regard. Under symmetry, inference on group sample averages is not compromised when data collection is adaptively driven by asymptotically even functionals of the data, like sample variances. Asymptotic normality is preserved without discarding or adjusting the data. Beyond my algorithms, this principle guarantees valid inference for popular adaptive sampling procedures such as Neyman allocations after sample variance estimation with pilot data. Finally, I show that the algorithms are minmax optimal up to logarithmic factors and improve upon any misspecified static policy. These results provide a foundation for adaptive sampling in clustered surveys and randomized controlled trials.

1 Introduction

Over the last few years, surveys and Randomized Control Trials (RCT) have become increasingly larger, not only in size but in scope, spanning several regions in the quest for external validity and efficiency (Muralidharan and Niehaus (2017)).¹ Multi-cluster designs are prevalent in the Survey and Applied Economics literature, including applications to development economics, education, health, finance, trade, gender and more (Duflo and Banerjee (2017)). For instance, Mbiti et al. (2019) deploy an experiment at the school level consisting of a combination of school grants and teacher incentives. Because of logistical concerns, their implementation featured random selection of 10 districts from mainland Tanzania and then, random treatment allocation to 35 schools in each district. Analogously, Sedgwick (2015) designed a clustered survey to learn about the preferences of British households on

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¹Regions can be physical or figurative. Examples of physical regions include countries, provinces, villages, etc. and examples of figurative regions include demographic strata, markets, industries, etc.

medical data privacy. Because the interviews were conducted in person, the designer reduced costs by randomly selecting a few postcodes in the UK and then randomly selecting a few households in each postcode.² I refer broadly to these experimental designs as clustered experiments, where the design is given by the width (number of clusters, K) and depth (number of units per cluster, N).

While the sample average of the estimators obtained via clustered experiments remain unbiased conditional on sequential randomization, their variance depend critically on the design. From a statistical perspective, the inclusion of multiple clusters is always desirable. If the researcher believes that observations/treatment effects are more correlated within clusters than across clusters, she can always improve on the variance of the resulting estimator by sampling from several clusters. However, the inclusion of additional clusters is often costly, especially in the context of developing countries. Some examples of cluster-level fixed costs are travel costs, training of additional staff, time-costs of partnering with different local authorities and companies, increased coordination efforts, location-specific knowledge, etc. Importantly, these costs are usually *sunk* and take place before the actual implementation of the experiment/survey.³ As a result, the optimal depth vs width balances statistical accuracy and implementation costs.

The relevant statistical component entering the design problem is the intra-cluster correlation (ICC), i.e. how much individuals vary within their cluster compared to how much they vary overall. In particular, the marginal value of an additional observation from the same cluster (in terms of variance reduction) is lower whenever the ICC is higher. As a result, the designer would prefer wide designs (many clusters, few units per cluster) in the presence of high ICC levels, and would prefer deep designs (few clusters, many units) in the presence of low ICC levels.⁴ While the cost structure is usually known to the designer, it is often difficult to make strong assumptions on the share of ICC, especially given its large variability across regions and domains. For reference, in the context of medical studies, [Knox and Chondros \(2004\)](#) encounter ICC estimates varying an order of magnitude across covariates within the same population.⁵ The variability of the ICC and the design’s sensitivity to this parameter—see [Figure 1](#)—make its imputation a first order concern for experimentalists.

While the ICC is empirically relevant, its estimation may be quite challenging. To see why, observe that the ICC can be written as a function of the within- and between-cluster variances. The estimation of the latter requires sampling from many clusters, which is, precisely, the expensive element of the design. This dynamic places the designer at a conundrum: Optimal design identification requires knowing the ICC but ICC estimation requires sampling from many clusters. Because cluster exploration often requires bearing fixed costs that cannot be undone, ICC estimation damages optimal design implementation.

To the author’s knowledge, this is the first study to investigate the depth-vs-width problem

²Some additional examples include: surveys to CEOs across different industries [Datta and Rajagopalan \(1998\)](#), land surveys across different regions [Lin and Ho \(2003\)](#), treatment allocation to public clinics across different districts [Karing \(2024\)](#), treatment allocation to villages across different wards [Heß et al. \(2021\)](#), and treatment allocation to districts across different provinces [Ashraf et al. \(2020\)](#).

³For instance, [Meriggi et al. \(2024\)](#) need to train different social-mobilization teams in rural Sierra Leone, and [Li et al. \(2026\)](#) need to build an algorithm around the hiring infrastructure of a big company. These costs are nearly invariable to the number of treated units in each region/company, and must be covered before the actual implementation of the experiment.

⁴In the extreme case where observations are perfectly correlated within clusters, the analyst extracts all the cluster’s information by sampling a single individual. In this extreme setting, the analyst would sample a single unit from as many clusters as the budget allows for.

⁵[Knox and Chondros \(2004\)](#) encounter ICC estimates ranging from 0.055 to 0.451 on demographic covariates, and from 0.005 to 0.059 on morbidity covariates, both set of covariates showing an order of magnitude variation.

with unknown intra-cluster correlation. To do so, I consider a minimal model with nested errors, linear costs and a hard budget. I then characterize the oracle policy (the variance minimizing design) by solving the designer’s problem under known ICC. Secondly, I develop two algorithms which solve the problem efficiently without any previous knowledge of the ICC. Both algorithms follow the same structure: they first run a pilot where they learn the ICC, and then, they allocate the remaining budget between units and clusters according to plug-in rules. The first algorithm implements the design in just two stages by taking a worst-case approach in the pilot. I refer to this algorithm as Explore-then-Commit Cluster Selection (**EtC-CS**). The second algorithm implements an adaptive pilot by sequentially sampling clusters until it stops according to a data-dependent rule and commits to the final design. I refer to this algorithm as Adaptive Cluster Selection (**ACS**).

In terms of performance, I first show that the variance of the oracle design—the variance minimizing solution to the problem under known ICC—decreases linearly with the budget B , i.e. it is $O(1/B)$. I then show that both algorithms implement designs which induce negligible excess variance on the estimators compared to the optimal design. In particular, they achieve worst-case expected excess variance of order $O(\ln B/B^2)$, much smaller than the order of the variance itself. Although both algorithms display bounds of the same order, empirically, **ACS** outperforms **EtC-CS** at the expense of larger adaptability requirements.

Even if the algorithms induce estimators with lower variance, design-conditional inference is potentially compromised, provided the design depends on pilot data. After all, the literature on adaptive data collection—as in [Nie et al. \(2018\)](#), [Hadad et al. \(2021\)](#) and [Wager \(2024\)](#)—is full of negative results which suggest that adaptive data collection invalidates design-conditional inference on the resulting average estimators. I show a strong result on this regard. Under a symmetry assumption on the errors, inference on sample averages is invariant to adaptive data collection procedures which depend on asymptotically even functionals of the data such as sample variances. This means that asymptotic normality is preserved without further correction nor (pilot) data wastage. The theory applies to **EtC-CS** and **ACS** as both algorithms are functions of (sequences of) ICC estimators, which are ratios of within- and between-cluster variances. However, the theory extends far beyond the clustered sampling problem and includes, among others, popular sampling procedures like Neyman allocations with preliminary cluster variance estimation. The inference invariance result hinges on a simple heuristic: under error symmetry, error signs and error magnitudes are independent. As a result, by conditioning on the even σ -algebra (a σ -algebra generated by even functionals of the errors), the analyst freezes the magnitudes while leaving the signs unrestricted. It follows that inference is invariant to conditioning on events induced by evenly-measurable policies. A final complication is worth discussing. In practice, policies do not have access to the (squared) errors, so exactly error-even policies are rarely available. At best, sampling policies may depend on asymptotically even statistics like (sequences of) sample variances. Luckily, results on conditional inference can be extended to the feasible class of asymptotically even policies.

Finally, I show that the excess variance of **EtC-CS** and **ACS** is minmax optimal by deriving matching lower bounds (up to logarithmic terms) of $O(1/B^2)$. I also argue that their excess variance is non-trivial, at least when compared to canonical static policies. For reference, any static policy that takes a misspecified ICC as an input (including the minmax optimal static policy) will incur excess variance of $O(1/B)$, that is of the same order than the oracle variance and much larger than the excess variance of the proposed algorithms. A large array of simulations confirm the improved performance

of my algorithms compared to static policies, the superior performance of ACS compared to EtC-CS and the design conditional asymptotic normality of the induced estimators under symmetry.

Empirical Application - TBC

Extensions - TBC

The rest of the manuscript is organized as follows: Section 2 provides a brief literature review on survey theory, adaptive experimental design and inference under adaptive data collection, Section 3 introduces the cost-aware variance minimization problem under known ICC, Section 4 discusses estimation in the context of unknown ICC and presents the two candidate algorithms, Section 5 provides upper bounds on the expected excess of the algorithms, Section 6 discusses design-conditional inference for asymptotically even policies and specializes the results to the K vs N problem, Section 7 derives matching lower bounds on the problem and compares the performance of the algorithms to misspecified static policies, Section 8 collects simulation evidence confirming performance and inference guarantees, and Section 9 concludes. All proofs including a standalone technical note on the general inference invariance principle can be found in the Appendix.

2 Literature Review

Survey Theory. The problem of optimal multi-stage survey design under a budget is a well-studied problem in the field of survey theory dating back to the seminal work of Kish (1965).⁶ On its simplest form, a multi-stage survey is deployed by first splitting the population of interest in a set of (non-overlapping) clusters, then, randomly selecting a number of clusters K (proportionally to their size), and, finally, randomly sampling N units from each of the selected clusters. The process may be towered by first drawing K_1 super-clusters, then K_2 sub-clusters, etc. The goal in this literature is very close to the spirit of this paper: to minimize the variance of the resulting population estimators by careful definition and/or selection of clusters and surveyed units. The literature can be divided into three broad categories: (i) sampling designs that take the clusters as given and optimize for unit selection, (ii) sampling designs that optimize cluster definition, and (iii) sampling designs that jointly optimize for cluster definition and unit selection (Ballin and Barcaroli (2013)). This manuscript does take cluster partitions as given, thus I discuss the first category in further detail.

Stratified Survey Design. This sub-literature finds its most prominent representative on Bethel’s algorithm (Bethel (1989)), an extension of Neyman’s allocation to budget constraints and generic convex costs (Tschprow (1923), Neyman (1992)). Since the popularization of Bethel’s algorithm, the literature has focused on extending its logic to multi-domain, multi-variable and multi-objective settings, with a particular focus on algorithmic performance (Barcaroli et al. (2022)). Surprisingly, despite (i) the field’s impressive technical machinery and (ii) its emphasis on the role of intra-cluster correlation for optimal design,⁷ to the author’s knowledge there has been no attempt at introducing ICC estimation as part of the design optimization problem. Recent treatments as Barcaroli et al. (2022) assume that the cluster correlation can always be estimated from auxiliary data. In practice, the design selection seems driven by *feel*, broad logistic considerations and common misconceptions. To

⁶see Biemer and Lyberg (2003) for a more recent textbook treatment, and Sedgwick (2015) for a brief discussion and policy guide.

⁷See for instance Cicchitelli et al. (1992) for an extended discussion on the role of ICC and the inclusion of several nested clusters. Knox and Chondros (2004) presents compelling empirical evidence on the variability of the ICC across domains. Campbell et al. (2004) collect a methodological note on the usage and reporting of ICC estimates.

name a few, the UN Handbook of Household Sample Surveys in Developing and Transition Economies [Tam \(2005\)](#) recommends to always sample as many clusters as logistically feasible. Other sources like [Sedgwick \(2015\)](#) argue that the number of clusters must be fixed in advance. I will argue that none of these considerations are necessarily optimal.

The Econometrics of Stratified Experiments. My work is closely related in spirit to the econometric literature on optimal stratification and optimal covariate adjustment, albeit the designs in this literature are not really adaptive. The goal of this literature is to minimize the variance of an ATE estimator. Some of it even considers the role of sampling costs. Some examples include [Cytrynbaum \(2021\)](#), [Cytrynbaum \(2024\)](#), [Bai \(2022\)](#) or [Bai et al. \(2024\)](#). Despite being closely related theme-wise, this paper does not look into the problem of strata definition nor covariate adjustment, but at the trade-off between cluster and unit selection under a budget. Moreover, on its simplest form, the proposed solution does not rely on external covariates for its implementation.

Adaptive Experimental Design. Formally, my work resonates with the adaptive experimental design and online policy learning literature. For instance, in the context of best arm identification, [Adusumilli \(2026\)](#) considers the problem of optimal stopping in sampling experiments, [Imbens et al. \(2025\)](#) shows the inadmissibility of uniform sampling rules in multi-arm settings, and [Kasy and Sautmann \(2021\)](#) discuss sequential treatment allocation. I further encounter theoretical and applied papers on bandit learning in Economics with a similar notion of regret to the one described in this paper such as [Cesa-Bianchi et al. \(2025\)](#) and [Li et al. \(2026\)](#).

Inference on Adaptive Designs. The literature on inference with adaptive designs is relatively thin and full of negative results: In general, adaptive data selection compromises estimation and invalidates inference. [Nie et al. \(2018\)](#) argue that most adaptive data collection procedures deliver negatively biased sample average estimates. Similarly, [Hadad et al. \(2021\)](#) show that IPW estimators are not asymptotically normal and suffer from fat tails.⁸ Unlike most data collection procedures in this literature, my algorithms do not depend on the levels but on the second moments of the data. This difference allows us to escape from the malaise of adaptive data collection (at least under homogeneous clusters with balanced designs). Asymptotic independence is established under an error symmetry assumption, which resembles results in randomization inference as the ones in [Canay et al. \(2017\)](#).

Others. Finally, a shallow connection exists between my work and the bandit literature on optimal sampling designs, going back to the equivalence result of D-optimal and G-optimal problems in [Kiefer and Wolfowitz \(1960\)](#).⁹ Nonetheless, the tools and scope in this literature are quite removed from the analysis in this manuscript.

3 Design Problem under Known ICC

3.1 Survey Design Problem

Consider a population with $\overline{K}$ equally sized clusters containing $\overline{N}$ units each. The goal of the researcher is to minimize the MSE of an estimator of a population average by deciding how many units $N^k \ll \overline{N}$ sample for each cluster, where $K = \sum \mathbb{1}(N^k > 0) \ll \overline{K}$ is the number of sampled clusters. Clusters

⁸See [Wager \(2024\)](#) for a recent textbook treatment.

⁹See [Lattimore and Szepesvári \(2020\)](#) for a more recent textbook treatment.

are indexed by $k = 1, \dots, K$ and units are indexed by $i = 1, \dots, N^k$ for each cluster k . Let y_{ki} follow an additive model with nested errors,¹⁰

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \quad (1)$$

where μ is the population mean and parameter of interest, u_k is the cluster-level error, ε_{ki} is the individual-level error.

Assumption 3.1. Error Distributions. $(u_k)_k$ is an iid sequence with $\mathbb{E}[u_k] = 0$, $\text{Var}(u_k) = \tau^2$; $(\varepsilon_{ki})_{k,i}$ is an iid sequence with $\mathbb{E}[\varepsilon_{ki}] = 0$, $\text{Var}(\varepsilon_{ki}) = \sigma^2$; and $u_k \perp\!\!\!\perp \varepsilon_{ki}$ for all k, i .

Define $\lambda_0 = \sigma^2/\tau^2$ as the **variance ratio**. λ_0 is the main object entering the analysis, and is connected one-to-one to the intra-cluster correlation, $\text{ICC} = \tau^2/(\tau^2 + \sigma^2) = 1/(1 + \lambda_0)$.¹¹ Let $\omega = (\tau^2, \lambda_0)$ such that

Assumption 3.2. Compact Parameter Space. ω lives in a strict subset of a known compact set $\Omega = [\tau_m, \tau_M] \times [\lambda_m, \lambda_M]$ with $\tau_m > 0$, i.e. $(\tau^2, \lambda_0) \in [\tau_m + \delta, \tau_M - \delta] \times [\lambda_m + \delta, \lambda_M - \delta]$ for some $\delta > 0$.

For simplicity in the exposition, I focus on the case where $N^k \in \{0, N\}$ for all k . In this (homogeneous) setting this is without loss, as the variance minimizing allocation is indeed a balanced one.¹² The general $(N^k)_k$ setting is discussed in Appendix A. For fixed (K, N) , the natural estimator of the population parameter μ is simply the **average of averages**,

$$\hat{\mu}_{K,N} = \frac{1}{KN} \sum_{k=1}^K \sum_{i=1}^N y_{ki}$$

which is unbiased for μ under multi-stage randomization (and mean-zero noise).

3.2 Variance Minimization under Linear Costs

The goal of the learner is to minimize the MSE of the estimator $\hat{\mu}_{K,N}$ subject to linear costs and a hard budget B . Under two-stage randomization, $\hat{\mu}$ is unbiased for μ , hence, the problem reduces to a constrained variance minimization problem

$$\min_{K \geq 2, N \geq 2} W(K, N) = \frac{1}{K} \tau^2 \left(1 + \frac{\lambda_0}{N} \right) \quad \text{s.t.} \quad FK + VKN \leq B \quad (2)$$

where $W(K, N) = \text{Var}(\hat{\mu}_{K,N})$, F is the fixed cost of sampling an additional cluster, V is the variable cost of sampling an additional unit and $B \geq 2(F + 2V)$ is the hard budget.¹³ The

¹⁰The variance minimization problem and excess variance analysis only rely on certain moment restrictions implied by Equation 1. In that sense, the analysis extends beyond this additive model. However, the additive error structure of the model plays an important role in the inference analysis, thus, for internal consistency and simplicity, I take Equation 1 as the model of reference.

¹¹The model in Equation 1 implicitly assumes the following moment restrictions: (i) cross-cluster independence, i.e. $\text{CoVar}(y_{ki}, y_{lj}) = 0$ for all i, j and $k \neq l$; (ii) homogeneous total variance, i.e. $\text{Var}(y_{ki}) = \tau^2(1 + \lambda_0)$ for all k, i ; and (iii) within-cluster equicorrelation, i.e. $\text{CoVar}(y_{ki}, y_{kj}) = \tau^2$ for all i, j , and all k .

¹²The lower bounds in Section 7 do not restrict to balanced allocation policies. In that sense, the proposed balanced allocation algorithms are near-optimal within a very general class of policies.

¹³The lower bound $K \geq 2, N \geq 2$ has been imposed for technical reasons that affect the estimation of the parameters in Section 4. In practice, this restriction is with little loss as *all* clustered surveys/experiments consider larger samples anyway.

equality follows from simple algebra under cross-cluster independence. For simplicity, when discussing performance guarantees, I will consider the continuous relaxation of the problem above with $(K, N) \in \mathbb{R}^2 \geq \mathbf{2}$.¹⁴ Solving for the optimal design as a function of parameters $(K^*(\omega), N^*(\omega))$ from the first order conditions returns

$$K^*(\omega) = \frac{B}{F + VN^*(\omega)}, \quad N^*(\omega) = \min \left\{ \max \left\{ 2, \sqrt{\lambda_0 \frac{F}{V}} \right\}, \frac{B - 2F}{2V} \right\} \quad (3)$$

The map $\omega \mapsto (K^*(\omega), N^*(\omega))$ depends on ω only through the variance ratio λ_0 . Accordingly, for any $\lambda \in [\lambda_m, \lambda_M]$ define the **optimal map** as

$$\lambda \mapsto (K^*(\lambda), N^*(\lambda))$$

Remark 3.3. Observe that for any $\tau^2 > 0$, $\hat{\mu}_{K,N}$ is consistent for μ iff $K \rightarrow \infty$. Intuitively, large N can only reduce the individual level noise ε_{ki} , so large K remains necessary to erase cluster level noise u_k . On this note, observe that $\lim_B K^*(\lambda) = \infty$ (for every $\lambda > 0$), while $N^*(\lambda) \perp B$ (for some fixed λ).

Remark 3.4. Despite its name, the optimal map is only really optimal if the designer selects as input $\lambda = \lambda_0$. In general, applying the optimal map to a misspecified λ can have sizable implications on the design and the variance of the estimator induced by this design. For $\lambda \in [\lambda_m, \lambda_M]$, Figure 1 plots $(\lfloor K^*(\lambda) \rfloor, \lfloor N^*(\lambda) \rfloor)$, as well as the excess variance of a policy which selects $(K = \lfloor K^*(\lambda) \rfloor, N = \lfloor N_B^*(\lambda) \rfloor)$ compared to the oracle policy $(\lfloor K^*(\lambda_0) \rfloor, \lfloor N^*(\lambda_0) \rfloor)$. The figure suggests that small λ misspecification can have large implications on the induced design and the resulting excess variance.¹⁵

¹⁴Final integer allocation in all simulation results concerning performance of own and competitor algorithms has been conducted under the floor mapping $(\lfloor K^\pi \rfloor, \lfloor N^\pi \rfloor)$. This rounding criterion guarantees implementability, eases cross-algorithm comparisons and removes differences stemming from integer approximations. In practice, alternative terminal budget allocation may achieve superior performance. Replication code can be found in the Supplementary Material.

¹⁵See Section 8 for further discussion.

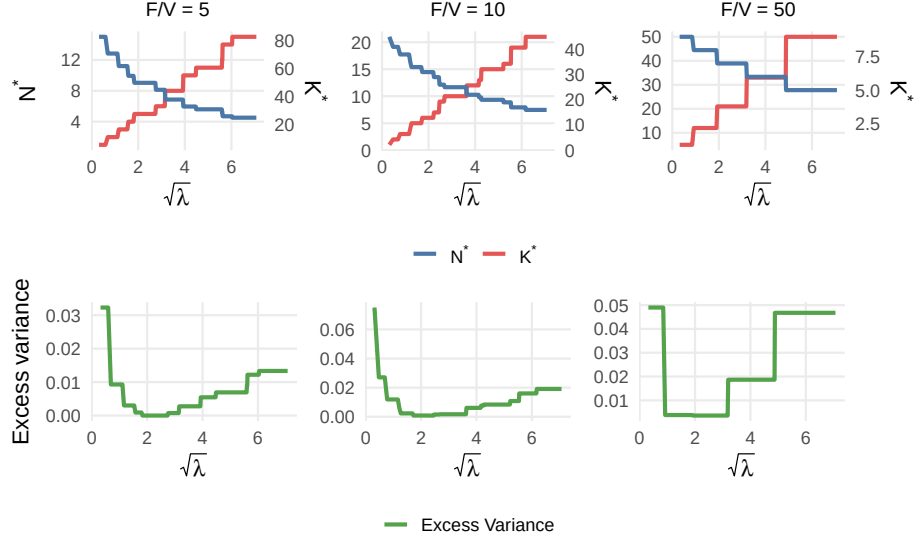


Figure 1: $\lfloor N^*(\lambda) \rfloor$ (top-left), $\lfloor K^*(\lambda) \rfloor$ (top-right) and Excess Variance (bottom) for different F/V ratios. $\mu = 0$, $\omega = (1, 5)$, $B = 1000$, $V = 2$

The variance of the oracle policy in the continuous relaxation of the problem is given by

$$W(K^*(\lambda_0), N^*(\lambda_0)) = \frac{\tau^2 \left(\sqrt{V\lambda_0} + \sqrt{F} \right)^2}{B} \sim O(1/B)$$

$(K^*(\lambda_0), N^*(\lambda_0))$ is an oracle policy as it depends on the unknown variance ratio λ_0 . Any sampling policy with no access to the true variance ratio will find in the integer version of the **oracle variance** a natural benchmark to compete against. Finally, I include an additional assumption to avoid (uninteresting) corner solutions in the analysis.

Assumption 3.5. *Interior Solution.* Given a compact set Ω such that $\lambda_0 \in [\lambda_m, \lambda_M]$, for any fixed $B \geq 2(F + 2V)$, $\lfloor N^*(\lambda_m) \rfloor \geq 2$ and $\lfloor K^*(\lambda_M) \rfloor \geq 2$

3.3 Experimental Design

The problem in Equation 2 can be extended to a general class of problems including two canonical experimental designs: experiments with clusters as blocks (under no interference) and experiments with treatment assignment at the cluster level.

3.3.1 Clusters as Blocks

Suppose the researcher samples $K \ll \bar{K}$ clusters and $N \ll \bar{N}$ units per cluster. She then randomly allocates treatment to $N/2$ units per cluster. Let $D_{ki} \in \{0, 1\}$ denote the treatment status of unit i in cluster k . Using standard potential-outcomes notation under no interference, let $Y_{ki}(1)$ and $Y_{ki}(0)$ denote the treated and untreated potential outcomes of unit i in cluster k . Assume that each potential-outcome random variable follows the same nested-error structure as in Equation 1, i.e. $Y_{ki}(d) = \mu_d + u_k^d + \varepsilon_{ki}^d$ for $d \in \{0, 1\}$, where u_k^d is iid with $\mathbb{E}[u_k^d] = 0$, $\mathbb{E}[(u_k^d)^2] = \tau_d^2$ and $\text{Cov}(u_k^1, u_k^0) = \tau_{01}$. Similarly, ε_{ki}^d is iid with $\mathbb{E}[\varepsilon_{ki}^d] = 0$ and $\mathbb{E}[(\varepsilon_{ki}^d)^2] = \sigma_d^2$. The observed outcome can

be written compactly via

$$Y_{ki}(D_{ki}) = D_{ki}Y_{ki}(1) + (1 - D_{ki})Y_{ki}(0) \quad (4)$$

The object of interest is the average treatment effect

$$\text{ATE} = \mathbb{E}[Y_{ki}(1) - Y_{ki}(0)] = \mu_1 - \mu_0$$

Given the balanced randomization within each sampled cluster, the natural estimator of the average treatment effect is the **average of differences in means** estimator,

$$\hat{\text{ATE}}_{K,N} = \frac{2}{KN} \sum_{k=1}^K \left(\sum_{i:D_{ki}=1}^N y_{ki} - \sum_{i:D_{ki}=0}^N y_{ki} \right)$$

Under random cluster sampling and random treatment allocation, $\hat{\text{ATE}}(K, N)$ is unbiased for the ATE. The variance of $\hat{\text{ATE}}$ is given by

$$W(K, N) = \frac{1}{K} \tau^2 \left(1 + \frac{\lambda}{N} \right) \quad \tau^2 = (\tau_1^2 + \tau_0^2 - 2\tau_{01}), \quad \sigma^2 = 2(\sigma_0^2 + \sigma_1^2), \quad \lambda = \sigma^2 / \tau^2$$

Remark 3.6. *In the especial case where $u_k^1 = u_k^0$, within-cluster differencing removes the cluster level noise entirely, therefore, the variance is minimized by selecting a single cluster and as many units as possible. The researcher has no incentive to sample different clusters.*

3.3.2 Clusters as Treatment Level

Alternatively, consider the setting where the researcher is not willing to assume away interference at the cluster level, i.e. $Y_{ki} = Y_{ki}(D_{ki}, D_{ki-})$. Let $Y_{ki}(d, \mathbf{d}) = \mu_d + u_k^d + \varepsilon_{ki}^d$ where $\mathbf{d} = (d, \dots, d)$. Then, the designer can still recover an unbiased estimate of the total average treatment effect

$$\text{TATE} = \mathbb{E}[Y_{ki}(1, \mathbf{1}) - Y_{ki}(0, \mathbf{0})]$$

by randomizing at the cluster level, i.e. assigning $K/2$ clusters to treatment and $K/2$ clusters to control with N units in each cluster such that $D_{ki} = D_k$. The natural estimator is given by the **difference in means estimator**

$$\hat{\text{TATE}}_{K,N} = \frac{2}{K} \left(\sum_{k:D_k=1}^K \frac{1}{N} \sum_{i=1}^N y_{ki} - \sum_{k:D_k=0}^K \frac{1}{N} \sum_{i=1}^N y_{ki} \right)$$

Under random cluster sampling, random treatment allocation at the cluster level and random unit selection within clusters, $\hat{\text{TATE}}(K, N)$ is unbiased for the TATE with variance

$$W(K, N) = \frac{2}{K} \tau^2 \left(1 + \frac{\lambda}{N}\right) \quad \tau^2 = (\tau_1^2 + \tau_0^2), \quad \sigma = (\sigma_0^2 + \sigma_1^2), \quad \lambda = \sigma^2 / \tau^2$$

4 Design Problem under Unknown ICC

4.1 λ Estimation

Optimal design under unknown model parameters will involve some form of parameter estimation, therefore, I start this section by characterizing the relevant estimators. Fix (K, N) with $K \geq 2, N \geq 2$ and define

$$\begin{aligned} \hat{\mu}_N^k &= \frac{1}{N} \sum_{i=1}^N y_{ki} & W_{k,N}^2 &= (y_{ki} - \hat{\mu}_N^k)^2, & \hat{w}_{K,N}^2 &= \frac{1}{K(N-1)} \sum_{k=1}^K \sum_{i=1}^N W_{k,N}^2 \\ \hat{b}_{K,N}^2 &= \frac{1}{K-1} \sum_{k=1}^K (\hat{\mu}_N^k - \hat{\mu}_{K,N})^2 \end{aligned}$$

where $\hat{\mu}^k$ is the sample average for cluster k , $\hat{w}^2$ is the within-cluster sample variance, and $\hat{b}^2$ is the between-cluster sample variance. Now define the clip operator $\Pi_{[a,b]}(x) = \min\{b, \max\{a, x\}\}$ such that

$$\hat{\lambda}_{K,N} = g(\hat{w}_{K,N}^2, \hat{b}_{K,N}^2) = \Pi_{[\lambda_m, \lambda_M]} \left(\frac{\hat{w}_{K,N}^2}{\Pi_{[\tau_m, \tau_M]} \left(\hat{b}_{K,N}^2 - \frac{\hat{w}_{K,N}^2}{N} \right)} \right)$$

where $\hat{\lambda}_{K,N}$ is the (clipped) sample estimator of λ . Lemma 4.1 shows consistency of these estimators.

Lemma 4.1. *Estimator Properties. Let $\xrightarrow[T]{x}$ denote convergence in x under $T \rightarrow \infty$ asymptotics. Under Assumption 3.1 and Assumption 3.2, (i) $\hat{w}_{K,N}^2 \xrightarrow[N]{a.s.} \sigma^2$ (with fixed $K \geq 1$) and $\hat{w}_{K,N}^2 \xrightarrow[K]{a.s.} \sigma^2$ (with fixed $N \geq 2$); (ii) $\hat{b}_{K,N}^2 \xrightarrow[N]{a.s.} \tau^2 + \frac{\sigma^2}{N} = \theta_N^2$ (with fixed $N \geq 2$); and (iii) $\hat{\lambda}_{K,N} \xrightarrow[K]{a.s.} \lambda_0$ (with fixed $N \geq 2$).*

Remark 4.2. *From the definitions and Lemma above it follows that, while the within-variance can be estimated consistently under both N and K asymptotics, the between-variance can only be consistently estimated under K asymptotics. As a result, K asymptotics delivers consistent estimation of μ and consistent estimation of λ_0 .*

Lemma 4.1 has an important implication for pilot-based designs that I summarize in Lemma 4.4. This Lemma will substantiate the heuristics leading to the proposed algorithms, and will play a central role in the proofs of performance guarantees and conditional inference.

Definition 4.3. *Pilot-Based Policies. Consider some (possibly random) pilot design $(K_{pB}, N_{pB}) \in \mathbb{N}^2 \geq 2$ depending on budget B . A policy π is said to be pilot-based if*

$$(K_B^\pi, N_B^\pi) = (K^*(\hat{\lambda}_{K_{pB}, N_{pB}}), N^*(\hat{\lambda}_{K_{pB}, N_{pB}}))$$

Lemma 4.4. *Convergence to Oracle Policy under K -Asymptotic Pilots. Let $\nu_B = K_B/B$ and $\nu_0 = K^*(\lambda_0)/B$. ν_B^π or ν_{pB} are defined as expected. Let $N_0 = N^*(\lambda_0)$. Consider a pilot-policy with pilot*

design $(K_{pB}, N_p) \in \mathbb{N}^2 \geq \mathbf{2}$ where N_p is fixed and $\mathbb{P}(\nu_{pB} \geq c) \rightarrow 1$ for some $c > 0$, then, under Assumptions 3.1, 3.2 and 3.5, $|\nu_B^\pi - \nu_0| \xrightarrow{p} 0$ and $|N_B^\pi - N_0| \xrightarrow{p} 0$.

Remark 4.5. This Lemma suggest that pilot-policies which sample an asymptotically large number of clusters as $B \rightarrow \infty$ induce designs converging in probability to the oracle policy, where convergence is exact in the N -dimension, and B -scaled in the K -dimension. The Lemma leverages the consistency of the $\hat{\lambda}$ estimator along the K -dimension as given in Lemma 4.1.

4.2 Explore-then-Commit Cluster Sampling

In this section I propose a simple solution to the problem of the designer under unknown ICC using an Explore-then-Commit structure. As anticipated in Lemma 4.4, consider a pilot-based sampling policy which first conducts a pilot with size (K_p, N_p) , then constructs an estimator of the form $\hat{\lambda}_p = \hat{\lambda}_{K_p, N_p}$, and, finally, implements the optimal map $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$. This intuitive protocol, however, presents *a priori* a major flaw: There is no guarantee that $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$ is budget implementable provided clusters (and units) cannot be unsampled. More precisely, a pilot-based policy is implementable iff $K^\pi \geq K_p$ and $N^\pi \geq N_p$. In other words, the policy is **implementable** iff in every dimension (K, N) no more resources are devoted to learning than those predicted for optimal sampling after learning. This circular reference places the learner at a conundrum. Sampling a large number of clusters favors learning in a way that benefits the identification of the optimal sampling policy, but threatens its implementation.

Under Assumptions 3.2 and 3.5, a safe way to guarantee the implementability of the policy is setting $K_p = \lfloor K^*(\lambda_M) \rfloor$ and $N_p = \lfloor N^*(\lambda_m) \rfloor$. Then, by definition of the clipped estimator, for any $\hat{\lambda}_p$, it holds that $K_p \leq K^*(\lambda_M) \leq K^*(\hat{\lambda}_p) = K^\pi$ and $N_p \leq N^*(\lambda_m) \leq N^*(\hat{\lambda}_p) = N^\pi$, so the policy is trivially implementable. This two-stage protocol is referred to as Explore-then-Commit Cluster Selection (EtC-CS). Algorithm 1 presents its pseudo-code.

Algorithm 1 EtC-CS

Input $B, F, V, \lambda_m, \lambda_M$

Sample $K_p = \lfloor K^*(\lambda_M) \rfloor$ clusters and $N_p = \lfloor N^*(\lambda_m) \rfloor$ units per cluster

observe $(y_{ki})_{i \leq N_p, k \leq K_p}$

compute $\hat{\lambda}_p = \hat{\lambda}_{K_p, N_p}$

Set $(K^{\text{EtC-CS}}, N^{\text{EtC-CS}}) = (K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$

Remark 4.6. For some fixed budget B , the EtC-CS policy satisfies the assumptions in Lemma 4.4. Namely, N_p is fixed and $K_{pB}/B = 1/(F + VN^*(\lambda_M)) > 0$, deterministically. Therefore, EtC-CS converges to the integer oracle policy, i.e. $(\nu_B^{\text{EtC-CS}}, N_B^{\text{EtC-CS}}) \xrightarrow{p} (\nu_0, N_0)$.

4.3 Adaptive Cluster Selection

While theorists emphasize the superior performance of adaptive designs, practitioners often encounter logistical challenges that prevent their full implementation (Chow and Corey (2011)). In that sense, the simplicity of EtC-CS is praiseworthy as it achieves near-optimal performance guarantees (as per Theorem ??) with minimal adaptability—just a two-stage design. However, the reader may wonder what could be achieved if the designer was given access to further adaptability.

Example 4.7. A researcher must select a design that samples something between 20 and 60 clusters, i.e. given a budget B , $\lfloor K^*(\lambda_M) \rfloor = 20$ and $\lfloor K^*(\lambda_m) \rfloor = 60$. **EtC-CS** would then recommend sampling 20 clusters as part of the pilot, compute $\hat{\lambda}_p = \hat{\lambda}_{20, N_p}$, and implement $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$. Now, imagine that the design prescribed by the pilot is $K^*(\hat{\lambda}_p) = 50$. It would seem unreasonable to commit straight-away to this design, as the designer could learn more about λ_0 by exploring a few more clusters before committing, with minimal risk of incurring cluster oversampling.

Based on this heuristic, I propose an improved algorithm which samples clusters sequentially until it stops according to a data-dependent rule and implements the final design. I refer to this algorithm as Adaptive Cluster Selection (**ACS**). The size of the pilot in **ACS** is determined by the stopping time s , where

$$s = \inf \{k \in \mathbb{N} : k + 1 > K^*(\hat{\lambda}_{k, N_p})\} \quad (5)$$

Observation 4.8. By construction of the clipped estimator, $s \in [\lfloor K^*(\lambda_M) \rfloor, \lfloor K^*(\lambda_m) \rfloor]$

In other words, the adaptive algorithm explores N_p units from an additional cluster until sampling the new cluster would entail oversampling according to the online estimator of variance ratio. By treating N_p as fixed, and exploring only in the cluster dimension, the algorithm combines simplicity with statistical guarantees (recall Lemma 4.4).

Remark 4.9. In principle, the learner may consider alternative iterative sampling designs. In full generality, every period the learner could sample a new unit from an already explored cluster or from a new cluster to maximize the (cost adjusted) expected marginal return to exploration. Unfortunately, this allocation process is not only computationally infeasible but also highly impractical in real applications. In addition, as per the discussion above, reducing exploration to the K dimension is the correct simplification. Although sampling additional units from already explored clusters (increasing N) may reduce the variance of the $\hat{\lambda}$ estimator in small samples, asymptotically, its variance is independent of N and only strictly decreasing in K . It follows that, when it comes to λ estimation, the learner has an asymptotic incentive to commit all the exploration resources to sample additional clusters.

Algorithm 2 ACS

Input $B, F, V, \lambda_m, \lambda_M$
Initialize $K = 0, N_p = \lfloor N^*(\lambda_m) \rfloor$
compute $\lfloor K^*(\lambda_M) \rfloor \geq 2$, **explore** clusters $k \leq \lfloor K^*(\lambda_M) \rfloor$ and **observe** $(y_{ki})_{i \leq N_p, k \leq \lfloor K^*(\lambda_M) \rfloor}$
set $k = \lfloor K^*(\lambda_M) \rfloor, \hat{\lambda}_k = \hat{\lambda}_{k, N_p}$
while $k + 1 \leq K^*(\hat{\lambda}_k)$
 explore an additional cluster k and **observe** $(y_{ki})_{i \leq N_p}$
 update $k = k + 1$
Set $s = k$
Recover $\hat{\lambda}_{\text{ACS}} = \hat{\lambda}_{\max\{s-1, \lfloor K^*(\lambda_M) \rfloor\}, N_p}$ using $(y_{ki})_{i \leq N_p, k \leq \max\{s-1, \lfloor K^*(\lambda_M) \rfloor\}}$
Set $(K^{\text{ACS}}, N^{\text{ACS}}) = (K^*(\hat{\lambda}_{\text{ACS}}), N^*(\hat{\lambda}_{\text{ACS}}))$

ACS replicates **EtC-CS** for the first $K^*(\lambda_M)$ periods, but then, extends exploration sequentially under its data-driven stopping rule. At the expense of further adaptability requirements, **ACS** makes use of weakly more information without any additional risk. In fact, **ACS** concentrates (nearly) all cluster exploration within the pilot, while **EtC-CS** may conduct a large part of cluster sampling outside

the pilot. The estimation of λ_0 improves with K , which, incidentally, is the expensive element of the design, hence, the inclusion of many clusters is particularly desirable for optimal design identification.

Remark 4.10. *The use of information from $s - 1$ clusters for λ estimation is a minor technical concern. Given a stopping time s , $s + 1 > K^*(\hat{\lambda}_s)$ (by definition of the stopping time), but also $s \leq K^*(\hat{\lambda}_{s-1})$, otherwise the algorithm would have not explored cluster s . Therefore, $K^*(\hat{\lambda}_s) - 1 \leq s \leq K^*(\hat{\lambda}_{s-1})$. From these conditions alone one cannot infer the natural condition $s \leq K^*(\hat{\lambda}_s)$, except for $s = \lfloor K^*(\lambda_M) \rfloor$. Under Assumptions 3.2 and 3.5, it follows that while the design $(K^*(\hat{\lambda}_{s-1}), N^*(\hat{\lambda}_{s-1}))$ is implementable a.s., the design $(K^*(\hat{\lambda}_s), N^*(\hat{\lambda}_s))$ may not be. Clearly, for large pilots $\hat{\lambda}_s \approx \hat{\lambda}_{s-1}$, so the bounds on s will be tight and at most one additional cluster will be explored outside the pilot.*

5 Upper Bounds on Excess Variance

This Section presents the main results in terms of algorithmic performance. In particular, I provide upper bounds on the expected excess variance bounds of both algorithms of order $O(\ln B/B^2)$. For any policy π , define the excess variance of the policy as the random variable governing the difference between the variance of the design (K^π, N^π) induced by policy π and the variance of the integer oracle policy, i.e.

$$\Delta_\pi = W(K^\pi, N^\pi) - W(K^*(\lambda_0), N^*(\lambda_0))$$

I now impose a standard subgaussian assumption on the distribution of errors, which guarantees sufficient concentration.¹⁶

Assumption 5.1. *Errors Subgaussianity. Let u_k be κ_u -subgaussian and ε_{ki} be κ_ε -subgaussian.*

Theorem 5.2. *Expected Excess Variance. Under Assumptions 3.1, 3.2, 3.5 and 5.1, there exists constants $C_1, C_2 < \infty$ independent of B such that*

$$\begin{aligned} \mathbb{E}_{\text{EtC-CS}}[W(K, N)] - W(K^*(\lambda_0), N^*(\lambda_0)) &\leq C_1 \frac{\ln(B)}{B^2} \\ \mathbb{E}_{\text{ACS}}[W(K, N)] - W(K^*(\lambda_0), N^*(\lambda_0)) &\leq C_2 \frac{\ln(B)}{B^2} \end{aligned}$$

Remark 5.3. *Theorem 5.2 is constructed in three steps. I first derive bounds of order $O(\ln(1/\alpha)/B^2)$ that hold with high probability $(1 - \alpha)$ for both algorithms (Theorem C.8 in Appendix C). I then derive an unconditional worst-case bound of order $O(1)$ for any policy that satisfies the budget constraint. Finally, by fine-tuning the $(1 - \alpha)$ probability of the good event, I recover upper bounds on the expected excess variance of order $O(\ln B/B^2)$, i.e. the stronger performance guarantee comes at the expense of a $\ln B$ factor compared to the high-probability bounds*

The main takeaway of Theorem 5.2 is that the (expected) excess variances of **EtC-CS** and **ACS** are of order $O(\ln B/B^2) \ll O(1/B)$, thus, both algorithms induce asymptotically negligible excess variance compared to the oracle variance. In other words, a designer that implements a design according to **EtC-CS** or **ACS** is asymptotically not worse than an oracle designer who knows λ_0 ex-ante. I now explain the mechanics and intuition behind this result, and, in doing so, I argue that **ACS** is likely to outperform **EtC-CS**, despite sharing the same order of excess variance.

¹⁶Similar results hold under a weaker subexponential assumption.

From the discussion above, both algorithms are pilot-based with pilot-estimated $\hat{\lambda}$, where $\hat{\lambda}_p = \hat{\lambda}_{\lfloor K^*(\lambda_M) \rfloor, N_p}$ in the case of **EtC-CS**, and $\hat{\lambda}_p = \hat{\lambda}_{\max\{s-1, \lfloor K^*(\lambda_M) \rfloor\}, N_p}$ in the case of **ACS**. Under a second order approximation of $W(K^*(\lambda), N^*(\lambda))$ around $W(K^*(\lambda_0), N^*(\lambda_0))$, the excess variance is governed by deviations of the form $(\hat{\lambda}_p - \lambda_0)^2$. Hence, deviations are expected to be smaller under **ACS** provided it has a weakly larger sample size entering λ_0 estimation. The expected superior performance of **ACS** compared to **EtC-CS** is verified consistently in the simulation evidence (see Section 8). Yet, the increase in precision does not modify the order of the upper bounds because the sample sizes of the pilots share the same order a.s., i.e. $K_p^{\text{ACS}}/K_p^{\text{EtC-CS}} = O(1)$ in B .¹⁷ In Section 7, I argue that the rate in the excess variance of these algorithms is not only negligible but also near-optimal—up to logarithmic terms. This means that no algorithm can substantially improve upon Algorithms 1 and 2. Finally, I show that their excess variance is also non-trivial, at least when compared to any misspecified static policy.

6 Inference

So far, I have argued vehemently in favor of the adaptive design of clustered experiments by proving negligible excess variance of two adaptive algorithms. However, it is well known to the literature that adaptive designs may interfere with classic inference on the resulting average estimators. On this regard, I prove a strong result which is of independent interest. In particular, I argue that under a symmetry assumption on the errors, inference on population means remains invariant after conditioning on designs induced by asymptotically even policies. Informally, a policy is even if it depends on even functionals of the errors, such as the squared errors; and a policy is asymptotically even if it depends on approximately even functionals of the errors, such as sample variances. This means that, under symmetry, terminal averages of adaptive nearly-even designs remain asymptotically normal without further correction nor pilot-data exclusion. Recall that the policies of interest, **EtC-CS** and **ACS**, are functionals of sequences of sample variances, therefore the general result applies nearly verbatim to the K vs N problem. This Section only discusses the results in the context of the K vs N problem, yet appropriate connections to the general theory are presented throughout.

6.1 Inference for Even Designs under Symmetry

For some design (K_B^π, N_B^π) ,¹⁸ define $M_{k,N} = u_k + \frac{1}{N} \sum_{i=1}^N \varepsilon_{ki}$, such that

$$Z^\pi(B) = \sqrt{K_B^\pi}(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu) = \frac{1}{\sqrt{K_B^\pi}} \sum_{k=1}^{K_B^\pi} M_{k, N_B^\pi}$$

The object of interest for inference purposes is the distribution of $Z^\pi(B)$ conditional on the design, i.e. $Z^\pi(B) \mid (K_B^\pi, N_B^\pi)$. To build towards this result, I first introduce the necessary assumptions and definitions.

Assumption 6.1. Error Symmetry. Let $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$, then $C_k \stackrel{d}{=} -C_k$.

¹⁷This result holds even under maximal deviations, namely $K_p^{\text{EtC-CS}} = \lfloor K^*(\lambda_M) \rfloor$ and $K_p^{\text{ACS}} = \lfloor K^*(\lambda_m) \rfloor$. In fact, from a technical standpoint, it is more challenging to recover the desired rate in the adaptive case as the analyst must upper bound the deviations for a pilot with random size—as opposed to a deterministic size $K^*(\lambda_M)$. I circumvent this issue by noting (i) that $\hat{\lambda}_k - \lambda_0$ is, asymptotically, a function of martingales, therefore amenable to a maximal version of Bernstein’s martingales inequality, and (ii) that the stopping time is uniformly bounded in a window of size $O(1) \perp B$.

¹⁸The subindex B emphasizes the main asymptotic regime of this Section.

Remark 6.2. For some fixed $2 \leq N$, under Assumptions [D.3](#) and [6.1](#), $(M_{k,N})_{k \geq 1}$ are iid, mean zero and symmetric ($M_{k,N} \stackrel{d}{=} -M_{k,N}$), with $\mathbb{E}[M_{k,N}^2] = \tau^2 + \frac{\sigma^2}{N} = \theta_N^2 < \infty$.

Assumption [6.1](#) is the load-bearing assumption driving the results as it allows for a decomposition of errors into independent signs and magnitudes. Formally,

Lemma 6.3. *Cluster Sign Factorization.* For some independent iid Rademacher sequence $(\zeta_k)_k$ and some fixed N , let

$$S_{k,N} = \mathbb{1}(M_{k,N} \neq 0) \text{sign}(M_{k,N}) + \mathbb{1}(M_{k,N} = 0) \cdot \zeta_k \quad R_{k,N} = |M_{k,N}|$$

such that $M_{k,N} = S_{k,N} R_{k,N}$. Under Assumptions [3.1](#) and [6.1](#), $(S_{k,N})_k$ is iid Rademacher with $(S_{k,N})_k \perp\!\!\!\perp (R_{k,N})_k$.

Next, consider a class of adaptive sampling policies which only depend on error magnitudes $(R_{k,N})_k$, such that the designs induced by these policies are only functions of the error magnitudes. For instance, a policy which only samples according to the observed squared errors satisfies this requirement. Then, imagine that the designer conditions on a σ -algebra generated by the error magnitudes. By doing so, she will “freeze” both the design and the magnitudes, while leaving the signs iid. Therefore, by conditioning on the magnitudes, the designer can get a CLT to go through the signs and preserve asymptotic normality of $Z^\pi(B) \mid (K_B^\pi, N_B^\pi)$. More precisely, define

Definition 6.4. *Even σ -algebra.* For some fixed N_p and some iid auxiliary sequence $(U_k)_k \sim \text{Unif}[0, 1] \perp\!\!\!\perp (C_k)_k$, the (full) even σ -algebra is defined via $\mathcal{G}_\infty := \sigma((W_{k,N_p}^2)_{k \geq 1}, (M_{k,N}^2)_{k \geq 1, N \geq 1}, (U_k)_{k \geq 1})$

Corollary 6.5. *Signs are Independent of the even σ -algebra.* For some fixed N , $(S_{k,N})_k \perp\!\!\!\perp \mathcal{G}_\infty$.

Definition 6.6. *Terminal Fraction Events.* Let (K_B^π, N_B^π) be the design induced by policy π and define $\nu_B^\pi = K_B^\pi / B$. A sequence $(A_B)_B$ is a terminal fraction event if $\{(\nu_B^\pi, N_B^\pi) \in C\}$ for some Borel subset C .

Definition 6.7. *Policy Class.* A policy π is

1. **Even:** if $(\nu_B^\pi, N_B^\pi) \in \mathcal{G}_\infty$;
2. **Non-Starving:** if $\mathbb{P}(\nu_B^\pi \geq c) \rightarrow 1$ for some constant $c > 0$;
3. **N -Consistent:** if $N_B^\pi \xrightarrow{p} N_0 \geq 1$

These definitions capture precisely the heuristics above. An even policy is a policy which induces an even terminal fraction event, i.e. an event measurable with respect to the σ -algebra generated by the (infinite) sequence of squared errors. By conditioning on $\mathcal{G}_\infty$, the design and magnitudes get frozen while the errors are left unrestricted. Non-starvation and N -consistency are technical requirements. While non-starvation is also an assumption of the general inference theory, N -consistency is specific to the K vs N problem. Clearly, any asymptotic analysis will require a divergent sequence. The particular nature of the divergence (linear in B) is selected to make possible the extension towards asymptotically even designs. N -consistency is assumed to obtain stability along the N -dimension while considering K asymptotics.¹⁹ Similarly, I consider the fractional event ν_B (as opposed to K_B) to es-

¹⁹Because $N_B, N_0 \in \mathbb{N}$, N -consistency is equivalent to $\mathbb{P}(N_B = N_0) \rightarrow 1$.

establish a meaningful notion of convergence between even and asymptotically even designs along the K dimension. Under the assumptions and definitions above, a strong mode of conditional convergence follows: For any even design (K_B^π, N_B^π) entering $Z^\pi(B)$, $Z^\pi(B) \implies N(0, \theta_{N_0}^2)$ uniformly-mixing over $\mathcal{G}_\infty$. Formally,

Theorem 6.8. *Uniform Mixing for Even Designs.* Let $(K_B^\pi, N_B^\pi, (A_B)_B)$, be a design and a sequence of terminal fraction events induced by an even, non-starving, N -consistent policy. Let $\phi_B(t) = \mathbb{E}[\exp(itZ^\pi(B)) \mid \mathcal{G}_\infty]$ be the conditional characteristic function, and $\phi(t) = \exp(-\sigma^2 t^2/2)$. Then, under Assumptions 3.1, 3.2, 3.5, and 6.1,

$$|\mathbb{E}[\phi_B(t)\mathbb{1}_{A_B}] - \phi(t)\mathbb{P}(A_B)| \rightarrow 0 \quad \forall t \in \mathbb{R}$$

Moreover, for some $A_B : \liminf_B \mathbb{P}(A_B) > 0$,

$$\mathcal{L}\left(\sqrt{K_B^\pi}(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu) \mid A_B\right) \implies N(0, \theta_{N_0}^2)$$

Finally,

$$\mathcal{L}\left(\frac{\sqrt{K_B^\pi}(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu)}{\hat{b}_{K^\pi, N^\pi}} \mid A_B\right) \implies N(0, 1)$$

Theorem 6.8 presents a very strong result for exactly even policies. In particular, it states that under a symmetry assumption on the errors, inference remains invariant when conditioning on any data-dependent design which samples according to even functionals of the data. Yet, this result remains restrictive as policies usually do not have access to the true (squared) errors. At best, policies may condition on sequences of nearly-even statistics such as sample variances. In the following, I argue that results extend nicely to this policy class.

6.2 Inference for Asymptotically Even Designs under Symmetry

Definition 6.9. *Asymptotically Even Policies.* A policy π is asymptotically even if it induces a terminal design (K_B^π, N_B^π) such that

$$|\nu_B^\pi - \tilde{\nu}_B| \xrightarrow{P} 0, \quad |N_B^\pi - \tilde{N}_B| \xrightarrow{P} 0, \quad \text{with } (\tilde{\nu}_B, \tilde{N}_B) \in \mathcal{G}_\infty$$

In other words, a policy π is asymptotically even if the scaled-design (ν_B^π, N_B^π) is $o_p(1)$ close to an even design. While the definition does not link the underlying even policy to the asymptotically even policy in any meaningful way, the analyst will generally find it easier to construct the even surrogate policy $\tilde{\pi}$ by replacing the estimated squared errors (usually of the form $(X_i - \hat{\mu})^2$) entering π by the true errors $M_{k,N}^2$, and then, show convergence in probability of the terminal designs induced by both policies.

Theorem 6.10. *Asymptotically Even Theorems in K vs N (Sketch).* Let π be an asymptotically even, non-starving, N -consistent policy. Consider any terminal fraction event A_B such that $\lim_{\eta \downarrow 0} \limsup_B$

$\mathbb{P}(\text{dist}((\nu_B^\pi, N_B^\pi), \partial C) \leq \eta) = 0$ and $\liminf_B \mathbb{P}(A_B) > 0$. Then, under Assumptions 3.1, 3.2, 3.5, and 6.1, the results in Theorem 6.8 apply verbatim for policy π .

The technical condition $\lim_{\eta \downarrow 0} \limsup_B \mathbb{P}(\text{dist}((\nu_B^\pi, N_B^\pi), \partial C) \leq \eta) = 0$ is a margin condition that guarantees that $\mathbb{P}(A_B \triangle \tilde{A}_B) \rightarrow 0$, i.e. that events induced by nearly-even policies and their even surrogate have converging probability. This section is now concluded by arguing that EtC-CS and ACS are asymptotically even. Propositions E.8 and E.9 verify constructively that this is the case, by building some surrogate policies EtC-CS and ACS which replace the nearly even statistics $\hat{b}_{K,N}^2$ by their exactly counterpart $\tilde{b}_{K,N}^2$.²⁰ The proof is straightforward for EtC-CS as it depends on a single statistic $\hat{\lambda}_p$. Some additional work is required for ACS as the surrogate depends on a random stopping time $\tilde{s}$ built on a sequence of even variance ratios $(\tilde{\lambda}_k)_{k \leq \tilde{s}}$. Convergence is established by noting that the number of clusters entering the pilot is almost surely divergent for both policies, i.e. $\mathbb{P}(K_{pB}^\pi/B \geq c) \rightarrow 1$ and $\mathbb{P}(\tilde{K}_{pB}/B \geq c) \rightarrow 1$, what erodes any differences between $\hat{\lambda}_s$ and $\tilde{\lambda}_{\tilde{s}}$ and ensures convergence in probability of both statistics to λ_0 .

Corollary 6.11. *Design Robust Inference.* Let (K_B^π, N_B^π) be the design induced by policy $\pi \in \{\text{EtC-CS}, \text{ACS}\}$. Let $A_B = \{(\nu_B, N_B) \in C\}$ for any C such that $\lim_{\eta \downarrow 0} \limsup_B \mathbb{P}(\text{dist}((\nu_B^\pi, N_B^\pi), \partial C) \leq \eta) = 0$ and $\liminf_B \mathbb{P}(A_B) > 0$. Then, under Assumptions 3.1, 3.2, 3.5, and 6.1,

$$\mathcal{L} \left(\frac{\sqrt{K_B^\pi}(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu)}{\hat{b}_{K_B^\pi, N_B^\pi}} \mid A_B \right) \Rightarrow N(0, 1)$$

6.3 Inference Outside Symmetry

TBC

7 Lower Bound and Policy Comparison

In this Section I show that EtC-CS and ACS are essentially unimprovable by deriving lower bounds on the problem of $O(1/B^2)$, i.e. lower bounds are matching up to logarithmic terms. I then prove that any static policy which optimizes with respect to a misspecified ICC guesstimate will incur excess variance of $O(1/B)$, that is, of the same order than the oracle variance and one order of magnitude larger than the excess variance of the proposed algorithms. I use this evidence to argue that the excess variance of the proposed algorithms is indeed non-trivial and that ICC learning is necessary for good asymptotic performance in survey and experimental designs.

7.1 Lower Bound

As anticipated, I derive lower bounds for the generalized game where the learner can select $(N^k)_k = \mathbf{N}$ arbitrarily subject to budget constraints.²¹ To do so, I consider an *easier* version of the problem where, the first time the learner samples from cluster k she observes the cluster noise u_k together with the outcome realizations. The problem is easier inasmuch as it contains weakly more information for the learner than the standard experiment. By construction, the lower bounds on the easier problem (the augmented experiment), hold immediately on the harder problem (the benchmark game). This

²⁰Observe that $\hat{w}_{K,N}^2$ is exactly even and feasible, so no adjustment is needed on this regard.

²¹See Appendix A for a complete characterization.

simplification avoids a major technical consideration: Because observations within the same cluster are correlated, when cluster k has already been visited, the new observations from that cluster are not identically distributed conditional on the history of observations. The lack of independence invalidates standard *divergence decomposition* arguments in the original experiment.²² By revealing the cluster noise u_k the first time cluster k is visited, history-conditional independence is restored and standard proof arguments go through. The rest of proof follows as expected. In particular, consider two data generating processes y_- , y_+ according to Equation (1) with the same expectation μ , and indexed by ω_- and ω_+ , respectively, such that

$$\omega_- = (\tau_-^2, \lambda_-) \quad \omega_+ = (\tau_+^2, \lambda_+) \quad \lambda_- = \sigma^2/\tau_-^2 \quad \lambda_+ = \sigma^2/\tau_+^2$$

For some $\lambda \in (\lambda_m, \lambda_M)$ and some small positive δ , τ_-^2 and τ_+^2 are selected such that $\lambda_- = \lambda - \delta$ and $\lambda_+ = \lambda + \delta$. In other words, the only difference across the two DGPs is some small variation in the cluster variance τ^2 . As a result, optimizing with respect to λ_- becomes optimal in environment y_- and optimizing with respect to λ_+ becomes optimal in environment y_+ . Figure 2 illustrates this intuition.

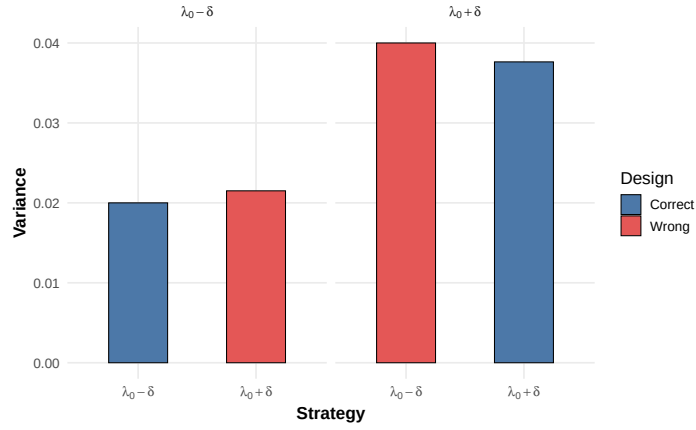


Figure 2: True Variance $W(K^*(\cdot), N^*(\cdot))$ under ω_- (columns 1, 2) and ω_+ (columns 3, 4) for policy $(K^*(\lambda_-), N^*(\lambda_-))$ (columns 1, 3) and policy $(K^*(\lambda_+), N^*(\lambda_+))$ (columns 2, 4). $\mu = 0$, $\omega = (1, 5 \pm \delta)$, $\delta = 0.5$, $B = 2000$, $F = 20$, $V = 2$.

Remark 7.1. δ is selected to balance a dual-objective: punishment and identification. A large δ makes it easier for the learner to distinguish across the two environments (and, therefore, to optimize accordingly), but it also penalizes the learner more heavily in case she fails to identify the correct instance. Conversely, a smaller δ makes it more difficult to tell instances apart, but it also makes the additional excess variance of selecting a suboptimal design smaller. By selecting $\delta \sim O(B^{-1/2})$ —as it is customary in learning problems—excess variance is maximized. Theorem 7.2 summarizes the main result in this Section. The proof of Theorem can be found in Appendix F.

Theorem 7.2. *Lower Bound on Excess Variance.* For any policy $\pi \in \Pi$, there exists a DGP y and a positive finite constant $C < \infty$ such that

$$\mathbb{E}_\pi [W(\mathbf{N})] - W(K^*(\lambda_0), N^*(\lambda_0)) \geq \frac{C}{B^2}$$

²²See Chapter 15 in Lattimore and Szepesvári (2020).

Theorem 7.2 establishes a lower bound for a very general policy class Π .²³ In particular, I allow policies to explore different number of units across different clusters, to revisit clusters freely, and to sample a different number of units every time the cluster is visited.

7.2 Policy Comparison

Theorem 7.2 suggest that Algorithms 1 and 2 are unimprovable up to logarithmic terms. However, it is worth asking if the rate in the excess variance is actually trivial, in the sense that default policies could achieve a similar rate. Because this is the first paper that fits learning within the K vs N problem, the adaptive algorithms can only be compared with static design policies. In particular, I compare EtC-CS and ACS to any static policy that takes a misspecified parameter λ as input and optimizes over it, including the minmax optimal policy. By doing so, I show that the rate in the excess variance of Algorithms 1 and 2 is indeed non-trivial as all these policies accrue (expected) excess variance of $O(1/B) \gg O(\ln B/B^2)$.²⁴

Lemma 7.3. *Misspecified Static Policies and Minmax Optimal Static Policy. Let $\lambda_b, \lambda_0 \in [\lambda_m, \lambda_M]$ such that $\lambda_b = \lambda_0 + b$ for some $b \in \mathbb{R}$. Define the static policy $\pi_b : (K^{\pi_b}, N^{\pi_b}) = (K^*(\lambda_b), N^*(\lambda_b))$, then for some positive finite constant C*

$$W(K^{\pi_b}, N^{\pi_b}, \omega) - W(K^*(\lambda_0), N^*(\lambda_0)) = C \frac{b^2}{B} + O\left(\frac{b^3}{B}\right) \sim O(1/B)$$

Moreover, define the optimal minmax static policy $\pi_{mM} : (K^{\pi_{mM}}, N^{\pi_{mM}}) = (K^*(\lambda_{mM}), N^*(\lambda_{mM}))$ where

$$\lambda_{mM} = \arg \min_{\lambda} \max_{\omega \in \Omega} W(K^*(\lambda), N^*(\lambda); \omega) - W(K^*(\lambda_0), N^*(\lambda_0); \omega)$$

Then,

$$W(K^{\pi_{mM}}, N^{\pi_{mM}}) - W(K^*(\lambda_0), N^*(\lambda_0)) = C \frac{(\lambda_{mM} - \lambda_0)^2}{B} + O\left(\frac{(\lambda_{mM} - \lambda_0)^3}{B}\right) \sim O(1/B)$$

The Lemma above shows that any misspecified static policy (including the minmax optimal) obtains a worst-case excess variance of the same order than the variance of the oracle policy itself and much larger than the worst-case excess variance of the learning protocols $O(\ln B/B^2)$. This result is not surprising. When there is no scope for learning, any deviation from the true λ_0 leads to a fixed second-order deviation. This preliminary result suggest that the excess variance of the policy is indeed non-trivial, at least when compared against static designs.

²³The exact definition of the policy class Π can be found in the Appendix. The restriction to the policy class Π essentially disregards policies which do not terminate in finite time almost surely or do not satisfy the budget constrain almost surely.

²⁴Conditional on the design, the variance of the estimator is not random. As a result, all performance guarantees (expected, high-probability, etc.) coincide for static policies that fix the design exogenously.

8 Simulations

8.1 Performance Guarantees

Figure 3 plots the distribution of designs (K^π, N^π) induced by EtC-CS and ACS. Both algorithms bunch around the oracle design $(K^*(\lambda_0), N^*(\lambda_0))$ with a slight better performance of ACS compared to EtC-CS. This is to be expected as both algorithms induce optimal designs in the limit (see Lemma 4.4), with ACS using weakly more information for estimation than EtC-CS.²⁵ The right-most panel in Figure 3 plots the distribution of the estimated $\hat{\lambda}_p$ entering the optimal map, i.e. the estimated $\hat{\lambda}_p$ at the end of the exploration stage. Although the estimation is relatively sharp, it displays a long right tail. While this is a finite sample artifact, it responds to a real phenomenon. Recall that $\hat{\lambda}$ is estimated as the ratio of the within variance $\hat{w}^2$ and the cluster variance $\hat{\tau}^2 = \Pi_{[\tau_m, \tau_M]}(\hat{b}^2 - \hat{w}^2/N_p)$. When τ^2 is small relative to σ^2 , $\hat{\tau}^2$ is estimated as the small difference of two large quantities, making the overall estimation quite noisy. Because $\hat{\tau}^2$ enters the estimator $\hat{\lambda}$ through a (very) convex map $\hat{\tau}^2 \mapsto 1/\hat{\tau}^2$, values of $\hat{\tau}^2$ near 0 will induce an unavoidable long right tail.

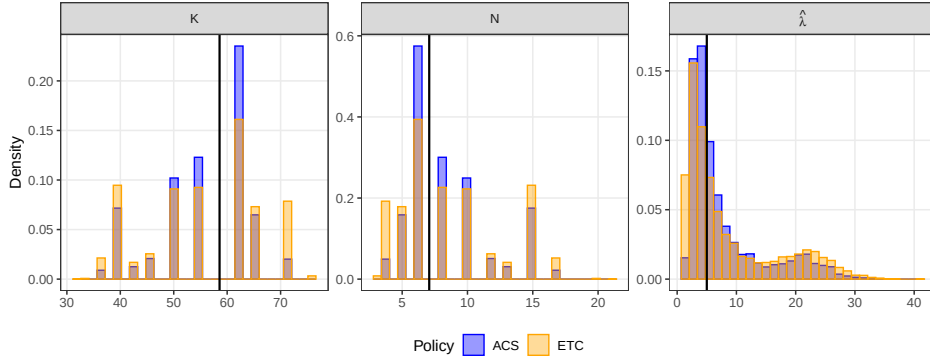


Figure 3: Design (K^π, N^π) and $\hat{\lambda}_p$ induced by EtC-CS (yellow) and ACS (blue). Solid black line represents oracle $K^*(\lambda_0), N^*(\lambda_0), \lambda_0$. 100,000 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2000$, $F = 20$, $V = 2$.

This long tail, however, is partially controlled by the design itself, as $\hat{\lambda}$ enters $N^*(\cdot)$ via a square-root which operates as a concave compressor. In addition, $K^*(\cdot)$ is decreasing in $\hat{\lambda}$, so, despite being convex, it is $\hat{\lambda}$ undershooting and not $\hat{\lambda}$ overshooting what could be problematic for the induced $K^*(\hat{\lambda})$. Overall, I expect some degree of N oversampling and K undersampling compared to the oracle for large values of λ_0 in finite samples. Figure 4 confirms this pattern. Overall, both algorithms track the optimal design relatively well on average.

²⁵Recall Observation 4.8.

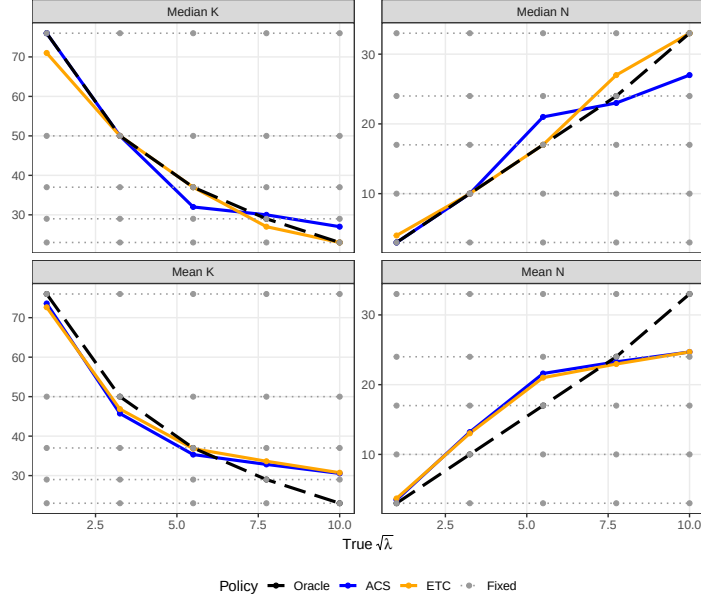


Figure 4: Median (top) and Mean (bottom) (K^π, N^π) across Policies and λ_0 . Fixed policies set at $\lambda \approx (1, 10, 30, 60, 100)$. 100,000 replications. $\mu = 0, \tau^2 = 1$. $B = 2000, F = 20, V = 2$.

Next, I evaluate the excess variance of the algorithms compared to the oracle design and an array of fixed (misspecified) policies. To do so, Figure 5 plots the mean and median excess variance of all policies in a λ grid. The red-dots in the pictures mark the worst median excess variance of each fixed policy—which naturally occur in the boundary—. One can extract several takeaways from this graph. Firstly, both EtC-CS and ACS perform very well compared to the oracle and the alternative fixed policies. Secondly, this improved performance is particularly salient when assessing the median excess variance, what is to be expected as the proposed algorithms are random by nature, therefore, they occasionally incur large excess variance compared to fixed designs. Thirdly, because of the stability considerations above, the learning algorithms behave particularly well for small values of λ (large values of τ^2). And, finally, it becomes clear that the proposed algorithms are the most robust to variations in λ . While fixed policies may outperform EtC-CS and ACS in small samples, their worst mean and median excess variance always improve on the worst mean and median excess variance of fixed competitors.

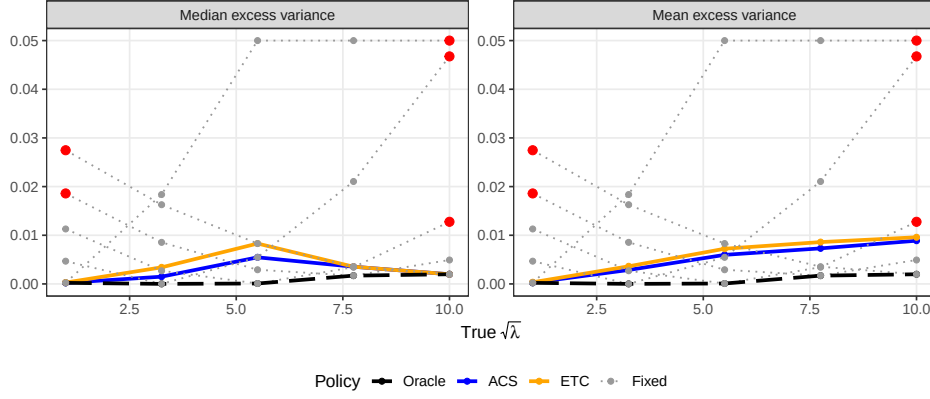


Figure 5: Median (left) and Mean (right) Excess Variance across Policies and λ_0 . Fixed policies set at $\lambda \approx (1, 10, 30, 60, 100)$. 100,000 replications. $\mu = 0, \tau^2 = 1$. $B = 2000, F = 20, V = 2$.

Finally, Figure 6 plots the evolution of excess variance as a function of budget, the main theoretical prescription in Section 4 (Theorem 5.2). For these particular parameter values, (i) both EtC-CS and ACS display very low excess variance compared to the oracle starting at relatively modest budgets, (ii) the excess variance of both EtC-CS and ACS converge faster to zero than fixed policies (see Lemma 7.3), and (iii) ACS outperforms EtC-CS, especially for small budgets.

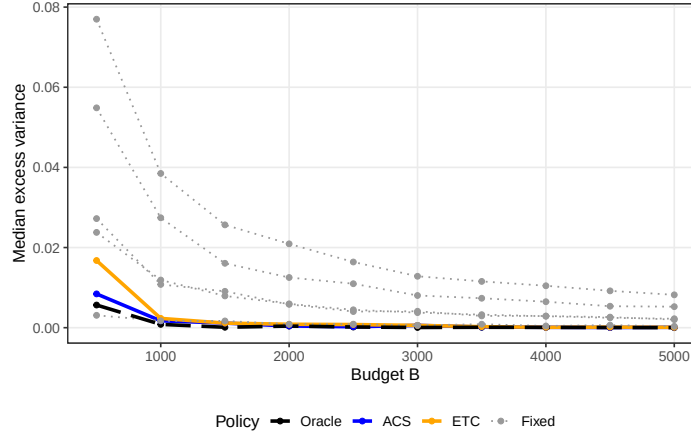


Figure 6: Median Excess Variance across Policies and Budget B . Fixed policies set at $\lambda \approx (1, 10, 30, 60, 100)$. 100,000 replications. $\mu = 0, \omega = (1, 5)$. $F = 20, V = 2$.

8.2 Conditional Inference

Next, I provide simulation evidence on conditional inference, with a particular emphasis on ACS.²⁶ In Figure 7, for a wide array of simulations satisfying symmetry, including bi-modal distributions and heavy-tailed (Laplace) distributions, I run a large number of simulations and plot the unconditional and the mode-design conditional density of $\hat{\mu}_{K,N}$.²⁷

²⁶ACS interacts with the data much more than EtC-CS, as (i) it performs nearly all cluster exploration within the pilot, therefore weakly more data enters the designs of ACS compared to EtC-CS, and (ii) its induced design depends on a sequence of statistics $(\hat{\lambda}_{k,N_p})_{k \leq s}$. As such, I consider the simulation evidence on ACS to be more suggestive of the theory results in Section 6.

²⁷Figure 12 in the Appendix displays the entire grid combination.

Additionally, for two of the distributions in the grid (the normal-normal and the bimodal-laplace), Figure 8 plots the share of observations within $\pm 1.96 \cdot \hat{b}_{K^\pi, N^\pi} / \sqrt{K^\pi}$.

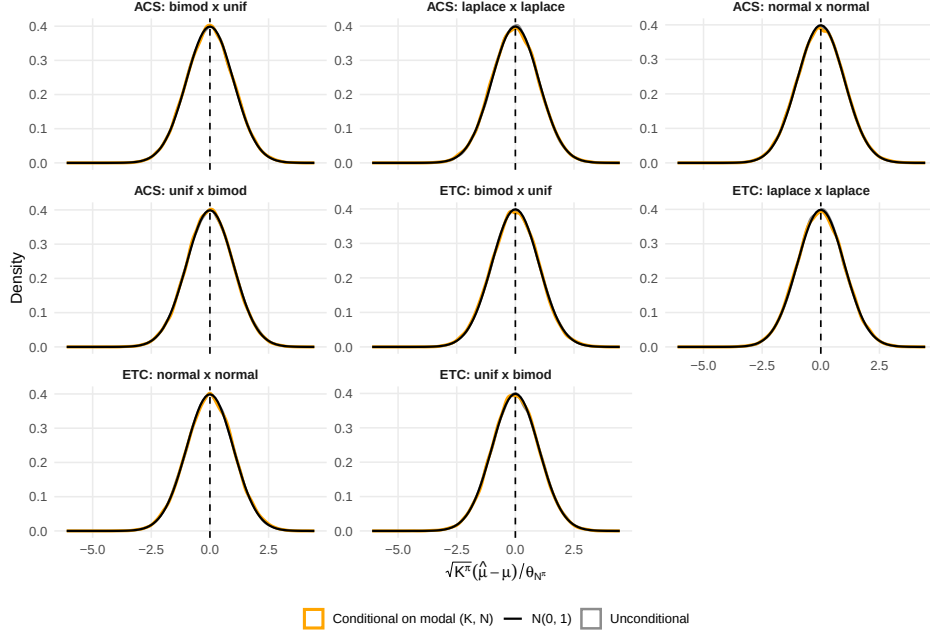


Figure 7: Empirical pdf for Modal Design Conditional Sample Average $\sqrt{K^\pi}(\hat{\mu}_{K^\pi, N^\pi} - \mu) / \theta_{N^\pi}$ across Symmetric Distributions. u_k distribution (left). ε_{ki} distribution (right). 100,000 replications. $\mu = 0$, $\omega = (1, 5)$. $B = 2000$, $F = 20$, $V = 2$.

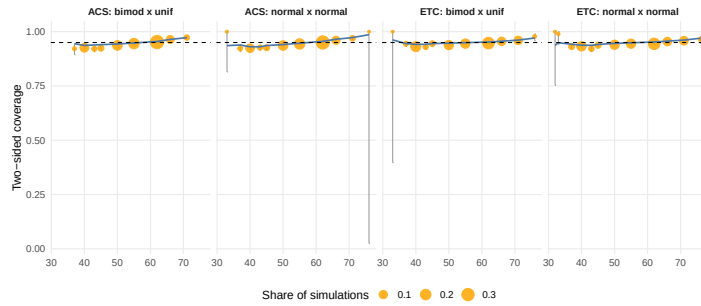


Figure 8: Coverage for Design Conditional Sample Averages $\sqrt{K^\pi}(\hat{\mu}_{K^\pi, N^\pi} - \mu) / \hat{b}_{K^\pi, N^\pi}$ for Normal-Normal and Bimodal-Laplace errors. Grey lines represent Clopper-Pearson 95% bars for each realized K^π . 100,000 replications. $\mu = 0$, $\omega = (1, 5)$. $B = 2000$, $F = 20$, $V = 2$.

Overall both graphs provide compelling evidence of design-conditional asymptotic normality in alignment with the results in Corollary 6.11. Although the coverage is fairly consistent around 95%, Figure 8 displays a slight tilt driven by finite sample concerns over the studentized variance $\hat{b}_{K^\pi, N^\pi}^2$. The tilt balances across designs to recover unconditional normality (as expected).²⁸ Next, I provide strong evidence on the necessity of (asymptotic) evenness entering Theorem 6.10. To do so, for a symmetric distribution of errors, I look at the conditional expectation $\mathbb{E}[Z^\pi(B) \mid A_B]$ induced by two asymptotically even policies (EtC-CS and ACS) and one odd policy. As per Corollary 6.5 and

²⁸Add footnote on θ_{N^π} vs $\hat{b}_{K^\pi, N^\pi}$

Theorem 6.10, $\mathbb{E}[Z^\pi(B) \mid A_B] = E[Z^\pi(B)] = 0$ only if π is (asymptotically) even. The odd policy is rather naive. It first runs a pilot where it samples $\lfloor K^*(\lambda_M) \rfloor$ and $\lfloor N^*(\lambda_m) \rfloor$ units, just like **EtC-CS**. Then, if $\bar{M}_{K_p, N_p} > 0$ it sets $K^o = \lfloor K^*(\lambda_M) \rfloor$, else $K^o = \lfloor K^*(\lambda_m) \rfloor$. N^o is selected to exhaust the budget. Crucially, $\bar{M}_{K_p, N_p}$ (the cluster error average after the pilot) is not even nor asymptotically even, hence, conditional and unconditional expectations may not be equal to each other. For each of the policies under consideration, Figure 9 pre-computes 5 quintiles for K^π , referred to as $(C_j)_j$, and then plots the conditional expectation for each of the quintiles. Figure 9 clearly shows that the conditional expectations are nearly zero for the asymptotically even policies, but are radically different for the odd policy. This evidence may be interpreted as a falsification benchmark for policy evenness.

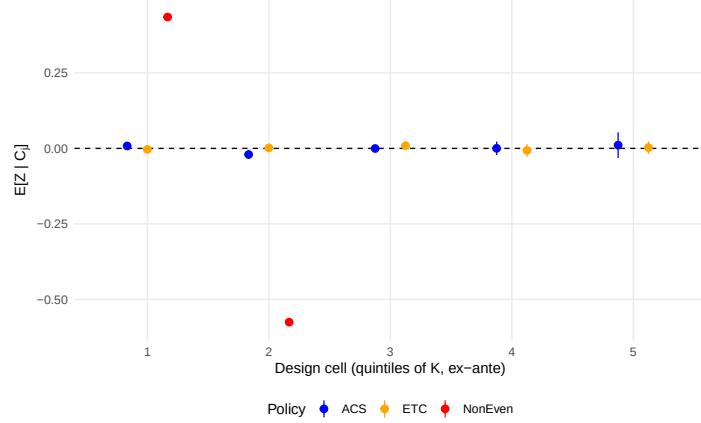


Figure 9: Conditional Expectation $\mathbb{E}[Z^\pi(B) \mid C_j]$ for the pre-computed K^π quintiles, with $\pi \in \{\text{ACS}, \text{EtC-CS}, \text{Odd}\}$. 100,000 replications. $\mu = 0$, $\omega = (1, 5)$. $B = 2000$, $F = 20$, $V = 2$.

To continue, I provide direct evidence on Theorem 6.10 by plotting $D_B = \max_j | \mathbb{E}[\exp(itZ^\pi(B)\mathbb{1}_{C_j}) - \phi(t)\mathbb{P}(C_j)] |$ as a function of budget B . I do so for a grid of four policies. The odd policy under symmetric errors, **ACS** under asymmetric errors ($u_k \sim \Gamma_u(\alpha = 1, \beta)$, such that $\text{Skew}(u_k) = 2/\sqrt{\alpha} = 2$), **ACS** under Laplace symmetric errors, and **ACS** under Gaussian symmetric errors. Figure 10 carries several takeaways. First, it shows that either policy oddness or error asymmetry break the results in 6.10. In that sense, the theory of asymptotic design independence is exhaustive in the grid $\{\text{policy distribution}\} \times \{\text{error symmetry}\}$: it only holds under policy evenness and error symmetry. Second, the simulation evidence confirms convergence of the conditional characteristic function for asymptotically even policies (**ACS**) under error symmetry. Convergence is most notable for Laplace errors. Third, it shows that exact error normality bypasses the notion of asymptotic evenness. Under Gaussian errors, sample variances are independent of sample averages. Therefore, the design induced by **ACS** is independent in finite samples of $\hat{\mu}_{K,N}$. The simulation evidence confirms that D_B cannot be told apart from 0 under exact normality for designs depending on functionals of sample variances.²⁹

²⁹For each B , grey bands are computed by first assigning a multinomial probability of design (K, N) according to the empirical distribution of designs given by **ACS** under Gaussian errors. Then, I simulate ~ 200 Gaussian errors $z_1, \dots, z_{200}$ and construct $Z^*(B)$. The grey band plots the 95% distribution of D_B under the simulated normal draws.

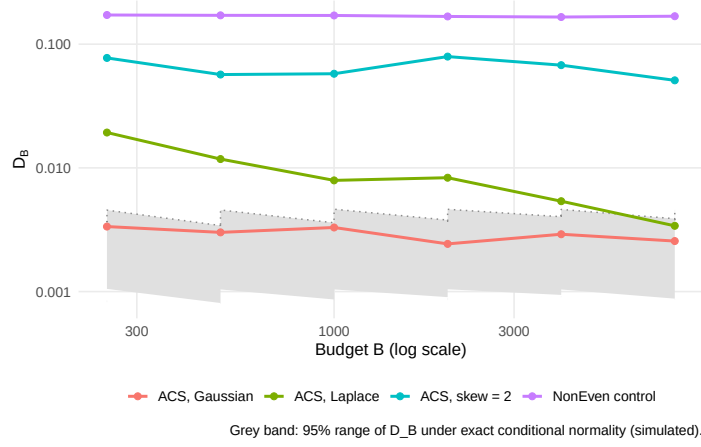


Figure 10: Maximal Difference between Normal and Conditional Characteristic Functions as a Function of B . $(C_j)_j$ is the set of pre-computed K^π quintiles. $\pi \in \{\text{ACS}, \text{Odd}\}$. Symmetric Distribution is $N(0, 1)$. Asymmetric distribution is $\Gamma(1, 1)$. 100,000 replications. $\mu = 0$, $\omega = (1, 5)$. $B = 2000$, $F = 20$, $V = 2$.

Finally, Figure 11 provides further evidence on the role of symmetry by plotting quintile-conditional expectations $\mathbb{E}[Z^\pi(B) | C_j]$ as a function of $\text{Skew}(u_k)$. The results are exactly as prescribed. For exact symmetry all design-conditional expectations are nearly 0. As skewness grows, deviations from the unconditional expectation grow as well. As expected, violations of conditional independence are particularly salient at extreme designs (cells 1 and 5)—as these carry more information about the underlying distribution of errors entering $\hat{\mu}$ —, and larger at ACS compared to EtC-CS—as the designs induced by the former depend on more data than the designs induced by the latter.

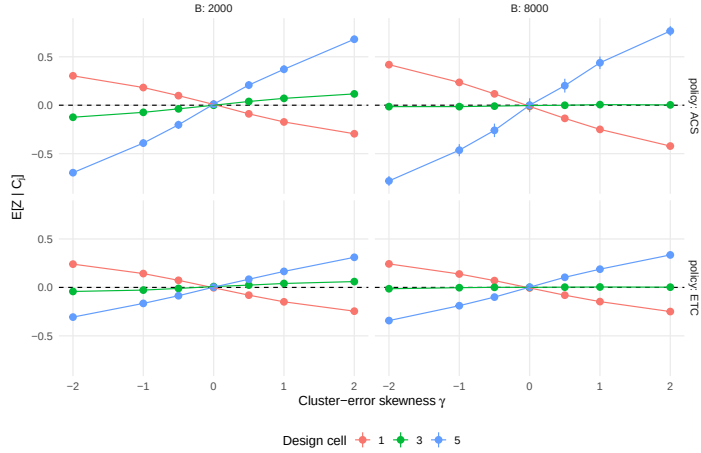


Figure 11: Conditional Expectation $\mathbb{E}[Z^\pi(B) | C_j]$ as a function of $\text{Skew}(u_k)$ and (C_j) , with $\pi \in \{\text{ACS}, \text{EtC-CS}\}$. 100,000 replications. $\mu = 0$, $\omega = (1, 5)$. $B = 2000$, $F = 20$, $V = 2$.

8.3 Debiasing outside Symmetry

9 Conclusion

This paper presents a first solution to the problem of clustered experimental and survey design under unknown intra-cluster correlation. In particular, it presents two algorithms with varying degrees of adaptability which turn-out to be minmax optimal in a setting with additive errors, linear costs and a fixed budget. The algorithms display negligible excess variance compared to the oracle, and significantly outperform every misspecified static policy. In addition, under a symmetry assumption, I show that inference on sample averages remains invariant after conditioning on asymptotically even designs. This strong result confirms that the variance reduction of the adaptive algorithms does not come at the expense of inference guarantees. In particular, the estimator remains asymptotically normal conditional on the design. Not without limitations, this is an encouraging result for adaptive data collection, and an attempt to push further adaptive experimentation and adaptive sampling in Economics, experimental sciences and public policy design.

References

- Adusumilli, K. (2026). How to sample and when to stop sampling: The generalised wald problem and minimax policies. *Review of Economic Studies*, 93(1):1–34.
- Ashraf, N., Bandiera, O., Davenport, E., and Lee, S. S. (2020). Losing prosociality in the quest for talent? sorting, selection, and productivity in the delivery of public services. *American Economic Review*, 110(5):1355–1394.
- Bai, Y. (2022). Optimality of matched-pair designs in randomized controlled trials. *American Economic Review*, 112(12):3911–3940.
- Bai, Y., Jiang, L., Romano, J. P., Shaikh, A. M., and Zhang, Y. (2024). Covariate adjustment in experiments with matched pairs. *Journal of Econometrics*, 241(1):105740.
- Ballin, M. and Barcaroli, G. (2013). Joint determination of optimal stratification and sample allocation using genetic algorithm. *Survey Methodology*, 39(2):369–393.
- Barcaroli, G., Fasulo, A., Guandalini, A., and Terribili, M. D. (2022). Two-stage sampling design and sample selection with the r package r2beat. *arXiv preprint arXiv:2209.06532*.
- Bethel, J. (1989). Sample allocation in multivariate surveys. *Survey methodology*, 15(1):47–57.
- Biemer, P. P. and Lyberg, L. E. (2003). *Introduction to survey quality*. John Wiley & Sons.
- Campbell, M. K., Grimshaw, J. M., and Elbourne, D. R. (2004). Intraclass correlation coefficients in cluster randomized trials: empirical insights into how should they be reported. *BMC Medical Research Methodology*, 4(1):9.
- Canay, I. A., Romano, J. P., and Shaikh, A. M. (2017). Randomization tests under an approximate symmetry assumption. *Econometrica*, 85(3):1013–1030.
- Cesa-Bianchi, N., Colomboni, R., and Kasy, M. (2025). Adaptive maximization of social welfare. *Econometrica*, 93(3):1073–1104.
- Chow, S.-C. and Corey, R. (2011). Benefits, challenges and obstacles of adaptive clinical trial designs. *Orphanet journal of rare diseases*, 6(1):79.
- Cicchitelli, G., Herzel, A., and Montanari, G. E. (1992). *Il campionamento statistico*. il Mulino.
- Cytrynbaum, M. (2021). Optimal stratification of survey experiments. *arXiv preprint arXiv:2111.08157*.
- Cytrynbaum, M. (2024). Covariate adjustment in stratified experiments. *Quantitative Economics*, 15(4):971–998.
- Datta, D. K. and Rajagopalan, N. (1998). Industry structure and ceo characteristics: An empirical study of succession events. *Strategic Management Journal*, 19(9):833–852.
- Duflo, E. and Banerjee, A. (2017). *Handbook of field experiments*, volume 1. Elsevier.

- Hadad, V., Hirshberg, D. A., Zhan, R., Wager, S., and Athey, S. (2021). Confidence intervals for policy evaluation in adaptive experiments. *Proceedings of the national academy of sciences*, 118(15):e2014602118.
- Heß, S., Jaimovich, D., and Schündeln, M. (2021). Development projects and economic networks: Lessons from rural gambia. *The Review of Economic Studies*, 88(3):1347–1384.
- Imbens, G., Qin, C., and Wager, S. (2025). Admissibility of completely randomized trials: A large-deviation approach. *arXiv preprint arXiv:2506.05329*.
- Karing, A. (2024). Social signaling and childhood immunization: A field experiment in sierra leone. *The Quarterly Journal of Economics*, 139(4):2083–2133.
- Kasy, M. and Sautmann, A. (2021). Adaptive treatment assignment in experiments for policy choice. *Econometrica*, 89(1):113–132.
- Kiefer, J. and Wolfowitz, J. (1960). The equivalence of two extremum problems. *Canadian Journal of Mathematics*, 12:363–366.
- Kish, L. (1965). Survey sampling. new york: John wesley & sons. *Am Polit Sci Rev*, 59(4):1025.
- Knox, S. A. and Chondros, P. (2004). Observed intra-cluster correlation coefficients in a cluster survey sample of patient encounters in general practice in australia. *BMC Medical Research Methodology*, 4(1):30.
- Lattimore, T. and Szepesvári, C. (2020). *Bandit algorithms*. Cambridge University Press.
- Li, D., Raymond, L., and Bergman, P. (2026). Hiring as exploration. *Review of Economic Studies*, 93(2):1200–1240.
- Lin, G. C. and Ho, S. P. (2003). China’s land resources and land-use change: insights from the 1996 land survey. *Land use policy*, 20(2):87–107.
- Mbiti, I., Muralidharan, K., Romero, M., Schipper, Y., Manda, C., and Rajani, R. (2019). Inputs, incentives, and complementarities in education: Experimental evidence from tanzania. *The Quarterly journal of economics*, 134(3):1627–1673.
- Meriggi, N. F., Voors, M., Levine, M., Ramakrishna, V., Kangbai, D. M., Rozelle, M., Tyler, E., Kallon, S., Nabieu, J., Cundy, S., et al. (2024). Last-mile delivery increases vaccine uptake in sierra leone. *Nature*, 627(8004):612–619.
- Muralidharan, K. and Niehaus, P. (2017). Experimentation at scale. *Journal of Economic Perspectives*, 31(4):103–124.
- Neyman, J. (1992). On the two different aspects of the representative method: the method of stratified sampling and the method of purposive selection. In *Breakthroughs in statistics: Methodology and distribution*, pages 123–150. Springer.
- Nie, X., Tian, X., Taylor, J., and Zou, J. (2018). Why adaptively collected data have negative bias

- and how to correct for it. In *International Conference on Artificial Intelligence and Statistics*, pages 1261–1269. PMLR.
- Rudelson, M. and Vershynin, R. (2013). Hanson-wright inequality and sub-gaussian concentration.
- Sedgwick, P. (2015). Multistage sampling. *Bmj*, 351.
- Tam, B. (2005). Household sample surveys in developing and transition countries.
- Tschprow, A. (1923). On the two different aspects of the representative method: the method of stratified son the mathematical expectation of the moments of frequency distributions in the case of correlated observation sampling and the method of purposive selection. *Metron*, 2(646-683):192.
- Wager, S. (2024). Causal inference: A statistical learning approach. URL https://web.stanford.edu/swager/causal_inf_book.pdf.

A Proofs Section 3

Generalized Problem. Consider the general problem where the learner can sample an arbitrary number of units N^k from each cluster k subject to budget constraints. Let $\mathbf{N} = (N^k)_k$ and define the set $\mathcal{K} = \{k : N^k > 0\}$ with $K = |\mathcal{K}|$. The estimator of interest the average of averages

$$\hat{\mu}_{\mathbf{N}} = \frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k} \sum_{i=1}^{N^k} y_{ki}$$

with variance

$$\begin{aligned} W(\mathbf{N}) &= \text{Var}(\hat{\mu}_{\mathbf{N}}) = \text{Var} \left(\frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k} \sum_{i=1}^{N^k} y_{ki} \right) = \text{Var} \left(\mu + \frac{1}{K} \sum_{k \in \mathcal{K}} u_k + \frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k} \sum_{i=1}^{N^k} \varepsilon_{ki} \right) \\ &= \frac{K}{K^2} \tau^2 + \frac{1}{K^2} \sum_{k \in \mathcal{K}} \frac{\sigma^2}{N^k} = \frac{\tau^2}{K} \left(1 + \frac{\lambda_0}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k} \right) = \frac{\tau^2}{K} \left(1 + \frac{\lambda_0}{N^H(\mathbf{N})} \right) \end{aligned}$$

where $N^H(\mathbf{N}) = \left(\frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k} \right)^{-1}$ is the harmonic mean. For $N^k = N$ for all $k \in \mathcal{K}$ the expression simplifies to Equation (2).

Variance Minimization. Analogously to Equation (2), the generalized variance minimization under a continuous relaxation can be written as

$$\min_{\mathbf{N}} W(\mathbf{N}) \quad \text{s.t.} \quad KF + V \sum_{k \in \mathcal{K}} N^k \leq B$$

Optimal Map. Define $S = \sum_{k \in \mathcal{K}} N^k$. For some fixed K , variance minimization reduces to minimizing $\sum_{k \in \mathcal{K}} \frac{1}{N^k}$. By convexity,

$$\frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k} \geq \frac{1}{\frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k}} = \frac{K}{S}$$

so $\sum_{k \in \mathcal{K}} \frac{1}{N^k} \geq \frac{K^2}{S}$, with equality iff $N^k = \frac{S}{K} = N$ for all $k \in \mathcal{K}$. So, for any fixed K , the variance is minimized by a balanced allocation $N^k = N$. Next, let $K(N) = \frac{B}{F+VN}$, such that

$$W(\mathbf{N} \mid N^k = N \ \forall k \in \mathcal{K}) = W(K(N), N) = \frac{(F+VN)\tau^2}{B} \left(1 + \frac{\lambda_0}{N} \right)$$

A first order condition returns

$$\frac{V\tau^2}{B} \left(1 + \frac{\lambda_0}{N} \right) = \frac{(F+VN)\tau^2}{B} \left(\frac{\lambda_0}{N^2} \right) \implies V(N+\lambda_0) = \frac{(F+VN)\lambda_0}{N} \implies N^*(\omega) = \sqrt{\lambda_0 F/V} = N^*(\lambda_0)$$

Further observe that $W(K(N), N)$ is strictly convex in N for $N > 0$, i.e.

$$W_{NN}(K(N), N) = -\frac{V\tau^2}{B} \frac{\lambda_0}{N^2} + \frac{V\tau^2\lambda_0}{BN^2} + \frac{(F+VN)\tau^2}{B} \frac{\lambda_0}{N^3} = \frac{(F+VN)\tau^2}{B} \frac{\lambda_0}{N^3} > 0$$

Thus the local point satisfying the FOC is indeed the global minimum of the function. Finally, because $N^*(\cdot)$ is strictly monotonic in λ with $\lim_{\lambda \rightarrow 0} N^*(\lambda) = 0$ and $\lim_{\lambda \rightarrow \infty} N^*(\lambda) = \infty$, the domain constraints on N lead to the final form of the optimal map

$$K^*(\lambda) = \frac{B}{F + VN^*(\lambda)}, \quad N^*(\lambda) = \min \left\{ \max \left\{ 2, \sqrt{\lambda \frac{F}{V}} \right\}, \frac{B-2F}{2V} \right\}$$

It follows that the optimal map (and oracle allocation) of the generalized problem is the same than the optimal map (and oracle allocation) of the simplified problem.

Oracle Variance. Under Assumption 3.5 (interior solution for $N^*(\lambda)$),

$$\begin{aligned} W(K^*(\lambda_0), N^*(\lambda_0)) &= \frac{\tau^2}{K^*(\lambda_0)} \left(1 + \frac{\lambda_0}{N^*(\lambda_0)} \right) = \frac{(F + \sqrt{FV\lambda_0})\tau^2}{B} \left(1 + \sqrt{\lambda_0 \frac{V}{F}} \right) \\ &= \frac{\tau^2}{B} \left(F + 2\sqrt{F\lambda_0 V} + \lambda_0 V \right) = \frac{\tau^2 \left(\sqrt{F} + \sqrt{\lambda_0 V} \right)^2}{B} \end{aligned}$$

B Proofs of Section 4

Proof of Lemma 4.1. Unbiasedness of the estimators. To be used for a LLN argument,

$$\begin{aligned} \mathbb{E}[\hat{w}_{K,N}^2] &= \frac{1}{K(N-1)} \sum_{k=1}^K \sum_{i=1}^N \mathbb{E}[(y_{ki} - \hat{\mu}_N^k)^2] = \frac{1}{K(N-1)} \sum_{k=1}^K \sum_{i=1}^N \text{Var}(y_{ki} - \hat{\mu}_N^k) \\ &= \frac{1}{K(N-1)} K(N-1) \sigma^2 = \sigma^2 \\ \mathbb{E}[\hat{b}_{K,N}^2] &= \frac{1}{K-1} \sum_{k=1}^K \mathbb{E}[(\hat{\mu}_N^k - \hat{\mu}_{K,N})^2] = \frac{1}{K-1} (K-1) \left(\tau^2 + \frac{\sigma^2}{N} \right) = \tau^2 + \frac{\sigma^2}{N} = \theta_N^2 \end{aligned}$$

Consistency under K asymptotics,

$$\hat{w}_{K,N}^2 = \frac{1}{K} \sum_{k=1}^K \frac{1}{N-1} \sum_{i=1}^N (y_{ki} - \hat{\mu}_N^k)^2 \xrightarrow{\text{a.s.}} \sigma^2$$

where convergence follows from the SLLN under *iid*-ness. I used cross-cluster independence to argue that $\sum_{i=1}^N (y_{ki} - \hat{\mu}_N^k)^2 \perp \sum_{i=1}^N (y_{li} - \hat{\mu}_N^l)^2$ for $k \neq l$. A similar argument holds for consistency of $\hat{w}_{K,N}^2$

under N asymptotics, namely

$$\begin{aligned}\hat{w}_{K,N}^2 &= \frac{1}{N-1} \sum_{i=1}^N \frac{1}{K} \sum_{k=1}^K (y_{ki} - \hat{\mu}_N^k)^2 = \frac{1}{N-1} \sum_{i=1}^N \frac{1}{K} \sum_{k=1}^K (\varepsilon_{ki} - \bar{\varepsilon}_{ki})^2 \\ &= \frac{N}{N-1} \left[\frac{1}{N} \sum_{i=1}^N \varepsilon_{ki}^2 - \bar{\varepsilon}_{ki}^2 \right] \xrightarrow[N]{\text{a.s.}} \sigma^2\end{aligned}$$

provided $\frac{1}{N} \sum_{i=1}^N \varepsilon_{ki}^2 \xrightarrow[N]{\text{a.s.}} \sigma^2$ under the SLLN given ε_{ki} is iid and $\bar{\varepsilon}_{ki}^2 \xrightarrow[N]{\text{a.s.}} 0$ also under the SLLN. Applying CMT completes the proof. Similarly,

$$\hat{b}_{K,N}^2 = \frac{1}{K-1} \sum_{k=1}^K (\hat{\mu}_N^k - \hat{\mu}_{K,N})^2 \xrightarrow[N]{\text{a.s.}} \tau^2 + \frac{\sigma^2}{N} = \theta_N^2$$

where I used the same decomposition and argued that $\bar{u}_k + \bar{\varepsilon}_{ki} \xrightarrow[N]{\text{a.s.}} 0$. The denominator entering $\hat{\lambda}_{K,N}$ converges almost surely (under the CMT) to τ^2 . Additionally, note that any projection $\Pi_{[a,b]}(x)$ onto a closed interval $[a, b]$ is 1-Lipschitz, therefore the denominator preserves convergence after clipping provided λ_0 in a compact subset of $[\lambda_m, \lambda_M]$. By applying the CMT once again, and noting that the denominator is strictly larger than zero (after clipping) under Assumption 3.2, $\hat{\lambda}_{K,N} \xrightarrow[N]{\text{a.s.}} \lambda_0$. $\square$

C Proofs of Section 5.

This Section is structured as follows. I first present a set of definitions and preliminary lemmas that will play an important role in the derivation of high probability upper bounds and inference for the K vs N problem. I then discuss an intermediate theorem on high probability bounds by first presenting a general lemma from which the theorem will follow as a simple corollary. Finally, I transform the high probability bounds into bounds in expectation by deriving worst-case upper bounds, and in doing so, proving Theorem 5.2 in the main text.

C.1 Definitions

Define

$$\varepsilon_k = (\varepsilon_{1k}, \varepsilon_{2k}, \dots) \quad \bar{\varepsilon}_{k,N} = \frac{1}{N} \sum_{i=1}^N \varepsilon_{ki} \quad W_{k,N}^2 = \frac{1}{N-1} \sum_{i=1}^N (\varepsilon_{ki} - \bar{\varepsilon}_{k,N})^2 \quad \hat{w}_{K,N}^2 = \frac{1}{K-1} \sum_{k=1}^K W_{k,N}^2$$

where $\mathbb{E}[W_{k,N}^2] = \sigma^2$ for all N and $A_{k,N}^2 = W_{k,N}^2 - \sigma^2$. In particular, note that $W_{k,N}^2$ is the usual within-cluster k sample variance and $\hat{w}_{K,N}^2$ is the average of within-cluster sample variances, which identifies the within-cluster variance along the K -dimension (as per Lemma 4.1). Now, observe that

$$M_{k,N} = u_k + \bar{\varepsilon}_{k,N} \quad \hat{\mu}_{K,N} - \mu = \bar{M}_{K,N} = \frac{1}{K} \sum_{k=1}^K M_{k,N}$$

where $M_{k,N}$ is the average of cluster k errors such that $\mathbb{E}[M_{k,N}] = 0$. Similarly, mimicking the structure of the within-cluster variance $\hat{w}_{K,N}^2$,

$$\hat{b}_{K,N}^2 = \frac{1}{K-1} \sum_{k=1}^K (M_{k,N} - \bar{M}_{K,N})^2$$

where $\hat{b}_{K,N}^2$ is the between-cluster variance (as in Equation 4.1) such that $\mathbb{E}[\hat{b}_{K,N}^2] = \tau^2 + \sigma^2/N = \theta_N^2$. In addition, define the centered second moment of the average of cluster k errors

$$B_{k,N}^2 = M_{k,N}^2 - \theta_N^2$$

Unlike $\hat{w}_{K,N}^2$, $\hat{b}_{K,N}^2$ cannot be written as the K -average of $B_{k,N}^2$ because $\hat{b}_{K,N}^2$ demeans $M_{k,N}$ by the (residuals of the) grand mean $\bar{M}_{K,N}$, yet an approximate decomposition is possible via

$$\begin{aligned} \hat{b}_{K,N}^2 - \theta_N^2 &= \frac{1}{K-1} \sum_{k=1}^K (M_{k,N} - \bar{M}_{K,N})^2 - \theta_N^2 \\ &= \frac{1}{K-1} \left(\sum_{k=1}^K M_{k,N}^2 - K \bar{M}_{K,N}^2 \right) - \theta_N^2 \\ &= \frac{1}{K-1} \left(K \theta_N^2 + \sum_{k=1}^K (M_{k,N}^2 - \theta_N^2) - K \bar{M}_{K,N}^2 \right) - \theta_N^2 \\ &= \frac{1}{K-1} \sum_{k=1}^K B_{k,N}^2 + \frac{\theta_N^2}{K-1} - \frac{K}{K-1} \bar{M}_{K,N}^2 \end{aligned}$$

C.2 Preliminary Lemmas and Definitions

Lemma C.1. *Implementation of Optimal Designs.* Let (K^π, N^π) be the design induced by policy π , and define $K_M = \lfloor K^*(\lambda_M) \rfloor$, $K_m = \lfloor K^*(\lambda_m) \rfloor$. N_M and N_m are defined as expected. Then, under Assumption 3.2 and 3.5, $(K^{\text{EtC-CS}}, N^{\text{EtC-CS}}) = (K^*(\hat{\lambda}_{K_M, N_m}), N^*(\hat{\lambda}_{K_M, N_m}))$ and $(K^{\text{ACS}}, N^{\text{ACS}}) = (K^*(\hat{\lambda}_{\max\{K_M, s-1\}, N_m}), N^*(\hat{\lambda}_{\max\{K_M, s-1\}, N_m}))$, where s is defined as in Equation 5.

Proof of Lemma C.1. In the case of EtC-CS, the Lemma is a simple restatement of the algorithm under the assumptions. In the case of ACS, I simply provide some further context on Observation 4.8 and Remark 4.9. By the estimator's clipping, $K^*(\hat{\lambda}_k) \geq K^*(\lambda_M) \geq K_M$, therefore for any period $k \leq K_M - 1$, $k+1 \leq K^*(\hat{\lambda}_k)$, hence cluster K_M will always be explored. Similarly, $K^*(\hat{\lambda}_k) \leq K^*(\lambda_m) \leq K_m + 1$, therefore for any period $k \geq K_m$, $k+1 \geq K^*(\hat{\lambda}_k)$ and cluster $K^*(\lambda_m) + 1$ will never be explored. It follows that $s \in [\lfloor K^*(\lambda_M) \rfloor, \lfloor K^*(\lambda_m) \rfloor]$. Finally, as noted in Remark 4.9, the learner cannot assert that for any $k > K_M$, $k \leq K^*(\hat{\lambda}_k)$, yet, by construction, for $k = K_M$, $k \leq K^*(\hat{\lambda}_{K_M})$. Because the learner may stop in step K_M with positive probability (i.e. if $K_M + 1 > K^*(\hat{\lambda}_{K_M})$), using the stopping time $s - 1$ turns out being overly conservative for the corner case $k = K_M$. ■

Lemma C.2. *Second Order Approximation.* Fix $(K, N) \geq 2$. For any policy π which induces $(K^\pi, N^\pi) = (K^*(\hat{\lambda}_{K,N}), N^*(\hat{\lambda}_{K,N}))$ a.s., then

$$\Delta_\pi \mid (K, N) = \frac{1}{2} W_{\lambda\lambda} \mid_{\lambda=\lambda_0} (\hat{\lambda}_{K,N} - \lambda_0)^2 + R_p$$

where $W_{\lambda\lambda}|_{\lambda=\lambda_0} = W_{\lambda\lambda}(K^*(\lambda_0), N^*(\lambda_0)) \sim O(1/B)$ and $R_p \sim O((\hat{\lambda}_{K,N} - \lambda_0)^3)$.

Proof of Lemma C.2. First note that Δ_π conditional on the design is not random. Lemma follows from a simple second order approx around λ_0

$$\begin{aligned}\Delta_\pi | (K, N) &= W(K^\pi, N^\pi) - W(K^*(\lambda_0), N^*(\lambda_0)) \\ &= W(K^*(\hat{\lambda}_{K,N}), N^*(\hat{\lambda}_{K,N})) - W(K^*(\lambda_0), N^*(\lambda_0)) \\ &= W_\lambda|_{\lambda=\lambda_0}(\hat{\lambda}_{K,N} - \lambda_0) + \frac{1}{2}W_{\lambda\lambda}|_{\lambda=\lambda_0}(\hat{\lambda}_{K,N} - \lambda_0)^2 + O((\hat{\lambda}_{K,N} - \lambda_0)^3) \\ &= \frac{1}{2}W_{\lambda\lambda}|_{\lambda=\lambda_0}(\hat{\lambda}_{K,N} - \lambda_0)^2 + O((\hat{\lambda}_{K,N} - \lambda_0)^3)\end{aligned}$$

where the last equality follows from optimality of (K^*, N^*) at λ_0 and the Envelope Theorem. Finally, I show analytically that $W_{\lambda\lambda}|_{\lambda=\lambda_0} \sim O(1/B)$

$$\begin{aligned}W(K^*(\lambda), N^*(\lambda)) &= \tau^2 \frac{F + V\sqrt{\lambda \frac{F}{V}}}{B} \cdot \left(1 + \frac{\lambda_0}{\sqrt{\lambda \frac{F}{V}}}\right) = \frac{\tau^2}{B} \cdot \left(F + \lambda_0 V + \sqrt{VF} \left(\lambda_0/\lambda^{1/2} + \lambda^{1/2}\right)\right) \\ W_\lambda(K^*(\lambda), N^*(\lambda)) &= \frac{\tau^2}{2B} \sqrt{FV} \left(-\lambda_0 \lambda^{-3/2} + \lambda^{-1/2}\right) \\ W_{\lambda\lambda}(K^*(\lambda), N^*(\lambda)) &= \frac{\tau^2}{4B} \sqrt{FV} \left(3\lambda_0 \lambda^{-5/2} - \lambda^{-3/2}\right) \\ W_{\lambda\lambda}(K^*(\lambda_0), N^*(\lambda_0)) &= \frac{\tau^2}{4B} \sqrt{FV} \left(3\lambda_0^{-3/2} - \lambda_0^{-3/2}\right) = \frac{\tau^2 \sqrt{FV}}{2B\lambda_0^{3/2}}\end{aligned}$$

■

Lemma C.3. *Random-Index Composition. For some random variable X_M such that $X_M \xrightarrow[M]{a.s.} X$ with M a deterministic divergent sequence. Let $(M_N)_N$ be a random variable $\in \mathbb{N}$ such that $\mathbb{P}(M_N \geq cN) \rightarrow 1$, then $X_{M_N} \xrightarrow[N]{p} X$.*

Remark C.4. Lemma C.3 will be used repeatedly to jump between a.s. convergence over fixed divergent sequences and convergence in probability over random divergent sequences, both in performance and inference results.

Proof of Lemma C.3. Let $T_N = \sup_{M \geq cN} |X_M - X|$, by virtue of a.s. convergence, it follows that $T_N \xrightarrow{a.s.} 0$. Then, by event inclusion

$$\mathbb{P}(|X_{M_N} - X| > \epsilon) \leq \mathbb{P}(M_N < cN) + \mathbb{P}(T_N > \epsilon) \rightarrow 0$$

Proof of Lemma 4.4. By virtue of Lemma 4.1, $\hat{\lambda}_{K,N} \xrightarrow[K]{a.s.} \lambda_0$ for some diverging K and fixed $N \geq 2$. Next, by Lemma C.3, for some random variable K_B such that $\mathbb{P}(K(B)/B \geq c) \rightarrow 1$, it follows that $\hat{\lambda}_{K_B,N} \xrightarrow[B]{p} \lambda_0$. Applying the CMT over the continuous map $K^*(\cdot), N^*(\cdot)$ completes the proof.

Corollary C.5. *EtC-CS and ACS are K -asymptotic Pilot Policies. Namely,*

$$\mathbb{P}(\nu_{pB}^{\text{EtC-CS}} \geq c) \rightarrow 1, \quad \mathbb{P}(\nu_{pB}^{\text{ACS}} \geq c) \rightarrow 1$$

with $N_p^{\text{EtC-CS}} = N_p^{\text{ACS}} = N^*(\lambda_m)$ fixed.

Proof of Corollary C.5. The result follows immediately by noting that for any finite λ_M , $N^*(\lambda_M) < \infty$, hence $\lim_{B \rightarrow \infty} \nu_{pB}^{\text{EtC-CS}} = 1/(F + VN^*(\lambda_M)) > 0$. $\nu_{pB}^{\text{ACS}} \geq \nu_{pB}^{\text{EtC-CS}}$ almost surely, thus result follows.

Proof of Lemma 4.4. Lemma follows by noting that $\hat{\lambda}_{K_{pB}, N_p} \xrightarrow{P} \lambda_0$ provided $\hat{\lambda}_{K, N} \xrightarrow{\text{a.s.}} \lambda_0$ (under Lemma C.3). Applying the CMT over the optimal map $\lambda \mapsto (K^*(\lambda), N^*(\lambda))$ completes the proof.

Corollary C.6.

$$|\nu_B^\pi - \nu_0| \xrightarrow{P} 0, \quad |N_B^\pi - N_0| \xrightarrow{P} 0, \quad \text{for } \pi \in \{\text{EtC-CS}, \text{ACS}\}$$

C.3 Concentration Inequalities

Subgaussian and Subexponential Constants. I now discuss the concentration properties of $M_{k,N}, A_{k,N}^2, B_{k,N}^2$ in light of Assumption 5.1 for some fixed N . Let $\kappa_M = \sqrt{\kappa_u^2 + \kappa_\varepsilon^2/N}$. By independence of $u_k \perp \varepsilon_k$, $M_{k,N}$ is κ_M -subgaussian. Hence, it is subexponential with parameters $\kappa_M = \kappa_M^2, b_M = 0$. $B_{k,N}^2$ is the centered square of a κ_M -subgaussian variable M_k , therefore it is subexponential with parameters $\kappa_B = c_1 \kappa_M^4, b_B = c_2 \kappa_M^2$. Finally, note that

$$\mathbf{P}_N = I_N - \frac{1}{N} \mathbf{1}\mathbf{1}', \quad \tilde{\mathbf{P}}_N = \frac{1}{N-1} \mathbf{P}_N \quad A_{k,N}^2 = \frac{1}{N-1} (\varepsilon_k' \mathbf{P}_N \varepsilon_k - (N-1)\sigma^2) = \varepsilon_k' \tilde{\mathbf{P}}_N \varepsilon_k - \sigma^2$$

such that $\mathbb{E}[\varepsilon_k' \mathbf{P}_N \varepsilon_k] = \sigma^2 \text{tr}(\mathbf{P}_N) = \sigma^2(N-1)$ and $A_{k,N}^2$ is centered. Then, by Hanson-Wright (see Rudelson and Vershynin (2013)), $A_{k,N}^2$ is subexponential with parameters $\kappa_A = c_3 \kappa_\varepsilon^2 \|\tilde{\mathbf{P}}_N\|_F^2$ and $b_A = c_4 \kappa_\varepsilon \|\tilde{\mathbf{P}}_N\|$, where $\|\cdot\|_F$ is the Frobenius norm and $\|\cdot\|$ is the operator norm. Because $\mathbf{P}_N$ is an orthogonal projection of rank $N-1$, $\|\mathbf{P}_N\|_F^2 = (N-1)$ and $\|\mathbf{P}_N\| = 1$. After scaling, $A_{k,N}^2$ is subexponential with parameters $\kappa_A = c_3 \frac{\kappa_\varepsilon^2}{N-1}$ and $b_A = c_4 \frac{\kappa_\varepsilon}{N-1}$.

Maximal Bernstein. I now present without proof Bernstein's maximal inequality for sums of iid subexponential variables. This maximal inequality will play a central role in the derivation of Lemma C.7. Let $\{\xi_k\}$ be a sequence of iid (κ, b) -subexponential random variables. Then, for every $K_+ \geq 1$ and $t > 0$,

$$\mathbb{P} \left(\max_{1 \leq K \leq K_+} \sum_{k=1}^K \xi_k \geq t \right) \leq 2 \exp \left(-\frac{1}{2} \min \left(\frac{t^2}{K_+ \kappa}, \frac{t}{b} \right) \right)$$

C.4 High Probability Upper Bounds

Lemma C.7. Generalized Theorem C.8. Assume that a policy π induces a pilot of random size (K_{pB}^π, N_p^π) with $K_{pB}^\pi \in \mathcal{K} = [K_{B-}, K_{B+}]$ a.s. with $K_{B+}/K_{B-} \leq C_0 \sim O(1) \perp B$ and $N_p^\pi = N_p$ fixed.

Under Assumptions 3.1, 3.2, 3.5 and 5.1, there exists a positive finite constant C_1 not depending on B such that

$$\mathbb{P}(|\hat{\lambda}_{K_{pB}, N_p}^\pi - \lambda_0| \leq D^\lambda(\alpha)) \geq 1 - \alpha, \quad D^\lambda(\alpha) = C_1 \sqrt{\frac{\ln(1/\alpha)}{K_{B-}}}$$

Proof of Lemma C.7. Take $A_{k,N}^2$ as an example (with subexponential constants (κ_A, b_A)). The same argument will go through for $B_{k,N}^2$ and $M_{k,N}$. Define $\delta(\kappa, \alpha) = \sqrt{\frac{2C_0\kappa \ln(6/\alpha)}{K_{B-}}}$, then

$$\begin{aligned} \mathbb{P}(|\hat{w}_{K_{pB}, N_p}^2 - \sigma^2| \geq \delta(\kappa_A, \alpha)) &\leq \mathbb{P}\left(\sup_{K_{B-} \leq K \leq K_{B+}} |\hat{w}_{K, N_p}^2 - \sigma^2| \geq \delta(\kappa_A, \alpha)\right) \\ &\leq \mathbb{P}\left(\frac{1}{K_{B-}} \max_{K_{B-} \leq K \leq K_{B+}} \left|\sum_{k=1}^K A_{k, N_p}^2\right| \geq \delta(\kappa_A, \alpha)\right) \\ &\leq 2 \exp\left(-\frac{1}{2} \min\left(\frac{\delta(\kappa_A, \alpha)^2 K_{B-}^2}{K_{B+} \kappa_A}, \frac{\delta(\kappa_A, \alpha) K_{B-}}{b_A}\right)\right) \\ &\leq 2 \exp\left(-\frac{1}{2} \frac{\delta(\kappa_A, \alpha)^2 K_{B-}^2}{K_{B+} \kappa_A}\right) \\ &\leq 2 \exp\left(-\frac{1}{2} \frac{\delta(\kappa_A, \alpha)^2 K_{B-}}{C_0 \kappa_A}\right) \leq \alpha/3 \end{aligned}$$

where the main inequality follows from Maximal Bernstein's Inequality applied over $t = \delta(\kappa_A, \alpha) K_{B-}$. For consistency with the notation in the online learning literature, define $\delta(\kappa_A, \alpha) = D^w(\alpha) \sim O\left(\sqrt{\frac{\ln(1/\alpha)}{K_{B-}}}\right)$. A similar argument shows that $\mathbb{P}\left(\left|\frac{1}{K_{pB}^\pi} \sum_{k=1}^{K_{pB}^\pi} M_{k, N_p}\right| \geq \delta(\kappa_M, \alpha)\right) \leq \alpha/3$ and $\mathbb{P}\left(\left|\frac{1}{K_{pB}^\pi} \sum_{k=1}^{K_{pB}^\pi} B_{k, N_p}^2\right| \geq \delta(\kappa_B, \alpha)\right) \leq \alpha/3$. Define the good event $\mathcal{E}$ as the intersection of the negation of the 3 events above, then $\mathbb{P}(\mathcal{E}) \geq 1 - \alpha$. Moreover, under event $\mathcal{E}$,

$$\begin{aligned} |\hat{b}_{K_{pB}, N_p}^2 - \theta_{N_p}^2| &= \left| \frac{1}{K_{pB}^\pi} \sum_{k=1}^{K_{pB}^\pi} B_{k, N_p}^2 + \frac{\theta_{N_p}^2}{K_{pB}^\pi - 1} - \frac{K_{pB}^\pi}{K_{pB}^\pi - 1} \left(\frac{1}{K_{pB}^\pi} \sum_{k=1}^{K_{pB}^\pi} M_{k, N_p} \right)^2 \right| \\ &\leq \delta(\kappa_B, \alpha) + C_0 \delta^2(\kappa_M, \alpha) + \left| \frac{\theta_{N_p}^2}{K_{B-} - 1} \right| \leq D^b(\alpha) \end{aligned}$$

where $D^b(\alpha) \sim O\left(\sqrt{\frac{\ln(1/\alpha)}{K_{B-}}}\right)$ (provided $\frac{\theta_{N_p}^2}{K_{B-} - 1} \sim O(1/K_{B-})$).

As a next step, define $\tilde{\tau}_{K, N}^2 = \hat{b}_{K, N}^2 - \hat{w}_{K, N}^2/N$. Applying the triangle inequality in event $\mathcal{E}$,

$$\begin{aligned} |\tilde{\tau}_{K_{pB}, N_p}^2 - \tau^2| &= |\hat{b}_{K_{pB}, N_p}^2 - \hat{w}_{K_{pB}, N_p}^2/N_p - \tau^2| = |\hat{b}_{K_{pB}, N_p}^2 - (\tau^2 + \sigma^2/N_p) - 1/N_p(\hat{w}_{K_{pB}, N_p}^2 - \sigma^2)| \\ &\leq |\hat{b}_{K_{pB}, N_p}^2 - (\tau^2 + \sigma^2/N_p)| + \frac{1}{N_p} |\hat{w}_{K_{pB}, N_p}^2 - \sigma^2| \leq D^b(\alpha) + \frac{1}{N_p} D^w(\alpha) = D^\tau(\alpha) \end{aligned}$$

with $D_\alpha^\tau \sim O\left(\sqrt{\frac{\ln(1/\alpha)}{K_{B-}}}\right)$. The same bound is mechanically preserved for $\hat{\tau}_{K_{pB}, N_p}^2 = \Pi_{[\tau_m, \tau_M]}(\tilde{\tau}_{K_{pB}, N_p}^2)$, i.e. the denominator entering the $\hat{\lambda}_{K, N_p}$ estimator. Similarly, define the unclipped version of $\hat{\lambda}$ as $\tilde{\lambda}$, then

$$\begin{aligned} |\tilde{\lambda}_{K_{pB}, N_p} - \lambda| &= \left| \frac{\hat{w}_{K_{pB}, N_p}^2 - \sigma^2}{\hat{\tau}_{K_{pB}, N_p}^2} - \frac{\sigma^2}{\tau^2} \right| = \left| \frac{\tau^2 \hat{w}_{K_{pB}, N_p}^2 - \sigma^2 \hat{\tau}_{K_{pB}, N_p}^2}{\tau^2 \hat{\tau}_{K_{pB}, N_p}^2} \right| = \left| \frac{\tau^2 (\hat{w}_{K_{pB}, N_p}^2 - \sigma^2) - \sigma^2 (\hat{\tau}_{K_{pB}, N_p}^2 - \tau^2)}{\tau^2 \hat{\tau}_{K_{pB}, N_p}^2} \right| \\ &\leq \left| \frac{\tau^2 (\hat{w}_{K_{pB}, N_p}^2 - \sigma^2) - \sigma^2 (\hat{\tau}_{K_{pB}, N_p}^2 - \tau^2)}{\tau^2 \tau_m} \right| \leq \frac{1}{\tau_m} (D^w(\alpha) + \lambda_M D^\tau(\alpha)) = D^\lambda(\alpha) \sim O\left(\sqrt{\frac{\ln(1/\alpha)}{K_{B-}}}\right) \end{aligned}$$

where the first inequality follows from clipping of $\hat{\tau}_{K_{pB}, N_p}^2$, and the second inequality follows from $\lambda \in [\lambda_m, \lambda_M]$. Clipping of $\tilde{\lambda}_{K_{pB}, N_p}$ leaves the bound unchanged. $\blacksquare$

Theorem C.8. *High Probability Bounds on EtC-CS and ACS. Under Assumptions 3.1, 3.2, 3.5 and 5.1, with probability at least $1 - \alpha$, there exists constants $D_3, D_4 < \infty$ independent of B such that*

$$\Delta_{\text{EtC-CS}} \leq D_3 \frac{\ln(1/\alpha)}{B^2} \quad \Delta_{\text{ACS}} \leq D_4 \frac{\ln(1/\alpha)}{B^2}$$

Proof of Theorem C.8. Observe that both EtC-CS and ACS satisfy the assumptions in Lemma C.7. In particular, for EtC-CS, $K_p^\pi = \lfloor K^*(\lambda_M) \rfloor$, so assumption holds with $C_0 = 1$, and for ACS $K_p^\pi = \max\{\lfloor K^*(\lambda_M) \rfloor, s - 1\} \in [\lfloor K^*(\lambda_M) \rfloor, \lfloor K^*(\lambda_m) \rfloor]$ a.s. with $C_0 = \lfloor K^*(\lambda_m) \rfloor / \lfloor K^*(\lambda_M) \rfloor = O(1) \perp B$. Then, under event $\mathcal{E}$

$$\begin{aligned} \Delta_\pi &= \frac{D_1}{B} (\hat{\lambda}_{K_{pB}, N_p} - \lambda_0)^2 + R_p \leq \frac{D_1}{B} D^\lambda(\alpha)^2 + R_p \\ &\leq \frac{D_2}{B} \frac{\ln(1/\alpha)}{\lfloor K^*(\lambda_M) \rfloor} + R_p \leq D_\pi \frac{\ln(1/\alpha)}{B^2} \end{aligned}$$

where the first equality follows from Lemma C.1, the second and third inequalities follow from Lemma C.7 under the good event $\mathcal{E}$, and the fourth inequality from noting that $\lfloor K^*(\lambda_M) \rfloor = \Theta(B) = \left\lfloor \frac{B}{F + V N^*(\lambda_M)} \right\rfloor$. The final inequality absorbs the term $R_p \sim O(B^{-5/2})$ within the constant D_π . All three constants (D_1, D_2, D_π) depend on N_p, C_0 and C_1 , but not on B . Theorem follows from noting that $\mathbb{P}(\mathcal{E}) \geq 1 - \alpha$ as per Lemma C.7. $\square$

Interpretation of Theorem C.8. In a nutshell, the theorem says that both algorithms implement optimal maps for some $\hat{\lambda}_{K_{pB}, N_p}^\pi$. By letting K_{pB}^π grow with budget, the learner can implement the oracle design in the limit. The proof is in some sense excessive for both EtC-CS and ACS but for different reasons. In the case of EtC-CS one needs to control deviations of $\hat{\lambda}$ for a deterministic pilot size, hence, a simple Bernstein inequality would have sufficed (without invoking maximal Bernstein nor the martingale properties of cumulative errors). This proof structure was in fact selected for ACS provided the pilot size is random under this policy. Yet, to effectively lower bound $\max\{\lfloor K^*(\lambda_M) \rfloor, s - 1\}$ by $\lfloor K^*(\lambda_M) \rfloor$ is overly conservative, provided $s \xrightarrow{B} K^*(\lambda_0)$ (possibly much bigger than $\lfloor K^*(\lambda_M) \rfloor$). Still,

the order of these bounds is otherwise unimprovable as maximal deviations in a window of length $O(1)$ do not induce any order loss.

C.5 Worst Case and Expected Excess Variance

Lemma C.9. *Worst Case Excess Variance.* Under Assumption 3.1 and 3.2, for any policy π such that $FK^\pi + VK^\pi N^\pi = B$ with $K^\pi \geq 2, N^\pi \geq 2$ and sufficiently large B , there exists a small positive δ such that

$$W(K^\pi, N^\pi) - W(K^*(\lambda_0), N^*(\lambda_0)) \leq \frac{\tau_M^2}{2} + \delta$$

Interpretation of Lemma C.9. As discussed in Section 3, estimator's variance is only driven to 0 under K asymptotics. As a result, any design which samples a finite number of clusters will incur non-vanishing excess variance. In particular, the lower bound in excess variance of these policies is $\tau^2/2$. Mathematically, the worst case excess variance is uniformly bounded by a constant.

Proof of Lemma C.9. For all policies satisfying the budget constraint, the variance minimization problem admits a representation as a single variable unconstrained minimization exercise of the form $\min_{2 \leq N \leq (B-2F)/(2V)} W(K(N), N)$ (see Appendix A). Additionally, $W(K(N), N)$ is convex in N , therefore the maximum is achieved at the boundaries, i.e. $W(K(2), 2)$ or $W(2, (B-2F)/(2V))$. The conclusion remains unaltered if the designer could pick from the extended set $N \in \{1, (B-F)/V\}$. Now, a simple calculation shows that

$$\begin{aligned} W(K(2), 2) - W(K^*(\lambda_0), N^*(\lambda_0)) &= \frac{\tau^2}{2B} \left(\sqrt{\lambda_0 F} - 2\sqrt{V} \right)^2 \\ W(2, (B-2F)/(2V)) - W(K^*(\lambda_0), N^*(\lambda_0)) &= \frac{\tau^2}{2} \left(1 + \frac{2\lambda_0 V}{B-2F} \right) - \frac{\tau^2}{B} \left(\sqrt{F} + \sqrt{\lambda_0 V} \right)^2 \end{aligned}$$

The first term converges $\xrightarrow{B} 0$, and the second $\xrightarrow{B} \frac{\tau^2}{2}$ from above (both deterministically). Therefore, the worst case excess variance is asymptotically upper bounded by $\frac{\tau^2}{2} + \delta$ for some small positive δ . Under Assumption 3.2, $\tau^2 < \tau_M^2$ completing the proof. $\square$

Proof of Theorem 5.2. Let $\alpha = \frac{C_\alpha}{B^2}$ for some finite C_α , then for any policy $\pi \in \{\text{EtC-CS}, \text{ACS}\}$,

$$\begin{aligned} \mathbb{E}_\pi [\Delta_\pi] &= \mathbb{P}_\pi(\bar{\mathcal{E}}) \cdot \mathbb{E}_\pi [\Delta_\pi \mid \bar{\mathcal{E}}] + \mathbb{P}_\pi(\mathcal{E}) \cdot \mathbb{E}_\pi [\Delta_\pi \mid \mathcal{E}] \\ &\leq \alpha \left(\frac{\tau_M^2}{2} + \delta \right) + C_\pi \frac{\ln(1/\alpha)}{B^2} \\ &= \frac{C_\alpha}{B^2} + C_3 \frac{\ln(B^2/C_\alpha)}{B^2} \leq C'_\pi \frac{\ln(B)}{B^2} \end{aligned}$$

where the key inequality follows from Theorem C.8 and Lemma C.9. α has been selected such that the second expression is minimized. Replacing C_π by C_3 and $C_{\pi'}$ by C_1 in the case of EtC-CS and C_π by C_4 and C'_π by C_2 in the case of ACS completes the proof. $\square$

D Design-Robust Inference for Asymptotically Even Policies

This Section can be read as a self-contained technical note on a general theory of inference for asymptotically even policies under symmetry. The main takeaway of this section is that, under a symmetry assumption on the errors, inference on sample averages remains unaffected when conditioning on designs induced by even or asymptotically even policies. As a disclaimer, this section uses N as the main asymptotic regime (indexed by i), and K as the number of arms (indexed by k), as it is customary on the literature of multi-armed adaptive policy.

D.1 Definitions

Definition D.1. *Random Sequences.* For each arm $k \in [K] = \{1, \dots, K\}$ define an infinite iid error sequence $(\varepsilon_s^k)_{s \geq 1}$ where s is the number of times arm k has been played. Consider two tie-breaking auxiliary random sequences $(U_i)_{i \geq 1}$ iid Unif $[0, 1]$ and $(\zeta_s^k)_{s \geq 1}$ iid Rademacher.

Definition D.2. *Sampling Protocol.* Every period $i = 1, 2, \dots, N$, (i) the learner selects an arm $k_i \in [K]$ according to some $\mathcal{F}$ -adapted policy π , (ii) the pull-count updates $N^k(i) = N^k(i-1) + \mathbb{1}(k_i = k)$ with $N^k(0) = 0$, and (iii) the learner observes

$$X_i = \mu_{k_i} + \varepsilon_{N^k(i)}^{k_i}$$

Let $\hat{\mu}_i^k = \hat{\mu}^k(N^k(i)) = N^k(i)^{-1} \sum_{j=1}^i \mathbb{1}(k_j = k) X_j$, defined on $\{N^k(i) > 0\}$. For its especial relevance write $N^k(N) = N_N^k$

Assumption D.3.

- i) *Moments.* $\mathbb{E}[\varepsilon_s^k] = 0$ and $\mathbb{E}[(\varepsilon_s^k)^2] = \sigma_k^2$ for all k, s .
- ii) *Independence.* The collections $(\varepsilon_s^k)_{k,s}, (U_i)_i, (\zeta_s^k)_{k,s}$ are mutually independent and independent across arms.
- iii) **Symmetry.** $\varepsilon_s^k \stackrel{d}{=} -\varepsilon_s^k$ for all k, s .

Definition D.4. *Even σ -algebra.* The (full) even σ -algebra is defined via $\mathcal{E}_\infty := \sigma\left((\varepsilon_s^k)_{k,s}^2, (U_i)_i\right)$

D.2 Signs and Magnitudes

Definition D.5. *Signs and Magnitudes.* For each (k, s) , let $\varepsilon_s^k = S_s^k R_s^k$ where

$$S_s^k = \mathbb{1}(\varepsilon_s^k \neq 0) \cdot \text{sign}(\varepsilon_s^k) + \mathbb{1}(\varepsilon_s^k = 0) \cdot \zeta_s^k, \quad R_s^k = |\varepsilon_s^k|$$

Lemma D.6. *Signs are Independent of Magnitudes.* Under Assumption D.3, for each (k, s) , $S_s^k \sim \text{Rad}$ and $S_s^k \perp\!\!\!\perp R_s^k$.

Proof of Lemma D.6. Fix (k, s) and drop subscripts and superscripts. Let $\Omega_S = \{-1, 1\}$ and $\mathcal{B}(\Omega_R)$ the Borel σ -algebra over some well behaved space $\Omega_R \subseteq \mathbb{R}_+$. Then, for any Borel Set $C \in \mathcal{B}(\Omega_R)$,

$$\mathbb{P}(S = 1, R \in C) = \mathbb{P}(\varepsilon > 0, \varepsilon \in C) + \frac{1}{2}\mathbb{P}(\varepsilon = 0)\mathbb{1}(0 \in C)$$

where equality follows from $\zeta \sim \text{Rad} \perp\!\!\!\perp \varepsilon$. By symmetry, $\mathbb{P}(\varepsilon > 0, \varepsilon \in C) = \mathbb{P}(-\varepsilon > 0, -\varepsilon \in C) = \mathbb{P}(\varepsilon < 0, \varepsilon \in C)$. Because the two events $\{\varepsilon < 0, \varepsilon \in C\}$ and $\{\varepsilon > 0, \varepsilon \in C\}$ partition $\{\varepsilon \in C \setminus 0\}$, $\mathbb{P}(\varepsilon > 0, \varepsilon \in C \setminus 0) = \frac{1}{2}\mathbb{P}(\varepsilon \in C \setminus 0)$, such that, together with the $\zeta \sim \text{Rad}$, $\mathbb{P}(S = 1, R \in C) = \frac{1}{2}\mathbb{P}(R \in C)$. Same argument holds for $\mathbb{P}(S = -1, R \in C)$. It follows $S_s^k \sim \text{Rad}$ and $S_s^k \perp\!\!\!\perp R_s^k$ (the joint law factorizes).

Lemma D.7. *Signs are Independent of the Even σ -algebra. Under Assumption D.3, $(S_s^k)_{k \geq 1, s \geq 1} \perp\!\!\!\perp \mathcal{E}_\infty$.*

Proof of Lemma D.7. The blocks $(R_s^k, S_s^k)_{k,s}$ are functions of $(\varepsilon_s^k, \zeta_s^k)_{k,s}$ which are independent of $(U_i)_i$ by Assumption. Independence across k joint with $S_s^k \perp\!\!\!\perp R_s^k$ (as per Lemma D.6) achieves the factorization of the joint law, i.e. $(S_s^k)_{k,s} \perp\!\!\!\perp (R_s^k)_{k,s}$. Since $\mathcal{E}_\infty = \sigma((\varepsilon_s^k)_{k,s}^2, (U_i)_i) = \sigma((R_s^k)_{k,s}^2, (U_i)_i) = \sigma((R_s^k)_{k,s}, (U_i)_i)$, $(\zeta_s^k)_{k,s} \perp\!\!\!\perp \mathcal{E}_\infty$ and $(S_s^k)_{k,s} \perp\!\!\!\perp \mathcal{E}_\infty$.

D.3 Conditional CLT for Fixed Design

Throughout this section, I focus on a single fixed arm k , so it is removed from superscripts and subscripts for clarity. Moreover, replace the possible random N_N^k by some fixed M . Assumption D.3 is imposed throughout.

Lemma D.8. *Frozen Conditions. Let R_s be iid with $\mathbb{E}[R_s^2] = \sigma^2$, then there is an almost sure event $\Omega_0 \in \mathcal{E}_\infty$ such that, for every $\delta > 0$*

$$(i) \quad \frac{1}{M} \sum_{s=1}^M R_s^2 \xrightarrow{\text{a.s.}} \sigma^2, \quad (ii) \quad \frac{1}{M} \sum_{s=1}^M R_s^2 \mathbb{1}\{R_s > \delta\sqrt{M}\} \xrightarrow{\text{a.s.}} 0$$

Remark D.9. Lemma D.8 is referred to as “Frozen Conditions” as the squared errors entering the upcoming results will become frozen after conditioning on the even σ -algebra, i.e. $\mathbb{E}[\varepsilon_s^2 \mid \mathcal{E}_\infty] = R_s^2$.

Proof of Lemma D.8. For part (i), the proof is immediate under the strong law of large numbers with finite expectation (given by finite second moments of the errors). For the Lindeberg Condition in part (ii), fix $C > 0$. For all $M \geq (C/\delta^2)$ note that $\delta\sqrt{M} \geq C$, so $\mathbb{1}\{R_s > \delta\sqrt{M}\} \leq \mathbb{1}\{R_s > C\}$. Then,

$$\limsup_{M \rightarrow \infty} \frac{1}{M} \sum_{s=1}^M R_s^2 \mathbb{1}\{R_s > \delta\sqrt{M}\} \leq \limsup_{M \rightarrow \infty} \frac{1}{M} \sum_{s=1}^M R_s^2 \mathbb{1}\{R_s > C\} = \mathbb{E}[R_1^2 \mathbb{1}\{R_1 > C\}]$$

where the equality follows from SLLN over the integrable iid summands $R_s^2 \mathbb{1}\{R_s > C\}$. Letting $C \rightarrow \infty$, dominated convergence sends the bound to 0. Part (ii) then holds on the a.s. intersection over a rational δ and integer C . Let Ω_0 be the almost sure event which intersects the almost sure events where each of the conditions hold.

Theorem D.10. *L-1 Convergence of the Conditional Characteristic Function. Define $Z(M) :=$*

$\frac{1}{\sqrt{M}} \sum_{s=1}^M \varepsilon_s$ with $Z(0) = 0$. Let $\phi_M(t) := \mathbb{E}[\exp(itZ(M)) \mid \mathcal{E}_\infty]$ be the **conditional** characteristic function and $\phi(t) = \exp(-\sigma^2 t^2/2)$. Then,

$$\mathbb{E} \mid \phi_M(t) - \phi(t) \mid \rightarrow 0 \quad \forall t \in \mathbb{R}$$

Proof of Theorem D.10. *Step 1. Pathwise Lindeberg-Feller.* Fix ω in the almost sure event Ω_0 and $t \in \mathbb{R}$. Let $Y_{M,s} = \varepsilon_s/\sqrt{M} = R_s S_s/\sqrt{M}$ such that $Z(M) = \sum_s Y_{M,s}$. As per Lemma D.7, conditional on $\mathcal{E}_\infty$, $Y_{M,s}$ is mean-zero and has variance R_s^2/M (as R_s are fixed and S_s are independent iid Rademacher, therefore the $\phi_M(t)$ is the unconditional characteristic function of an array with frozen R_s). Moreover,

$$\text{Var}(Z(M) \mid \mathcal{E}_\infty) = \sum_s \text{Var}(Y_{M,s} \mid \mathcal{E}_\infty) = \frac{1}{M} \sum_s R_s(\omega)^2 \xrightarrow{\text{a.s.}} \sigma^2$$

as per Lemma D.8 part (i), and satisfies the conditional Lindeberg condition

$$\sum_s \mathbb{E}[Y_{M,s}^2 \mathbb{1}\{|Y_{M,s}| > \delta\}] = \frac{1}{M} \sum_s R_s(\omega)^2 \mathbb{1}\{R_s(\omega) > \delta\sqrt{M}\} \xrightarrow{\text{a.s.}} 0$$

as per Lemma D.8 part (ii). The Lindeberg-Feller CLT for independent, non-identically distributed triangular arrays applies along the diverging sequence $M \rightarrow \infty$ such that

$$\phi_M(t)(\omega) = \mathbb{E} \left[\exp \left(it \sum_s Y_{M,s} \right) \mid \mathcal{E}_\infty \right] \rightarrow \exp(-\sigma^2 t^2/2)$$

Result holds for every $\omega \in \Omega_0$, therefore convergence is almost surely. *Step 2. L-1 convergence.* Note that $\mid \phi_M(t) - \phi(t) \mid \leq 2$ (provided all ch.f. are absolutely bounded by 1). Boundedness plus a.s. convergence implies L-1 convergence of the form $\mathbb{E}[\mid \phi_M(t) - \phi(t) \mid] \rightarrow 0$.

Remark D.11. Observe that the entire proof apparatus relies on $\varepsilon_s^2 = R_s^2$ being $\mathcal{E}_\infty$ -measurable. Then, by conditioning on $\mathcal{E}_\infty$, the conditional variances and conditional Lindeberg function reduce to textbook conditions on the frozen array and the conditional CLT reduces to an unconditional CLT for the frozen array.

Corollary D.12. *Uniform Mixing.* Let $\mathbb{E} \mid \phi_M(t) - \phi(t) \mid \rightarrow 0$, then

$$\sup_{A \in \mathcal{E}_\infty} \mid \mathbb{E}[\phi_M(t) \mathbb{1}_A] - \phi(t) \mathbb{P}(A) \mid \rightarrow 0, \quad \forall t \in \mathbb{R}$$

Equivalently, $Z(M) \rightarrow N(0, \sigma^2)$ $\mathcal{E}_\infty$ -uniformly mixing.

Proof of Corollary D.12. To recover a uniform mixing statement, observe that, for any

$\mathcal{E}_\infty$ -measurable event A

$$\begin{aligned}
| \mathbb{E}[\exp(itZ_M)\mathbb{1}_A] - \phi(t)\mathbb{P}(A) | &= | \mathbb{E}[\mathbb{E}[\exp(itZ_M)\mathbb{1}_A \mid \mathcal{E}_\infty]] - \mathbb{E}[\phi(t)\mathbb{1}_A] | \\
&= | \mathbb{E}[\mathbb{E}[\exp(itZ_M) \mid \mathcal{E}_\infty]\mathbb{1}_A - \phi(t)\mathbb{1}_A] | \\
&= | \mathbb{E}[(\phi_M(t) - \phi(t))\mathbb{1}_A] | \\
&\leq \mathbb{E}[| \phi_M(t) - \phi(t) |] \rightarrow 0
\end{aligned}$$

where the second equality follows from $\mathcal{E}_\infty$ -measurability of $\mathbb{1}_A$, and the convergence statement follows from L^1 convergence of $\phi_M(t) \rightarrow \phi(t)$. The result holds uniformly over A , hence it holds for $\sup_A \in \mathcal{E}_\infty$.

Remark D.13. *Uniform mixing implies convergence in distribution conditional on a worst case $A \in \mathcal{E}_\infty$. The uniform mixing statement allows the A events to depend on the diverging index M , i.e. $(A_M)_M$ in a way that standard pointwise mixing cannot handle.*

D.4 Even Policies and Terminal Fraction Events

Definition D.14. *Terminal Fraction Events.* For some set $\mathcal{K} \subseteq [K]$, let $\nu_N^\mathcal{K} = (N_N^k/N)_{k \in \mathcal{K}}$. A sequence $(A_N^\mathcal{K})_N$ is a terminal fraction event of the arm set $\mathcal{K}$ if $A_N^\mathcal{K} = \{\nu_N^\mathcal{K} \in C\}$ for some Borel subset $C \subseteq [0, 1]^\mathcal{K}$.

Definition D.15. *Policy Class.* A policy π is

1. **Even:** if $\nu_N^{[K]} \in \mathcal{E}_\infty$;
2. **Non- $\mathcal{K}$ -Starving:** if $\mathbb{P}(\nu_N^\mathcal{K} \geq c) \rightarrow 1$ for some set $\mathcal{K} \subseteq [K]$ and some constant $c > 0$;

Remark D.16. *Comment on Definition D.15.* Even policies are the policy class of interest in this technical note. If the policy could select k_i based on raw errors ε_s^k , then the terminal counts (and associated terminal fraction events) would not be $\mathcal{E}_\infty$ -measurable. Note that evenness does not immediately guarantee that the policy is valid in a bandit sense, as it may pick into the future. The analyst must surely verify that their policy is valid, yet that feature is immaterial for the terminal inference conducted in this document. As per non-starvation, clearly, as I move into asymptotic considerations, results can only hold for arms which are sampled infinitely often. Nothing prevents even policies from starving (stop playing) an arm, hence the assumption is necessary. Weaker forms of asymptotic divergence of the counts—not necessarily linear in N —are possibly sufficient for most results in this document, yet, linear non-starvation plays an important role when discussing asymptotically even policies. For simplicity, the assumption on asymptotic counts is presented consistently in this way.

D.5 Conditional CLT for (Random) Even Designs

Observation D.17. *Inference on Terminal Averages is Inference on $Z_k(N_N^k)$.*

$$\sqrt{N_N^k} (\hat{\mu}^k(N_N^k) - \mu_k) = \frac{1}{\sqrt{N_N^k}} \sum_{i=1}^N \mathbb{1}(k_i = k)(X_i - \mu_k) = \frac{1}{\sqrt{N_N^k}} \sum_{s=1}^{N_N^k} \varepsilon_s^k = Z(N_N^k)$$

Theorem D.18. *Mixing CLT Conditional on the Design.* Define $\phi_k(t_k) = \exp(-\sigma_k^2 t_k^2/2)$. For some subset $\mathcal{K} \subseteq [K]$, let $((N_N^k)_{k \in [K]}, A_N^{\mathcal{K}})$ be a design and terminal fraction event induced by an even, non- $\mathcal{K}$ -starving policy. Then,

$$\left| \mathbb{E} \left[\exp \left(i \sum_{k \in \mathcal{K}} t_k Z_k(N_N^k) \right) \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{P}(A_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi_k(t_k) \right| \rightarrow 0 \quad \forall (t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$$

Moreover, for any terminal fraction event sequence $(A_N^{\mathcal{K}})_N$ such that $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$,

$$\mathcal{L} \left(\left(\sqrt{N_N^k} (\hat{\mu}_k(N_N^k) - \mu_k) \right)_{k \in \mathcal{K}} \mid A_N^{\mathcal{K}} \right) \Longrightarrow N(\mathbf{0}, \text{diag}(\sigma_k^2)_{k \in \mathcal{K}})$$

Proof of Theorem D.18. *Step 1. Decomposition of the Joint Conditional Characteristic Function.* For any $(t_k)_k \in \mathbb{R}^{\mathcal{K}}$

$$\begin{aligned} \mathbb{E} \left[\exp \left(i \sum_{k \in \mathcal{K}} t_k Z_k(N_N^k) \right) \mathbb{1}_{A_N^{\mathcal{K}}} \right] &= \mathbb{E} \left[\mathbb{E} \left[\exp \left(i \sum_{k \in \mathcal{K}} t_k Z_k(N_N^k) \right) \mathbb{1}_{A_N^{\mathcal{K}}} \mid \mathcal{E}_{\infty} \right] \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[\sum_{(M^k)_{k \in \mathcal{K}}} \mathbb{1}_{\{(N_N^k)_{k \in \mathcal{K}} = (M^k)_{k \in \mathcal{K}}\}} \exp \left(i \sum_{k \in \mathcal{K}} t_k Z_k(M^k) \right) \mid \mathcal{E}_{\infty} \right] \mathbb{1}_{A_N^{\mathcal{K}}} \right] \\ &= \mathbb{E} \left[\sum_{(M^k)_{k \in \mathcal{K}}} \mathbb{1}_{\{(N_N^k)_{k \in \mathcal{K}} = (M^k)_{k \in \mathcal{K}}\}} \mathbb{E} \left[\exp \left(i \sum_{k \in \mathcal{K}} t_k Z_k(M^k) \right) \mid \mathcal{E}_{\infty} \right] \mathbb{1}_{A_N^{\mathcal{K}}} \right] \\ &= \mathbb{E} \left[\sum_{(M^k)_{k \in \mathcal{K}}} \mathbb{1}_{\{(N_N^k)_{k \in \mathcal{K}} = (M^k)_{k \in \mathcal{K}}\}} \prod_{k \in \mathcal{K}} \phi_{M^k}(t_k) \mathbb{1}_{A_N^{\mathcal{K}}} \right] \\ &= \mathbb{E} \left[\prod_{k \in \mathcal{K}} \phi_{N_N^k}(t_k) \mathbb{1}_{A_N^{\mathcal{K}}} \right] \end{aligned}$$

where the second equality follows by $\mathcal{E}_{\infty}$ -measurability of $A_N^{\mathcal{K}}$, the third equality by $\mathcal{E}_{\infty}$ -measurability of the indicator functions and the fact that the sum runs over a finite grid of values for the count vector. The fourth equality is a load-bearing step. After conditioning on $\mathcal{E}_{\infty}$, the magnitudes entering $Z_k(M^k)$ are frozen and the signs are *iid* and independent across arms, therefore the conditional expectation factorizes across k .

Step 2. L-1 Convergence along the diverging sequence. In Step 1 of Theorem D.10 it was shown that $\|\phi_M(t) - \phi(t)\| \xrightarrow[\text{M}]{\text{a.s.}} 0$ (in the almost sure event Ω_0). By applying Lemma C.3 to $\phi_M(t)$, it follows that $\|\phi_{N_N^k}(t) - \phi(t)\| \xrightarrow{P} 0$. A union bound over $k \in \mathcal{K}$ guarantees convergence in probability of the vector. The upgrade from convergence in probability to L -1 convergence is given by uniform boundedness. *Step 3. Uniform Mixing.* As per Corollary D.12, L -1 Convergence of the conditional characteristic function implies uniform mixing. In particular, the sequence $(A_N^{\mathcal{K}})_N$ is $\mathcal{E}_{\infty}$ -measurable under evenness. Finally, observe that $\mathbb{E}[\|\prod_k \phi_{N_N^k}(t_k) - \prod_k \phi_k(t_k)\|] \rightarrow 0$ provided $\phi_{N_N^k}(t_k) \rightarrow \phi_k(t_k)$ in L -1 for all k and all factors $(\phi_{N_N^k}(t_k), \phi(t_k))_{k \in \mathcal{K}}$ are uniformly bounded by 1. Result follows.

The second part of the theorem follows by dividing $\mathbb{P}(A_N^{\mathcal{K}})$ (which is asymptotically lower bounded under the Theorem statement) and applying Lévy's continuity theorem to switch between the (conditional) characteristic function and the (conditional) random variable.

Lemma D.19. *Variance Consistency. For some non-random diverging index M*

$$\hat{\sigma}_k^2(M) := M^{-1} \sum_{i=1}^M \mathbb{1}(k_i = k) (X_i - \hat{\mu}^k(M))^2 \xrightarrow[M]{a.s.} \sigma_k^2$$

Proof of Lemma D.19. As in Lemma D.8

$$\hat{\sigma}_k^2(M) = M^{-1} \sum_{s=1}^M (\varepsilon_s^k)^2 - \left(M^{-1} \sum_{s=1}^M \varepsilon_s^k \right)^2 \xrightarrow[M]{a.s.} \sigma_k^2$$

where convergence follows from the SLLN under *iid*-ness and finite second moments.

Corollary D.20. *Studentized Conditional Inference. Let $\sigma_k^2 > 0$ for all $k \in \mathcal{K}$. Let $((N^k)_{k \in [K]}, A_N^{\mathcal{K}})$ be a design and terminal fraction event induced by an even, non- $\mathcal{K}$ -starving policy, such that $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$. Then,*

$$\mathcal{L} \left(\left(\frac{\sqrt{N_N^k} (\hat{\mu}^k(N_N^k) - \mu_k)}{\hat{\sigma}_k(N_N^k)} \right)_{k \in \mathcal{K}} \mid A_N^{\mathcal{K}} \right) \Rightarrow N(\mathbf{0}, I)$$

Proof of Corollary D.20. Decompose

$$\left(\frac{Z_k(N_N^k)}{\hat{\sigma}_k(N_N^k)} \right)_{k \in \mathcal{K}} = \left(\frac{Z_k(N_N^k)}{\sigma_k} \right)_{k \in \mathcal{K}} \odot \left(\frac{\sigma_k}{\hat{\sigma}_k(N_N^k)} \right)_{k \in \mathcal{K}}$$

Next, recall from Lemma C.3 that $|\hat{\sigma}_k^2(M) - \sigma_k^2| \xrightarrow{a.s.} 0$, implies $\mathbb{P}(|\hat{\sigma}_k^2(N_N^k) - \sigma_k^2| \geq \eta) \rightarrow 0$, under non-starvation. Moreover,

$$\begin{aligned} \mathbb{P}(\|(\sigma_k / \hat{\sigma}_k(N_N^k))_{k \in \mathcal{K}} - \mathbf{1}\| > \delta \mid A_N^{\mathcal{K}}) &= \frac{\mathbb{P}(\{ \|(\sigma_k / \hat{\sigma}_k(N_N^k))_{k \in \mathcal{K}} - \mathbf{1}\| > \delta \} \cap A_N^{\mathcal{K}})}{\mathbb{P}(A_N^{\mathcal{K}})} \\ &\leq \frac{\mathbb{P}(\|(\sigma_k / \hat{\sigma}_k(N_N^k))_{k \in \mathcal{K}} - \mathbf{1}\| > \delta)}{\mathbb{P}(A_N^{\mathcal{K}})} \rightarrow 0 \end{aligned}$$

The numerator is well-defined because $\mathbb{P}(\hat{\sigma}_k(N_N^k)_{k \in \mathcal{K}} > \mathbf{0}) \rightarrow 1$. By Theorem D.18, $\mathcal{L}((Z_k(N_N^k)/\sigma_k)_{k \in \mathcal{K}} \mid A_N^{\mathcal{K}}) \Rightarrow N(\mathbf{0}, I)$. Then, the (vectorial) Slutsky theorem under the conditional law implies convergence of the pair. Observe that $(\hat{\sigma}_k^2)_{k \in \mathcal{K}}$ are not themselves $\mathcal{E}_\infty$ -measurable (as they demean by the group sample means $(\hat{\mu}^k)_{k \in \mathcal{K}}$ and not the true group means $(\mu_k)_{k \in \mathcal{K}}$). The result was only possible due to convergence in probability of the variance term to a constant.

D.6 Asymptotically Even Designs

Even policies are usually not feasible as the learner does not get access to the errors nor the squared errors but only to the random realizations X_i (see the definition of the Sampling Protocol in D.2). In practice, the learner may rely at best on quasi-even surrogates such as the sample variance $\hat{s}_k^2(i) = (N^k(i))^{-1} \sum_{j=1}^i \mathbb{1}(k_j = k)(X_j - \hat{\mu}^k(i))^2 = (N^k(i))^{-1} \sum_{j=1}^i \mathbb{1}(k_j = k)(\varepsilon_{N^k(j)}^{k_j} - \bar{\varepsilon}_{N^k(i)})^2$. The example below shows that $\hat{s}_k^2$ is not exactly even as it *leaks* relative signs.

Example D.21. Let $\bar{\varepsilon}_i = i^{-1} \sum_{s \leq N(i)} \varepsilon_s$ and $\hat{s}_i^2 = i^{-1} \sum_{s \leq N(i)} \varepsilon_s^2 - \bar{\varepsilon}_i^2$. Suppose a policy observes the sequence of sample variances $(\hat{s}_i^2)_i$ and the sequence of squared errors $(\varepsilon_s^2)_s \in \mathcal{E}_\infty$. Then it may recover $\bar{\varepsilon}_i^2 = i^{-1} \sum_{s \leq N(i)} \varepsilon_s^2 - \hat{s}_i^2$ for every i , and hence, from the identity $(i+1)\bar{\varepsilon}_{i+1} = i\bar{\varepsilon}_i + \varepsilon_{i+1}$

$$\bar{\varepsilon}_i \varepsilon_{i+1} = [(i+1)^2 \bar{\varepsilon}_{i+1}^2 - i^2 \bar{\varepsilon}_i^2 - \varepsilon_{i+1}^2] / (2i)$$

The policy can then recover the relative signs of $\text{sign}(\bar{\varepsilon}_i) \cdot \text{sign}(\varepsilon_{i+1})$ for every i , which are not $\mathcal{E}_\infty$ -measurable (as per Lemma D.7). A design built on $(\hat{s}_i^2)_i$ may therefore depend on information outside the σ -algebra and Theorem D.18 fails. The designer might think of conditioning on $\sigma((\varepsilon_s^k)_{k,s})$ instead of $\mathcal{E}_\infty$, yet, by doing so, $Z(N_N^k)$ becomes two-valued (as the signs $(S_s^k)_{k,s}$ are now degenerate), not Gaussian. The reason why the sample average is quasi-even, nearly even or asymptotically even is because the leakage is asymptotically negligible relative to the even component—i.e. $\bar{\varepsilon}_{N_N^k}^2 = O_p(1/N_N^k)$, while the even component is $(N_N^k)^{-1} \sum_{s \leq N_N^k} \varepsilon_s^2 = O_p(1)$. This gap licenses the notion of asymptotically even designs and asymptotically even events developed below.

Definition D.22. *Asymptotically Even Policies.* For some subset $\mathcal{K} \subseteq [K]$, a policy π is asymptotically even for $\mathcal{K}$ if it induces a terminal fraction design $\nu_N^\mathcal{K} = (N_N^k/N)_{k \in \mathcal{K}}$ such that

$$|\nu_N^\mathcal{K} - \tilde{\nu}_N^\mathcal{K}| \xrightarrow{P} 0, \quad \text{with } \tilde{\nu}_N^{[K]} \in \mathcal{E}_\infty$$

Remark D.23. A priori, the definition above does not restrict the conditioning set of the asymptotically even policy π in any meaningful way. In particular, π may condition on the raw errors $\varepsilon_s^k \notin \mathcal{E}_\infty$. Additionally, the definition above does not limit the underlying even policy which induces the terminal fractions $\nu_N^{[K]}$ in the same environment. However, in practice, the analyst will find it easiest to show asymptotic evenness of a candidate policy by considering a twin policy of π which has access to the true squared errors and then show that the terminal fraction designs of both policies converge.

Lemma D.24. *Asymptotically Even Events.* Let $(A_N^\mathcal{K})_N = \{\nu_N^\mathcal{K} \in C\}$ with $C \subseteq [0,1]^\mathcal{K}$ be a sequence of terminal fraction events generated by an asymptotically even policy π . Assume that $\lim_{\eta \downarrow 0} \limsup_N \mathbb{P}(\text{dist}(\nu_N^\mathcal{K}, \partial C) \leq \eta) = 0$, then, there exists a sequence of terminal fraction events $(\tilde{A}_N^\mathcal{K}) \in \mathcal{E}_\infty$ such that $\mathbb{P}(A_N^\mathcal{K} \triangle \tilde{A}_N^\mathcal{K}) \rightarrow 0$.

Proof of Lemma D.24. Consider the terminal fraction event $\tilde{A}_N^\mathcal{K} = \{\tilde{\nu}_N^\mathcal{K} \in C\}$, which is in $\mathcal{E}_\infty$ by evenness of $\tilde{\nu}_N^\mathcal{K}$. Recall, $A_N^\mathcal{K} \triangle \tilde{A}_N^\mathcal{K} = \{\mathbb{1}_C(\nu_N^\mathcal{K}) \neq \mathbb{1}_C(\tilde{\nu}_N^\mathcal{K})\} \subseteq \{\text{dist}(\nu_N^\mathcal{K}, \partial C) \leq |\nu_N^\mathcal{K} - \tilde{\nu}_N^\mathcal{K}|\}$. Then, by introducing a separating η

$$\mathbb{P}(A_N^\mathcal{K} \triangle \tilde{A}_N^\mathcal{K}) \leq \mathbb{P}(|\nu_N^\mathcal{K} - \tilde{\nu}_N^\mathcal{K}| > \eta) + \mathbb{P}(\text{dist}(\nu_N^\mathcal{K}, \partial C) \leq \eta)$$

where the first term goes to zero by definition of the asymptotically even policy. Taking $\lim \eta \downarrow 0$ sends the second term to zero, completing the proof.

Lemma D.25. *Increment Vanishes at Ratio-close Indices. Let $0 \leq (M_N), (\tilde{M}_N) \leq N$ be two integer-valued random indices such that $|M_N/N - \tilde{M}_N/N| \xrightarrow{P} 0$ and $\mathbb{P}(M_N/N \wedge \tilde{M}_N/N \geq c) \rightarrow 1$ for some $c > 0$. Then*

$$|Z(M_N) - Z(\tilde{M}_N)| \xrightarrow{P} 0 \quad (6)$$

Proof of Lemma D.25. Assume without loss (as the other case is symmetric) that $\tilde{M}_N \leq M_N$. Decompose

$$|Z(M_N) - Z(\tilde{M}_N)| \leq \left(\sqrt{\frac{\tilde{M}_N}{M_N}} - 1 \right) |Z(\tilde{M}_N)| + \frac{|S_{M_N} - S_{\tilde{M}_N}|}{\sqrt{M_N}}$$

where $S_M = \sum_{s=1}^M \varepsilon_s$. On $\{cN \leq \tilde{M}_N\}$, the term $|Z(\tilde{M}_N)| \leq \max_{cN \leq m \leq N} |S_m|/\sqrt{cN} = O_p(1)$ (under Kolmogorov's maximal inequality given *iid* errors with finite variance as in Assumption D.3). Moreover, under convergence in probability of the scaled indices and linear non-starvation, $|\sqrt{\tilde{M}_N/M_N} - 1| \xrightarrow{P} 0$, so their product is $o_p(1)$. As per the second term, fix $\delta \in (0, 1)$ and define the event $\{|M_N - \tilde{M}_N| \leq \delta N\}$ which has probability increasing to 1 by the conditions in the Lemma. Then,

$$\begin{aligned} \frac{|S_{M_N} - S_{\tilde{M}_N}|}{\sqrt{M_N}} &\leq \max_{\substack{cN \leq \tilde{m} \leq m \leq N, \\ |m - \tilde{m}| \leq \delta N}} \frac{|S_m - S_{\tilde{m}}|}{\sqrt{m}} \leq \max_{\substack{0 \leq \tilde{m} \leq m \leq N, \\ |m - \tilde{m}| \leq \delta N}} \frac{1}{\sqrt{c}} \frac{|S_m - S_{\tilde{m}}|}{\sqrt{N}} \\ &\leq \frac{1}{\sqrt{c}} \sup_{\substack{0 \leq s \leq t \leq 1 \\ |t-s| \leq \delta}} |W_N(t) - W_N(s)| \end{aligned}$$

where $W_N(t) = S_{\lfloor Nt \rfloor} / \sqrt{N}$ is the rescaled partial sum process. Under Assumption D.3, the sequence $(W_N)_N$ is tight in $D[0, 1]$ (Donsker's FCLT), which is equivalent to the local modulus-of-continuity property

$$\lim_{\delta \downarrow 0} \limsup_{N \rightarrow \infty} \mathbb{P} \left(\sup_{0 \leq s \leq t \leq 1, |t-s| \leq \delta} |W_N(t) - W_N(s)| \geq \eta \right) = 0 \quad \forall \eta > 0$$

Fix $\eta, \epsilon > 0$, and select $\delta > 0$ such that $\limsup_{N \rightarrow \infty} \mathbb{P} \left(\sup_{0 \leq s \leq t \leq 1, |t-s| \leq \delta} |W_N(t) - W_N(s)| \geq \eta \sqrt{c} \right) \leq \epsilon/2$. Since $\mathbb{P}(|M_N - \tilde{M}_N| > \delta N) \rightarrow 0$, for N large enough

$$\mathbb{P} \left(\frac{|S_{M_N} - S_{\tilde{M}_N}|}{\sqrt{M_N}} \geq \eta \right) \leq \mathbb{P} \left(|M_N - \tilde{M}_N| > \delta N \right) + \mathbb{P} \left(\sup_{\substack{0 \leq s \leq t \leq 1 \\ |t-s| \leq \delta}} |W_N(t) - W_N(s)| \geq \eta \sqrt{c} \right) \leq \epsilon$$

As η, ϵ were arbitrary, the term $|S_{M_N} - S_{\tilde{M}_N}| / \sqrt{M_N}$ is $o_p(1)$ and the claim follows.

Remark D.26. Lemma D.25 is purely a statement about the random walk $Z(\cdot)$. It does not condition on $\mathcal{E}_\infty$ and places no measurability requirement on M_N nor $\tilde{M}_N$. It will be used to bound the distance between the characteristic function of $Z(N_N^k)$ and $Z(\tilde{N}_N^k)$ with N_N^k non-measurable and the surrogate $\tilde{N}_N^k$ measurable.

Theorem D.27. *Asymptotically Even Theorems.* Let π be an asymptotically even, non- $\mathcal{K}$ -starving policy. Consider any terminal fraction event $A_N^{\mathcal{K}}$ such that $\lim_{\eta \downarrow 0} \limsup_N \mathbb{P}(\text{dist}(\nu_N^{\mathcal{K}}, \partial C) \leq \eta) = 0$. Then,

$$\left| \mathbb{E} \left[\exp \left(i \sum_{k=1}^{|\mathcal{K}|} t_k Z(N_N^k) \right) \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{P}(A_N^{\mathcal{K}}) \prod_{k=1}^{|\mathcal{K}|} \phi_k(t_k) \right| \rightarrow 0 \quad \forall (t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$$

Additionally, if $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$, then

$$\begin{aligned} \mathcal{L} \left(\left(\sqrt{N_N^k} (\hat{\mu}^k(N_N^k) - \mu_k) \right)_{k \in \mathcal{K}} \mid A_N^{\mathcal{K}} \right) &\Rightarrow N(\mathbf{0}, \text{diag}(\sigma_k^2)) \\ \mathcal{L} \left(\left(\frac{\sqrt{N_N^k} (\hat{\mu}^k(N_N^k) - \mu_k)}{\hat{\sigma}_k(N_N^k)} \right)_{k \in \mathcal{K}} \mid A_N^{\mathcal{K}} \right) &\Rightarrow N(\mathbf{0}, I) \end{aligned}$$

Remark D.28. The first statement cannot be written as $\mathcal{E}_\infty$ -uniform mixing, as $A_N^{\mathcal{K}}$ is not $\mathcal{E}_\infty$ -measurable. Accordingly, I do not define the conditional characteristic function $\phi_{N_N^k}$. In fact, I intentionally leave the σ -algebra entering this definition unspecified. Yet, the reader may assume that the design is measurable wrt to the filtration generated by the interaction of the policy and the environment.

Proof of Theorem D.27. I first argue that whenever an asymptotically even policy π is non- $\mathcal{K}$ -starving, then, the underlying even policy $\tilde{\pi}$ is also non- $\mathcal{K}$ -starving. For some $\eta > 0$

$$\mathbb{P}(\tilde{\nu}_N^{\mathcal{K}} < c) \leq \mathbb{P}(|\nu_N^{\mathcal{K}} - \tilde{\nu}_N^{\mathcal{K}}| > \eta) + \mathbb{P}(\nu_N^{\mathcal{K}} \leq c + \eta) \rightarrow 0$$

where the first term goes to 0 by the definition of asymptotic evenness and the second term goes to zero by non-starvation of π . Linear non-starvation plays a decisive role in this step by harmonizing the definition of asymptotically-even policies and non-starvation. Next, decompose

$$\begin{aligned} \left| \mathbb{E} \left[e^{i \sum_{k=1}^{|\mathcal{K}|} t_k Z(N_N^k)} \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{P}(A_N^{\mathcal{K}}) \prod_{k=1}^{|\mathcal{K}|} \phi_k(t_k) \right| &\leq \left| \mathbb{E} \left[e^{i \sum_{k=1}^{|\mathcal{K}|} t_k Z(N_N^k)} \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{E} \left[e^{i \sum_{k=1}^{|\mathcal{K}|} t_k Z(\tilde{N}_N^k)} \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}} \right] \right| \\ &\quad + \left| \mathbb{E} \left[e^{i \sum_{k=1}^{|\mathcal{K}|} t_k Z(\tilde{N}_N^k)} \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}} \right] - \prod_{k=1}^{|\mathcal{K}|} \phi_k(t_k) \mathbb{P}(\tilde{A}_N^{\mathcal{K}}) \right| \\ &\quad + \left| \prod_{k=1}^{|\mathcal{K}|} \phi_k(t_k) \right| \left| \mathbb{P}(A_N^{\mathcal{K}}) - \mathbb{P}(\tilde{A}_N^{\mathcal{K}}) \right| \end{aligned}$$

where $(\tilde{N}_N^k)_k$ is the even surrogate entering the definition of an asymptotic even policy, and $\tilde{A}_N^{\mathcal{K}}$ is defined accordingly. The third term immediately goes to 0 by Lemma D.24. For the second term, observe that $((\tilde{N}_N^k)_k, \tilde{A}_N^{\mathcal{K}})$ are $\mathcal{E}_\infty$ -measurable by construction and $(\tilde{N}_N^k)_k$ diverges with probability increasing to 1 (by non-starvation of the even policy π). Hence the freezing identity $\mathbb{E}[e^{i \sum_{k=1}^{|\mathcal{K}|} t_k Z(\tilde{N}_N^k)} \mid \mathcal{E}_\infty] = \prod_{k=1}^{|\mathcal{K}|} \phi_{\tilde{N}_N^k}(t)$ holds and Theorem D.18 applies verbatim to $((\tilde{N}_N^k)_{k \in \mathcal{K}}, \tilde{A}_N^{\mathcal{K}})$, sending the second term to 0. Decompose the first term further into

$$[1] \leq \left| \mathbb{E}[e^{i \sum_{k=1}^{|\mathcal{K}|} t_k Z(N_N^k)} - e^{i \sum_{k=1}^{|\mathcal{K}|} t_k Z(\tilde{N}_N^k)}] \mathbb{1}_{A_N^{\mathcal{K}}} \right| + \left| \mathbb{E}[e^{i \sum_{k=1}^{|\mathcal{K}|} t_k Z(\tilde{N}_N^k)} (\mathbb{1}_{A_N^{\mathcal{K}}} - \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}})] \right|$$

where the second term goes to zero provided it has modulus ≤ 1 and $\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \rightarrow 0$ (under Lemma D.24). For the first piece, recall that $|e^{ia} - e^{ib}| \leq |a - b|$, therefore

$$\leq \mathbb{E} \left| e^{i \sum_k t_k Z(N_N^k)} - e^{i \sum_k t_k Z(\tilde{N}_N^k)} \right| \leq \sum_{k=1}^{|\mathcal{K}|} \mathbb{E} \left[\min \left\{ 2, \left| t_k \cdot |Z(N_N^k) - Z(\tilde{N}_N^k)| \right| \right\} \right]$$

For every $k \in \mathcal{K}$, $\mathbb{P}(N_N^k/N \wedge \tilde{N}_N^k/N \geq c) \rightarrow 1$ and $N_N^k/\tilde{N}_N^k \xrightarrow{P} 1$, thus the indices satisfy the hypotheses in Lemma D.25. L -1 convergence follows from convergence in probability under uniform boundedness, i.e. the integrand is bounded by the uniform constant 2. Summing over a finite number of arms preserves convergence. The term then also goes to 0.

Once the first statement (analogue to Theorem D.18) has been established, the second and third statements follow as corollaries (without ever invoking measurability). In particular, the second statement follows by dividing by $\mathbb{P}(A_N^{\mathcal{K}})$ (which is asymptotically greater than 0) and applying Lévy's continuity theorem. Finally, studentization only requires convergence in probability of $\hat{\sigma}_k^2(N_N^k) \rightarrow \sigma_k^2$ (which holds under non-starvation) and $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$ (which holds by assumption). Measurability of N_N^k nor $A_N^{\mathcal{K}}$ were ever invoked in the proof.

Remark D.29. *In practice, many allocation rules based on second moments of the data (like sample variances) are further constrained by decision boundaries such as rounding functions, thresholds, clipping, argmaxes, etc. In principle, when the limit statistic sits at a positive distance from all boundaries, the decisions coincide with probability tending to one (satisfying Definition D.22). Early period disagreements are immaterial, provided the designer only conditions on asymptotic events such as terminal fraction events $\tilde{A}_N^{\mathcal{K}}$.*

D.7 Policy Examples

Proposition D.30. *Two-stage Neyman is Asymptotically Even. Let $\mathbf{0} < \sigma^2 = (\sigma_k^2)_k$. Let $0 < d \leq \min_k \sigma_k / \sum_l \sigma_l$ be a known constant to the designer, and set $M_N = dKN$. Assume $dN \in \mathbb{N}$. Consider a policy π which samples uniformly for M_N periods, so every arm is pulled at least $M_N^k = dN \in \mathbb{N}$ times. The policy then allocates the remaining $(1 - dK)N$ periods deterministically so the terminal fraction design $\nu_N^{[K]}$ equals the estimated Neyman target $\lfloor N f_k(\hat{\sigma}) \rfloor / N = \lfloor N \hat{\sigma}_k / \sum_\ell \hat{\sigma}_\ell \rfloor / N$, which I assume is implementable. Then, π is an asymptotically even and non- $[K]$ -starving with measurable surrogate $\tilde{\nu}_N^{[K]} = (\lfloor N f_k(q) \rfloor / N)_k$, where $q_k^2 = (M_N^k)^{-1} \sum_{s \leq M_N^k} (\varepsilon_s^k)^2$.*

Proof of Proposition D.30. Each arm receives a minimum of $M_N^k = dN$ samples, therefore, the policy π is non- $[K]$ -starving with constant $d - \eta$ for some $\eta > 0$. Under N asymptotics, both $q_k^2(M_N^k)$ and $\hat{\sigma}_k^2(M_N^k)$ to σ^2 a.s. under the SLLN. By Lipschitzness of the square root map, $|\hat{\sigma}_k - q_k|$ is $O_p(1/M_N^k)$. Further note that rounding is $O(1/N)$, in particular for any x , $|\lfloor Nx \rfloor / N - x| \leq 1/N$, thus for any x, y , $|\lfloor Nx \rfloor / N - \lfloor Ny \rfloor / N| \leq |x - y| + 1/N$. Finally, observe that for $\sum_l \sigma_l > 0$, hence f is Lipschitz in a neighborhood of σ . Bringing all results together for any $k \in [K]$

$$\begin{aligned} |N_N^k/N - \tilde{N}_N^k/N| &= |\lfloor Nf_k(\hat{\sigma}) \rfloor / N - \lfloor Nf_k(q) \rfloor / N| \leq |f_k(\hat{\sigma}) - f_k(q)| + O(1/N) \\ &\leq L |\hat{\sigma} - q| + O(1/N) = O_p(1/M_N^k) + O(1/N) = (O_p(1/N)) \end{aligned}$$

provided $(\hat{\sigma}, q)$ enter every neighborhood of σ with probability increasing to 1. Combining these results for every $k \in [K]$, it follows that $|\nu_N^{[K]} - \tilde{\nu}_N^{[K]}| \xrightarrow{p} 0$, thus satisfying Definition D.22.

Corollary D.31. *Design Conditional Inference for Two-Stage Neyman. Consider a set of K equally sized clusters with mean μ_k and variance σ_k^2 . Let $\mu = \frac{1}{K} \sum_k \mu_k$, and consider the estimators $\hat{\mu}_k = \frac{1}{N_N^k} \sum_s^{N_N^k} \mathbb{1}(k_s = k) X_{k,s}$ and $\hat{\mu} = \frac{1}{K} \sum_k \hat{\mu}_k(N_N^k)$, where N_N^k is given by protocol in Proposition D.30. Let $A_N^{[K]} = \{\nu_N^{[K]} \in C\}$ for any C such that $(f_k(\sigma))_k \notin \partial C$ and $\liminf_N \mathbb{P}(A_N^{[K]}) > 0$. Then, under the same assumptions as in Proposition D.30*

$$\mathcal{L} \left(\left(\frac{\sqrt{N_N^k}(\hat{\mu}_k(N_N^k) - \mu_k)}{\hat{\sigma}_k} \right)_{k \in [K]} \mid A_N^{[K]} \right) \rightarrow \mathcal{N}(\mathbf{0}, I)$$

Proof of Corollary D.31. By Proposition D.30, π is non- $[K]$ -starving and asymptotically even. $\nu_N^{[K]} \xrightarrow{p} (f_k(\sigma))_k$ (constant) provided $\hat{\sigma} \rightarrow \sigma$, continuity of f and rounding error of $O(1/N)$, therefore the condition $\lim_{\eta \downarrow 0} \limsup_N \mathbb{P}(\text{dist}(\nu_N^{[K]}, \partial C) \leq \eta) = 0$ is immediately satisfied. Theorem D.27 applies and the statement follows.

E Proofs of Section 6

The goal of this section is to translate the results from the standalone Appendix D into the K vs N setting, and, in doing so, present thoroughly and prove the results in Section 6. Although the theory is completely symmetric, the K vs N problem presents some specific features compared to the generic adaptive data-collection problem discussed in the Appendix. Firstly, the K vs N problem contains two sequences of errors $(u_k)_k$ and $(\varepsilon_{ki})_{k,i}$ and two degrees of freedom entering the terminal design, only one of which is divergent in B . Second, while showing that EtC-CS is asymptotically even will follow closely the constructive method for two-stage Neyman—showing that a single statistic $\hat{\lambda}_p$ and its even surrogate converge in probability—the argument for ACS is more involved. The design in ACS depends on a stopping time governed by a sequence of statistics, thus, one needs to show convergence between the statistic $\hat{\lambda}_s$ indexed by the random stopping time s and the statistic $\tilde{\lambda}_{\bar{s}}$ indexed by the even counterpart of s , which depends on the sequence of even statistics $(\tilde{\lambda}_k)_{k \leq \bar{s}}$.

The definitions and lemmas in this section have been reduced to a minimum, and appropriate

connections to the general theory, as well as proof devices, are made throughout. As such, I recommend reading this Section in light of Appendix D.

E.1 Definitions

Definition E.1. *Random Sequences in K vs N Problem.* For some fixed N , define an infinite iid error sequence $(M_{k,N})_{k \geq 1} = (u_k + \frac{1}{N} \sum_{i=1}^N \varepsilon_{ki})_k$, where k is the cluster's number. Consider two tie-breaking auxiliary random sequences $(U_k)_{k \geq 1}$ iid $\text{Unif}[0, 1]$ and $(\zeta_k)_{k \geq 1}$ iid Rademacher.

Under Assumption 3.1 and 6.1, $(M_{k,N})_{k \geq 1}$ are iid, mean zero and symmetric $(M_{k,N} \stackrel{d}{=} -M_{k,N})$, with $\mathbb{E}[M_{k,N}^2] = \theta_N^2 < \infty$, i.e. the sequence $(M_{k,N})_{k \geq 1}$ satisfies the equivalent to Assumption D.3 in Appendix D.

E.2 Signs and Magnitudes

Proof of Lemma 6.3 and Corollary 6.5. The proof follows the same structure as in Lemmas D.6 and D.7. Simply define the cluster-sign mapping $g : C_k \mapsto -C_k$. Then, for some fixed N , $M_{k,N} = M_{k,N}(C_k) = -M_{k,N}(g(C_k))$, while $W_{k,N_p}^2(C_k) = W_{k,N_p}^2(g(C_k))$ and $M_{k,N}^2(C_k) = (-M_{k,N})^2(g(C_k)) = M_{k,N}^2(g(C_k))$. The rest of the proof follows verbatim by replacing $(\varepsilon_s^2)_{k,s}$ by $(W_{k,N_p}^2, M_{k,N}^2)_{k \geq 1, N \geq 1}$.

E.3 Conditional CLT for Fixed Design

I now present a small section (not discussed in the main text), where I establish uniform mixing of $Z^\pi(B)$ over $\mathcal{G}_\infty$ when the design (K^π, N^π) is fixed. In particular, consider a deterministic design $(K_{0B}, N_{0B}) \in \mathbb{N}^2 \geq \mathbf{2}$ such that $\lim_{B \rightarrow \infty} K_{0B} = \infty$ and $\lim_{B \rightarrow \infty} N_{0B} = N_0$.

Corollary E.2. *Frozen Conditions in the K vs N Problem.* There is an almost sure event $\Omega_0 \in \mathcal{G}_\infty$ such that, for every $\delta > 0$

$$(i) \quad \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} R_{k,N_0(B)}^2 \xrightarrow[B]{a.s.} \theta_{N_0}^2 = \tau^2 + \frac{\sigma^2}{N_0}, \quad (ii) \quad \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} R_{k,N_0(B)}^2 \mathbb{1}\{R_{k,N_0(B)} > \delta \sqrt{K_{0B}}\} \xrightarrow[B]{a.s.} 0$$

Proof of Corollary E.2. Result follows by intersection of the almost sure events $\{N_{0B} = N_0\}$ (provided $N_{0B} \in \mathbb{N}$) and the convergence results for fixed N_0 as given by Lemma D.8, i.e. $\frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} R_{k,N_0}^2 \xrightarrow[B]{a.s.} \theta_{N_0}^2 = \tau^2 + \frac{\sigma^2}{N_0}$. Convergence for conditional Lindeberg is analogous.

Corollary E.3. *L-1 Convergence of the Conditional Characteristic Function for K vs N Problem.* Define $Z^0(B) := \frac{1}{\sqrt{K_{0B}}} \sum_{k=1}^{K_{0B}} M_{k,N_{0B}}$ under the deterministic map (K_{0B}, N_{0B}) . Let $\phi_B^0(t) := \mathbb{E}[\exp(itZ^0(B)) \mid \mathcal{G}_\infty]$ be the conditional characteristic function under the deterministic map and $\phi(t) = \exp(-t^2 \theta_{N_0}^2 / 2)$. Then,

$$|\phi_B^0(t) - \phi(t)| \xrightarrow[B]{a.s.} 0, \quad \mathbb{E} |\phi_B^0(t) - \phi(t)| \rightarrow 0 \quad \forall t \in \mathbb{R}$$

Proof of Corollary E.3. Corollary mimics Theorem D.10 in light of Corollary E.2.

E.4 Conditional CLT for (Random) Even Designs

From now on, I consider random (data-dependent) mappings (K_B^π, N_B^π) as those induced by $\pi \in \{\text{EtC-CS}, \text{ACS}\}$.

Proof of Theorem 6.8. Proof mirrors the structure of Theorem D.18 and Corollary D.20. In particular, for fixed B , Step 1 in Theorem D.18 goes through by summing over the finite grid of budget feasible designs (K_B^π, N_B^π) . Step 2 in Theorem D.18 goes through under a similar composition argument to the one identified in Lemma C.3. More precisely, let $T_B := \sup_{K_{0B} \geq cB} |\phi_B^0(t) - \phi(t)| \xrightarrow{\text{a.s.}} 0$ for every $c \in (0, 1]$. Then,

$$\mathbb{P}(|\phi_B(t) - \phi(t)| > \eta) \leq \mathbb{P}(N_B \neq N_0) + \mathbb{P}(K_B < cB) + \mathbb{P}(T_B > \eta) \rightarrow 0$$

where the first term converges to 0 in probability under design's N -consistency and the second term goes to 0 under non-starvation. Uniform mixing in Step 3 of Theorem D.18 follows under boundedness of the (conditional) characteristic function. Part (ii) of Theorem follows by dividing by $\mathbb{P}(A_B)$ (eventually lower bounded away from zero) and applying Lévy's Continuity Theorem. Finally, for the studentized extension in Part (iii), observe that the sample variance given by the deterministic map (K_{0B}, N_{0B}) converges a.s. to θ_{N_0} , i.e. $\hat{b}_{K_{0B}, N_{0B}}^2 \xrightarrow{\text{a.s.}} \theta_{N_0}^2$ (see Lemma 4.1). Convergence in probability under N -consistency and non-starvation follows the same composition argument as above.

E.5 Asymptotically Even Designs

Corollary E.4. *Asymptotically Even Events.* Let $(A_B)_B = \{(\nu_B^\pi, N_B^\pi) \in C\}$ for some Borel subset C be a sequence of terminal fraction events generated by an asymptotically even policy π . Assume that $\lim_{\eta \downarrow 0} \limsup_B \mathbb{P}(\text{dist}((\nu_B^\pi, N_B^\pi), \partial C) \leq \eta) = 0$, then, there exists a sequence of terminal fraction events $(\tilde{A}_B) \in \mathcal{G}_\infty$ such that $\mathbb{P}(A_B \triangle \tilde{A}_B) \rightarrow 0$.

Proof of Corollary E.4. Consider the terminal fraction event $\tilde{A}_B = \{(\tilde{\nu}_B, \tilde{N}_B) \in C\}$, which is in $\mathcal{G}_\infty$ by evenness of $(\tilde{\nu}_B, \tilde{N}_B)$. Recall, $A_B \triangle \tilde{A}_B = \{\mathbb{1}_C(\nu_B^\pi, N_B^\pi) \neq \mathbb{1}_C(\tilde{\nu}_B, \tilde{N}_B)\} \subseteq \{\text{dist}((\nu_B^\pi, N_B^\pi), \partial C) \leq \|(\nu_B^\pi, N_B^\pi) - (\tilde{\nu}_B, \tilde{N}_B)\|\}$. Then, by introducing a separating η

$$\begin{aligned} \mathbb{P}(A_B \triangle \tilde{A}_B) &\leq \mathbb{P}(\|(\nu_B^\pi, N_B^\pi) - (\tilde{\nu}_B, \tilde{N}_B)\| > \eta) + \mathbb{P}(\text{dist}((\nu_B^\pi, N_B^\pi), \partial C) \leq \eta) \\ &\leq \mathbb{P}(|\nu_B^\pi - \tilde{\nu}_B| > \eta) + \mathbb{P}(|N_B^\pi - \tilde{N}_B| > \eta) + \mathbb{P}(\text{dist}((\nu_B^\pi, N_B^\pi), \partial C) \leq \eta) \end{aligned}$$

where the first and second term go to zero by definition of the asymptotically even policy. Taking $\lim_{\eta \downarrow 0}$ sends the second term to zero, completing the proof.

Lemma E.5. *Increment Vanishes at Ratio-close Indices.* For some fixed B , let $0 \leq \tilde{K}_B, K_B^\pi \leq \bar{K}$ be two integer valued random indices such that $|\nu_B^\pi - \tilde{\nu}_B| \xrightarrow{\frac{p}{B}} 0$ and $\mathbb{P}(\nu_B^\pi \wedge \tilde{\nu}_B \geq c) \rightarrow 1$ for some $c > 0$. Additionally, let $0 \leq \tilde{N}_B, N_B^\pi \leq \bar{N} \in \mathbb{N}$, such that $\mathbb{P}(|N_B^\pi - \tilde{N}_B| \geq \eta) \rightarrow 0$. Define $\tilde{Z}(B)$ as the normalized error sum under the even policy. Then,

$$|Z^\pi(B) - \tilde{Z}(B)| \xrightarrow{\frac{p}{B}} 0$$

Proof of Lemma E.5.

$$|Z^\pi(B) - \tilde{Z}(B)| \leq \left| \frac{1}{\sqrt{K_B^\pi}} \sum_{k=1}^{K_B^\pi} M_{k,N_B^\pi} - \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k=1}^{\tilde{K}_B} M_{k,N_B^\pi} \right| + \left| \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k=1}^{\tilde{K}_B} M_{k,N_B^\pi} - \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k=1}^{\tilde{K}_B} M_{k,\tilde{N}_B} \right|$$

The terms ν_B^π, ν_B satisfy the conditions of Lemma D.25, therefore the first term $\xrightarrow{P} 0$. The second term cancels exactly with probability increasing to 1 under assumption $\mathbb{P}(|N_B^\pi - \tilde{N}_B| \geq \eta) \rightarrow 0$.

Theorem E.6. *Asymptotically Even Theorems in K vs N . Let π be an asymptotically even, non-starving, N -consistent policy. Consider any terminal fraction event A_B such that $\lim_{\eta \downarrow 0} \limsup_B \mathbb{P}(\text{dist}((\nu_B^\pi, N_B^\pi), \partial C) \leq \eta) = 0$. Then,*

$$|\mathbb{E}[\exp(itZ^\pi(B))\mathbb{1}_{A_B}] - \phi(t)\mathbb{P}(A_B)| \rightarrow 0 \quad \forall t \in \mathbb{R}$$

Additionally, if $\liminf_B \mathbb{P}(A_B^\pi) > 0$, then

$$\begin{aligned} \mathcal{L}\left(\sqrt{K_B^\pi}(\hat{\mu}(K_B^\pi, N_B^\pi) - \mu) \mid A_B\right) &\implies N(0, \theta_{N_0}^2) \\ \mathcal{L}\left(\frac{\sqrt{K_B^\pi}(\hat{\mu}(K_B^\pi, N_B^\pi) - \mu)}{\hat{b}_{K_B^\pi, N_B^\pi}} \mid A_B\right) &\implies N(0, 1) \end{aligned}$$

Proof of Theorem E.6. Theorem E.6 supersedes the sketch Theorem 6.10 in text. The proof mirrors the structure in Theorem D.27, with the additional step of arguing that whenever an asymptotically even policy π is N -consistent, then, the underlying even policy is also N -consistent (with the same N_0 constant limit). For some $\eta > 0$,

$$\mathbb{P}(\tilde{N}_B \neq N_0) \leq \mathbb{P}(\tilde{N}_B \neq N_B^\pi) + \mathbb{P}(N_B^\pi \neq N_0) \rightarrow 0$$

where convergence follows from asymptotic evenness and N -consistency of π , respectively. The rest of the proof follows as expected, invoking the renewed Lemma E.5 and the updated mixing theorems where needed.

E.6 Application to EtC-CS and ACS

Just like in the Policy Examples of Appendix, the argument is concluded by showing that the policies of interest (EtC-CS and ACS) are asymptotically-even, non-starving and consistent. To do so, I present a set of intermediate lemmas.

Lemma E.7. $\hat{\lambda}_{K,N_p}$ is Nearly Even. Fix N_p and consider some deterministic map K_{0B} such that $\lim_{B \rightarrow \infty} K_{0B}/B = c + \eta$ for some $c, \eta > 0$. Define $\tilde{b}_{K,N_p}^2 = \theta_{N_p}^2 + \frac{1}{K} \sum_{k=1}^K B_{k,N_p}^2$ and $\tilde{\lambda}_{K,N_p} = g(\hat{w}_{K,N_p}^2, \tilde{b}_{K,N_p}^2)$. Then,

$$i) (\hat{w}_{K_{0B},N_p}^2, \tilde{b}_{K_{0B},N_p}^2, \tilde{\lambda}_{K_{0B},N_p}) \in \mathcal{G}_\infty;$$

$$ii) \mid \hat{b}_{K_{0B}, N_p}^2 - \tilde{b}_{K_{0B}, N_p}^2 \mid \xrightarrow[B]{a.s.} 0$$

$$iii) \tilde{\lambda}_{K_{0B}, N_p} \xrightarrow[B]{a.s.} \lambda_0 \text{ and } \mid \hat{\lambda}_{K_{0B}, N_p} - \tilde{\lambda}_{K_{0B}, N_p} \mid \xrightarrow[B]{a.s.} 0$$

iv) Let $s, \tilde{s}$ be two possibly random stopping times such that $s, \tilde{s} \geq cB$ with $c > 0$ almost surely. Then, $\mid \hat{\lambda}_{s-1, N_p} - \tilde{\lambda}_{\tilde{s}-1, N_p} \mid \xrightarrow[B]{a.s.} 0$

Proof of Lemma E.7. (i) $\hat{w}_{K_{0B}, N_p}^2 = \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} W_{k, N_p}^2$ with $W_{k, N_p}^2 \in \mathcal{G}_\infty$, therefore $\hat{w}_{K_{0B}, N_p}^2 \in \mathcal{G}_\infty$. Similarly, $\tilde{b}_{K_{0B}, N_p}^2 = \theta_{N_p}^2 + \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} B_{k, N_p}^2$ with B_{k, N_p}^2 a function of $M_{k, N_p}^2 \in \mathcal{G}_\infty$, therefore $\tilde{b}_{K_{0B}, N_p}^2 \in \mathcal{G}_\infty$. Finally, $\tilde{\lambda}(\cdot, \cdot)$ is a (deterministic) measurable map of $\mathcal{G}_\infty$ -measurable arguments, hence, it is $\mathcal{G}_\infty$ -measurable.

(ii) Recall the expansion of $\hat{b}_{K_0(B), N_p}^2$ in the definitions of Appendix C,

$$\begin{aligned} \hat{b}_{K_{0B}, N_p}^2 &= \theta_{N_p}^2 + \frac{1}{K_{0B} - 1} \sum_{k=1}^{K_{0B}} B_{k, N_p}^2 + \frac{\theta_{N_p}^2}{K_{0B} - 1} - \frac{K_{0B}}{K_{0B} - 1} \bar{M}_{K_{0B}, N_p}^2 \\ &= \tilde{b}_{K_{0B}, N_p}^2 + \frac{\theta_{N_p}^2}{K_{0B} - 1} - \frac{K_{0B}}{K_{0B} - 1} \bar{M}_{K_{0B}, N_p}^2 \end{aligned}$$

Thus it suffices to show that the second and third terms in the last equation $\xrightarrow[B]{a.s.} 0$. As $B \rightarrow \infty$, the second term deterministically decreases to 0 for every finite $\theta_{N_p}^2$. For the final term, recall that $\bar{M}_{K_{0B}, N_p} \xrightarrow[B]{a.s.} 0$ under the SLLN. Therefore, the third term is $\xrightarrow[B]{a.s.} 0$.

(iii) Recall, $\tilde{\lambda}_{K_{0B}, N_p} = \Pi_{[\lambda_m, \lambda_M]} \left(\frac{\hat{w}_{K_{0B}, N_p}^2}{\Pi_{[\tau_m, \tau_M]} [\hat{b}_{K_{0B}, N_p}^2 - \hat{w}_{K_{0B}, N_p}^2 / N_p]} \right)$. From Lemma 4.1, $\hat{w}_{K_{0B}, N_p}^2 \xrightarrow[B]{a.s.} \sigma^2$, and by part (ii) $\tilde{b}_{K_{0B}, N_p}^2 \xrightarrow[B]{a.s.} \theta_{N_p}^2$. Applying the continuous map theorem, and noting that projection maps do not interfere with convergence (provided $\tau \in [\tau_m, \tau_M]$ and $\lambda_0 \in [\lambda_m, \lambda_M]$), then $\tilde{\lambda}_{K_{0B}, N_p} \xrightarrow[B]{a.s.} \lambda_0$. From Lemma 4.1, $\hat{\lambda}_{K_{0B}, N_p} \xrightarrow[B]{a.s.} \lambda_0$, therefore, by the triangle inequality/set inclusion, $\mid \hat{\lambda}_{K_{0B}, N_p} - \tilde{\lambda}_{K_{0B}, N_p} \mid \xrightarrow[B]{a.s.} 0$.

(iv) The proof is immediate

$$\mid \hat{\lambda}_{s-1, N_p} - \tilde{\lambda}_{\tilde{s}-1, N_p} \mid \leq \sup_{k, \tilde{k} \geq cB} \mid \hat{\lambda}_{k-1, N_p} - \tilde{\lambda}_{\tilde{k}-1, N_p} \mid \leq \sup_{k \geq cB} \mid \hat{\lambda}_{k-1, N_p} - \lambda_0 \mid + \sup_{k \geq cB} \mid \tilde{\lambda}_{k-1, N_p} - \lambda_0 \mid$$

But from Lemma C.3 and Part (iii) it follows that for any k such that the event $\{\lim_{B \rightarrow \infty} k_B = \infty\}$ holds a.s., both $\hat{\lambda}_{k, N_p}$ and $\tilde{\lambda}_{k, N_p} \rightarrow \lambda_0$ almost surely. Result follows.

Proposition E.8. *EtC-CS is Asymptotically Even. EtC-CS is an asymptotically even, non-starving and N-consistent policy with even surrogate EtC-CS = EtC-CS($\tilde{\lambda}_{K_p, N_p}$).*

Proof of Proposition E.8. By Corollary C.5, EtC-CS is non-starving. Moreover, by Lemma 4.4, $(K_B^{\text{EtC-CS}}/B, N_B^{\text{EtC-CS}}) = (K^*(\hat{\lambda}_{K_p, N_p})/B, N^*(\hat{\lambda}_{K_p, N_p})) \xrightarrow[B]{P} (K^*(\lambda_0)/B, N^*(\lambda_0)) := (\nu_0, N_0)$, by CMT. Hence, EtC-CS is a non-starving policy with constant $\nu_0 - \eta > 0$ (for some $\eta > 0$) and N-constant. Similarly, by Lemma E.7, $\tilde{\lambda}_{K_p, N_p} \xrightarrow[B]{P} \lambda_0$, therefore $(K^{\text{EtC-CS}}/B, N^{\text{EtC-CS}}) = (K^*(\tilde{\lambda}_{K_p, N_p})/B,$

$N^*(\tilde{\lambda}_{K_p, N_p}) \xrightarrow{B} (\nu_0, N_0)$. Because $\tilde{\lambda}_{K_p, N_p} \in \mathcal{G}_\infty$ (by Lemma E.7), then EtC-CS is even and EtC-CS is asymptotically even.

Proposition E.9. *ACS is Asymptotically Even. Let $\tilde{s}$ be the random stopping time induced by Equation (5) under the sequence of even statistics $(\tilde{\lambda}_{k, N_p})_k$. Then ACS is an asymptotically even, non-starving and N -consistent policy with even surrogate $\tilde{ACS} = ACS(\tilde{\lambda}_{\tilde{s}, N_p})$.*

Proof of Proposition E.9. By Corollary C.5, ACS is non-starving. Moreover, by Lemma 4.4, 4.4, $(K^{ACS}/B, N^{ACS}) = (K^*(\hat{\lambda}_{s-1, N_p})/B, N^*(\hat{\lambda}_{s-1, N_p})) \xrightarrow{B} (\nu_0, N_0)$, by CMT. So ACS is non-starving with constant $\nu_0 - \eta > 0$ (for some $\eta > 0$) and N -constant. Similarly, $(\tilde{K}^{ACS}/B, \tilde{N}^{ACS}) = (K^*(\tilde{\lambda}_{\tilde{s}-1, N_p})/B, N^*(\tilde{\lambda}_{\tilde{s}-1, N_p})) \xrightarrow{B} (\nu_0, N_0)$. Because $(\tilde{s}, (\tilde{\lambda}_{k, N_p})_{k \leq \tilde{s}}) = f((\tilde{\lambda}_{k, N_p})_{k \leq \tilde{s}})$ with $(\tilde{\lambda}_{k, N_p})_{k \leq \tilde{s}} \in \mathcal{G}_\infty$ (by Lemma E.7), then $\tilde{ACS}$ is even and ACS is asymptotically even.

Proof of Corollary 6.11. Proof follows under Theorem E.6, provided both EtC-CS and ACS are asymptotically-even, non-starving and N -consistent (as per Propositions E.8 and E.9). Observe that the margin condition $\lim_{\eta \downarrow 0} \limsup_B \mathbb{P}(\text{dist}((\nu_B, N_B), \partial C) \leq \eta) = 0$ could be replaced by the simpler $(\nu_0, N_0) \notin \partial C$ provided $(\nu_B^\pi, N_B^\pi) \xrightarrow{B} (\nu_0, N_0)$ by Lemma 4.4.

F Proofs of Section 7

F.1 Lower Bound Derivation

Policy Class. I first define the relevant competitor policy class Π . The policy class is fairly general and contains most reasonable policies as especial cases. In particular, I consider policies which, in every period t , select a cluster $k_t = k$ and sample at least one unit $N_t^{k_t} \geq 1$ until they run out of budget. As a result, the game contains at most $T = K_m N_M$ rounds with strictly positive sampling. If no additional observation can be sampled without violating the budget in period t_M , $N_t^{k_t} = 0$ for all $t \geq t_M$. Importantly, the same cluster k can be revisited several times. Let $N_T^k = \sum_{t=1}^T \mathbb{1}(k_t = k) N_t^{k_t}$ be the number of samples from cluster k up to period T . Define as well $K_T = \sum_k \mathbb{1}(N_T^k > 0)$. All policies must satisfy the budget constraint almost surely, i.e.

$$\mathbb{P}_\pi \left(\sum_{k \in [K_T]} (F + V N_T^k) \leq B \right) = 1$$

Data Generating Process. Consider two **normally** distributed Data Generating Processes (DGPs) y_-, y_+ indexed by ω_- and ω_+ , where $\omega_- = (\tau_-^2, \lambda_-)$ and $\omega_+ = (\tau_+^2, \lambda_+)$ where $\lambda_- = \lambda_0 - \delta$, $\lambda_+ = \lambda_0 + \delta$ for some $\lambda_0 \in (\lambda_m, \lambda_M)$ and a small positive δ . Additionally, $\tau_-^2 = \sigma^2/\lambda_-$ and $\tau_+^2 = \sigma^2/\lambda_+$. Under the additive errors model in Equation 1

$$y_{+ki} = \mu + u_{+k} + \varepsilon_{ki}$$

where $u_{+k} \sim N(0, \tau_+^2)$. y_{-ki} and u_{-k} are defined as expected.

Separating Events. As it is common practice in lower bound derivations, I characterize a separating event. Following the characterization of the variance for the generalized game (where $N^k \neq N$ for all k), I write the separating event in terms of deviations from the harmonic mean N^H . To do so, I

lower bound the excess variance in terms of the harmonic mean. By the linearity of the cost structure $K_T(F + \sum_{k \in [K_T]} V N_T^k) = K_T(F + V \bar{N}_T)$ where $\bar{N}_T = \frac{1}{K_T} \sum_{k \in [K_T]} N_T^k$ is the arithmetic mean. Recall that $N_T^H \leq \bar{N}_T$, hence $\frac{1}{K_T} \geq \frac{F + V N_T^H}{B}$. It follows

$$W(\mathbf{N}_T) = \frac{\tau^2}{K_T(\mathbf{N}_T)} \left(1 + \frac{\lambda}{N_T^H(\mathbf{N}_T)} \right) \geq \frac{\tau^2(F + V N_T^H(\mathbf{N}_T))}{B} \left(1 + \frac{\lambda}{N_T^H(\mathbf{N}_T)} \right) = W^H(N^H(\mathbf{N}_T))$$

Now, take the environment $\omega = \omega_+$, let $N_+ = N^*(\lambda_+)$ and consider the event $\{N^H \leq N_0\}$. Under the remarks above, the excess variance of a policy which selects $\mathbf{N}_T$ such that $N^H(\mathbf{N}_T) \leq N_0$ can be lower bounded by

$$W(\mathbf{N}_T, \omega_+) - W(N_+, \omega_+) \geq W^H(N_H, \omega_+) - W^H(N_+, \omega_+) \geq W^H(N_0, \omega_+) - W^H(N_+, \omega_+)$$

where I used the fact that $W(N_+, \omega_+) = W^H(N_+, \omega_+)$ because the oracle allocation is a balanced one (under a simple convexity argument). A symmetric result holds for ω_- . Finally, by observing the parallel between $W^H(N, \omega)$ and $W(N, \omega)$ in Appendix A, the second order Taylor approximation around λ_+ takes the form

$$W^H(N_0, \omega_+) - W^H(N_+, \omega_+) = \frac{\tau^2 \sqrt{FV}}{4B\lambda_0^{3/2}} (\lambda_0 - \lambda_+)^2 + O((\lambda_0 - \lambda_+)^3/B)$$

Augmented Experiment and Divergence Decomposition. For some policy $\pi \in \Pi$, let $(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the environments y_- and y_+ , respectively. Now, consider the augmented experiment where, every time the learner samples cluster k , she observes both $(y_{ki})_{N_k}$ **and** u_k . Let $(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the augmented environments. By data processing,

$$D_{\text{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$$

The augmented game simplifies the analysis of the KL-divergence significantly, as revisiting a cluster does not add any more *cluster-level* KL. This fact, together with the zero KL contribution of individual errors (provided they have the same distribution in both environments), allows us to decompose $D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$ in the usual *bandit style*,³⁰ i.e.

$$D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) = \mathbb{E}_{\pi-} [K_T] D_{\text{KL}}(N(0, \tau_-^2), N(0, \tau_+^2))$$

The expression above incorporates the fact that all clusters within each environment are equally distributed. Moreover, for all policies $\pi \in \Pi$, $FK_T + \sum_k V N_T^k \leq B$ a.s., thus $K_T \leq B/F$ a.s.. Next,

³⁰See Chapter 15 in [Lattimore and Szepesvári \(2020\)](#) for a complete treatment.

$$\begin{aligned}
D_{\text{KL}}(N(0, \tau_-^2), N(0, \tau_+^2)) &= \frac{1}{2} \left[\frac{\tau_-^2}{\tau_+^2} - 1 - \ln \left(\frac{\tau_-^2}{\tau_+^2} \right) \right] = \frac{1}{2} \left[\frac{\lambda_+}{\lambda_-} - 1 - \ln \left(\frac{\lambda_+}{\lambda_-} \right) \right] \\
&= \frac{1}{2} \phi \left(\frac{\lambda_+}{\lambda_-} \right) = \frac{1}{2} \phi \left(1 + \frac{2\delta}{\lambda_0 - \delta} \right)
\end{aligned}$$

where the third equality serves as definition of $\phi(x) = x - 1 - \ln(x)$. For $\delta \leq \lambda_0/2$, and using the fact that $\phi(1+x) \leq x^2/2$

$$\frac{1}{2} \phi \left(\frac{\lambda_+}{\lambda_-} \right) \leq \frac{1}{2} \phi \left(1 + \frac{4\delta}{\lambda_0} \right) \leq \frac{4\delta^2}{\lambda_0^2}$$

Therefore,

$$D_{\text{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq \frac{B}{F} \frac{4\delta^2}{\lambda_0^2} = C_1 B \delta^2$$

The argument is now completed through a standard Bretagnolle-Huber argument

$$\begin{aligned}
&\tilde{\mathbb{E}}_{\pi-} [W(\mathbf{N}, \omega_-)] - W(K^*(\lambda_-), N^*(\lambda_-), \omega_-) + \tilde{\mathbb{E}}_{\pi+} [W(\mathbf{N}, \omega_+)] - W(K^*(\lambda_+), N^*(\lambda_+), \omega_+) \geq \\
&\tilde{\mathbb{E}}_{\pi-} [W^H(N_H, \omega_-)] - W^H(N_-, \omega_-) + \tilde{\mathbb{E}}_{\pi+} [W^H(N_H, \omega_+)] - W^H(N_+, \omega_+) \geq \\
&\tilde{\mathbb{P}}_{\pi-} \left(N_H > \sqrt{\lambda_0 F/V} \right) \cdot \left(\mathbb{E}_{\pi-} [W^H(N_H, \omega_-) \mid N_H > \sqrt{\lambda_0 F/V}] - W^H(N_-, \omega_-) \right) + \\
&\quad \tilde{\mathbb{P}}_{\pi+} \left(N_H \leq \sqrt{\lambda_0 F/V} \right) \cdot \left(\mathbb{E}_{\pi+} [W^H(N_H, \omega_+) \mid N_H \leq \sqrt{\lambda_0 F/V}] - W^H(N_+, \omega_+) \right) = \\
&\frac{\tau^2 \sqrt{FV}}{4B\lambda_0^{3/2}} \delta^2 \left(\tilde{\mathbb{P}}_{\pi-} \left(N_H > \sqrt{\lambda_0 F/V} \right) + \tilde{\mathbb{P}}_{\pi+} \left(N_H \leq \sqrt{\lambda_0 F/V} \right) \right) + O(\delta^3/B) \geq \\
&\frac{\tau^2 \sqrt{FV}}{4B\lambda_0^{3/2}} \delta^2 \frac{1}{2} \exp \left(-D_{\text{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \right) - \frac{C_3 \delta^3}{B} \geq \\
&\frac{\tau^2 \sqrt{FV}}{4B\lambda_0^{3/2}} \delta^2 \frac{1}{2} \exp \left(-D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) \right) - \frac{C_3 \delta^3}{B} \geq \\
&\frac{\tau^2 \sqrt{FV}}{8B\lambda_0^{3/2}} \delta^2 \exp \left(-C_1 \delta^2 B \right) - \frac{C_2 \delta^3}{B}
\end{aligned}$$

Where the first inequality follows from the variance relaxation under the harmonic mean, the second inequality is a simple event partition, the third inequality follows from the second order Taylor expansion, the fourth inequality follows from Bretagnolle-Huber inequality, the fifth inequality from a data processing argument, and the final inequality from the derivations above. Finally, by selecting $\delta = \sqrt{1/(C_1 B)}$, which is smaller than $\lambda_0 \leq \lambda/2$ for sufficiently large B ,

$$\frac{\tau^2 \sqrt{FV}}{8B\lambda_0^{3/2}} \frac{1}{C_1 B} \exp(-1) - \frac{C_2 \delta^3}{B} \geq \frac{C_3}{B^2} - \frac{C_4}{B^{5/2}}$$

for some finite and strictly positive C_3 . Result follows from applying the inequality $\max(a, b) \geq \frac{a+b}{2}$ and absorbing it (as well as the higher order terms) into a constant C . $\square$

F.2 Policy Comparison

Proof of Lemma 7.3. The first part of the Lemma is immediate, and follows from a simple second order approximation around λ_0 . See Lemma C.2 for a complete derivation. For the minmax optimal policy, recall that

$$W(K^*(\lambda), N^*(\lambda)) - W(K^*(\lambda_0), N^*(\lambda_0)) = \frac{\tau^2}{B} \left[F + V\lambda_0 + \sqrt{FV} \left(\sqrt{\lambda} + \frac{\lambda_0}{\sqrt{\lambda}} \right) \right] - \frac{\tau^2(\sqrt{F} + \sqrt{V\lambda_0})^2}{B}$$

$$\frac{\tau^2\sqrt{FV}}{B} \cdot \frac{(\sqrt{\lambda} - \sqrt{\lambda_0})^2}{\sqrt{\lambda}}$$

Hence, the maximum is achieved at $\tau^2 = \tau_M$ and $\lambda_0 \in \{\lambda_m, \lambda_M\}$ (whichever yields a higher loss). Hence, the minmax policy is given by

$$\min_{\lambda} \max \left\{ \frac{(\sqrt{\lambda} - \sqrt{\lambda_M})^2}{\sqrt{\lambda}}, \frac{(\sqrt{\lambda} - \sqrt{\lambda_m})^2}{\sqrt{\lambda}} \right\}$$

which yields $\lambda_{mM} = \left(\frac{\sqrt{\lambda_m} + \sqrt{\lambda_M}}{2} \right)^2$. Under a second order approximation

$$W(K^*(\lambda_{mM}), N^*(\lambda_{mM})) - W(K^*(\lambda_0), N^*(\lambda_0)) = \frac{C(\lambda_{mM} - \lambda_0)^2}{B} + O\left(\frac{|\lambda_{mM} - \lambda_0|^3}{B}\right) \sim O(1/B)$$

□

G Additional Tables and Graphs

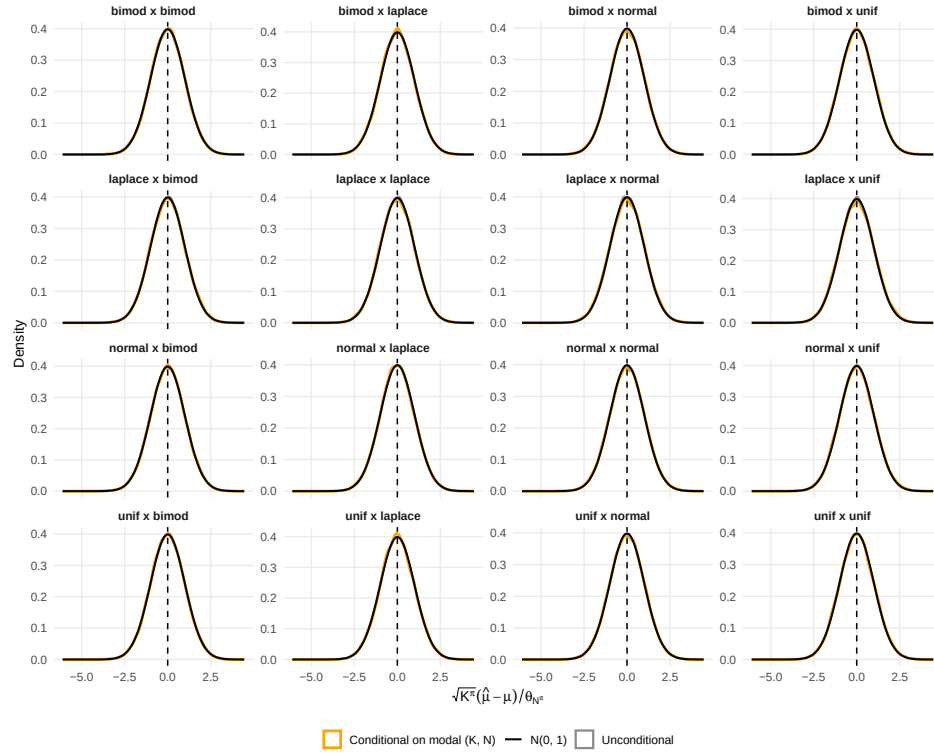


Figure 12: Empirical pdf for Modal Design Conditional Sample Average $\sqrt{K^\pi}(\hat{\mu}_{K^\pi, N^\pi} - \mu)/\theta_{N^\pi}$ across Symmetric Distributions. u_k distribution (left). ε_{ki} distribution (right). $\mu = 0$, $\omega = (1, 5)$. $B = 2000$, $F = 20$, $V = 2$. 100,000 simulations.

G.1 Identification of Symmetry beyond Skewness

TBC